

Solar Home Lighting

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FINAL REPORT

Change Record

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Table of Contents

Table of Contents.....	3
List of Tables.....	4
List of Figures.....	5
1. Executive Summary.....	6
2. Introduction.....	7
2.1. Background.....	7
2.2. Overview.....	8
2.3. Referenced Documents and Standards.....	9
3. Operating Concept.....	10
3.1. Scope.....	10
3.2. Operational Description and Constraints.....	10
3.3. System Description.....	10
3.4. Modes of Operations.....	11
3.4.1 Automatic Mode.....	11
3.4.2 Manual Mode.....	11
3.4.3 Security Mode.....	12
3.5. Users.....	12
3.6. Support.....	12
4. Scenarios.....	13
4.1. Airbnb Host #1.....	13
4.2. Independent/Off-Grid #2.....	13
4.3. Nighttime Security Monitoring.....	13
5. Analysis.....	13
5.1. Summary of Proposed Improvements.....	13
5.2. Disadvantages and Limitations.....	14
5.3. Alternatives.....	14
5.4. Impact.....	15
No table of figures entries found.....	5
1. Overview.....	5
2. References and Definitions.....	6
2.1. References.....	6
2.2. Definitions.....	6
3. Physical Interface.....	7
3.1. Weight.....	7
3.1.1. Physical Controls Subsystem Designed Weight.....	7
3.1.2. ESP32 Camera and Application Subsystem Designed Weight.....	7
3.1.3 MPPT Charge Controller Designed Weight.....	8
3.2. Dimensions.....	8
3.2.1. Physical Controls Subsystem Dimensions.....	8

3.2.2. ESP32 Camera and Application Subsystem Dimensions.....	8
3.2.3 MPPT Charge Controller Dimensions.....	9
3.3. Mounting Locations.....	9
3.3.1. Physical Controls Subsystem Panel.....	9
3.3.2. Physical Controls Subsystem Sensors.....	9
3.3.3. ESP32 Camera and Application Subsystem.....	9
3.3.4 MPPT Charge Controller.....	10
4. Thermal Interface.....	10
4.1. Physical Controls Subsystem.....	10
4.1.1. Physical Controls Thermal Parameters.....	10
4.2. ESP32 Camera and Application Subsystem.....	11
4.2.1. Camera and Application Thermal Parameters.....	11
4.3. MPPT Charge Controller Subsystem.....	11
4.3.1. MPPT Charge Controller Thermal Parameters.....	11
5. Electrical Interface.....	12
5.1. Primary Input Power.....	12
5.1.1. Physical Controls Subsystem.....	12
5.1.2. ESP32 Camera and Application Subsystem.....	12
5.1.3. MPPT Charge Controller Subsystem.....	12
5.2. Voltage and Current Ratings.....	12
5.2.1. Physical Controls Subsystem Internal Power Consumption.....	12
5.2.2. Physical Controls Subsystem Lighting Output Load.....	13
5.2.3. Camera Subsystem and Application.....	13
5.2.4 MPPT Charge Controller.....	13
5.3. Polarity Reversal.....	14
5.4. Signal Interfaces.....	14
5.4.1. Physical Controls Subsystem Analog and Digital Signals.....	14
5.4.2. MPPT Charge Controller Analog and Digital Signals.....	15
5.4.3. MPPT Charge Controller Communication Interface.....	16
6. Communications / Device Interface Protocols.....	17
6.1. Physical Controls Subsystem Communication Interface.....	17
6.1.1. Physical Controls Communication Parameters.....	17
6.2. Wireless Communications (WiFi).....	17
6.3. Host Device.....	17
6.4. Video Interface.....	17
List of Tables.....	4
List of Figures.....	5
1. Introduction.....	5
1.1. Purpose and Scope.....	5
Figure 1.1.1. Conceptual Product Layout.....	7
1.2. Responsibility and Change Authority.....	7

Final Report
Solar Home Lighting

2. Applicable and Reference Documents.....	8
2.1. Applicable Documents.....	8
2.1.1. Applicable Documents Table.....	8
2.2. Reference Documents.....	8
2.2.1. Reference Documents Table.....	9
2.3. Order of Precedence.....	9
3. Requirements.....	9
3.1. System Definition.....	9
Figure 3.1.1. Block Diagram of System.....	10
3.2. Characteristics.....	10
3.2.1. Functional / Performance Requirements.....	10
3.2.1.1. Lighting Duration.....	10
3.2.1.2. Motion Detection Accuracy.....	10
3.2.1.3. Camera Streaming Latency.....	11
3.2.1.4. Database Storage.....	11
3.2.1.5. User Control and Responsiveness.....	11
3.2.1.6. Motion Triggered Lighting.....	11
3.2.1.7. System Start-Up Sequence.....	11
3.2.1.8. System Start-Up Sequence.....	11
3.2.1.9. Hardware and Mobile Interface Interaction.....	12
3.2.2. Physical Characteristics.....	12
3.2.2.1. Subsystem Size.....	12
3.2.2.2. Battery Housing.....	12
3.2.2.3. Cable Management and Mounting.....	12
3.2.3. Electrical Characteristics.....	12
3.2.3.1. Inputs.....	12
3.2.3.1.1. Table of Inputs.....	13
3.2.3.1.1. Power Consumption.....	13
3.2.3.1.2. Input Voltage Range.....	13
3.2.3.1.3 DC/DC Conversion Efficiency.....	13
3.2.3.1.4. Overcurrent and Short Circuit Protection.....	14
3.2.3.1.4. Data and Control Signals.....	14
3.2.3.2. Outputs.....	14
3.2.3.2.1. Energy Output.....	14
3.2.3.2.2. Battery Status Output.....	14
3.2.3.2.3. Raw Video Output.....	14
3.2.3.2.4. Connectors.....	14
3.2.3.2.5. Wiring.....	15
3.2.4. Environmental Requirements.....	15
3.2.4.1. Thermal.....	15
3.2.4.2. Rain.....	15

Final Report
Solar Home Lighting

3.2.4.3. Solar Radiation.....	15
3.2.4.2. Wind.....	15
3.2.5. Failure Propagation.....	16
3.2.5.1. Failure Detection, Isolation, and Recovery (FDIR).....	16
3.2.5.1.1 Built-In Test (BIT).....	16
3.2.5.1.1.1. BIT Critical Fault Detection.....	16
3.2.5.1.1.2. BIT False Alarms.....	16
3.2.5.1.1.3. BIT Log.....	16
3.2.5.1.2 Isolation and Recovery.....	16
4. Support Requirements.....	16
Appendix A: Acronyms and Abbreviations.....	18
Appendix B: Definition of Terms.....	19
B. Definitions of Terms.....	19
Table of Contents.....	3
List of Tables.....	5
List of Figures.....	6
1. Introduction.....	7
2. Charge Controller Subsystem.....	7
2.1. Subsystem Introduction.....	7
2.2. Subsystem Overview.....	7
2.3. Solar Panel to Battery Buck Converter (18.8V to 14.4V - 13.6V).....	8
2.3.1. Operation.....	8
2.3.2. Simulation Validation.....	8
Figure 2.3.3 14.4V Buck conversion from OC Voltage.....	8
Figure 2.3.4 14.4V Buck conversion from Nominal 18.8V.....	9
Figure 2.3.5 14.4V Buck conversion from low voltage input.....	9
2.3.2.1. 18.8V to 13.6V Step down (ESP32 DAC implementation).....	9
Figure 2.3.2.2 Verification of ESP32 DAC implementation.....	10
2.4. 3.3V Buck Converter.....	10
2.4.1. Operation.....	10
2.4.2. Simulation Validation.....	11
Figure 2.1.3 3.3V LM3150 Buck converter during Bulk/Absorption Stage.....	11
2.5. ESP32 Perturb & Observe (P&O) Program.....	11
2.5.1. Operation.....	11
2.5.2. Program Validation.....	11
Figure 2.5.3 ESP32 adjusting DAC according to Battery SOC.....	12
Figure 2.5.4 ESP32 P&O according to efficiency of system.....	13
2.6. Subsystem Conclusion.....	13
3. Physical Controls Subsystem.....	13
3.1. Subsystem Introduction.....	13
3.2. Subsystem Overview.....	14

Figure 3.2.1. Physical Controls System Overview.....	14
3.3. 3.3V Buck Converter.....	15
3.3.1. Operation.....	15
3.3.2. Validation.....	15
3.3.2.1. Line Regulation.....	15
Figure 3.3.2.1. Line Regulation $I_{out} = 0A$	16
Figure 3.3.2.2. Load Regulation $I_{out} = 1.75A$	16
3.3.2.3. Load Regulation.....	17
Figure 3.3.2.3. Load Regulation.....	17
3.3.2.4. Efficiency.....	18
Figure 3.3.2.4. Efficiency at 3.3V.....	18
3.3.2.5. Output Voltage Ripple.....	19
Figure 3.3.2.5.1. No load.....	19
Figure 3.3.2.5.2. Nominal Load $I_{out} = 0.8$ Amps.....	20
3.3.2.6. Percent Error.....	20
Table 3.3.2.6. Percent Error Results.....	21
3.4. 12V Buck-Boost Converter.....	21
3.4.1. Operation.....	21
3.4.2. Partial Validation.....	22
3.4.2.1. Line Regulation.....	22
Figure 3.4.2.1. Line Regulation $I_{out} = 0A$	23
Figure 3.4.2.2. Load Regulation $I_{out} = 1.75A$	23
3.4.2.3. Load Regulation.....	23
Figure 3.4.2.3. Load Regulation.....	24
3.4.2.4. Efficiency.....	24
Figure 3.4.2.4. Efficiency at 3.3V.....	25
3.4.2.5. Output Voltage Ripple.....	25
Figure 3.4.2.5.1. No load.....	26
Figure 3.4.2.5.2. Nominal Load $I_{out} = 0.8$ Amps.....	26
3.4.2.6. Percent Error.....	27
Table 3.4.2.6. Percent Error Results.....	27
3.4.1. Failure Root Cause Analysis.....	28
Figure 3.4.1.1. Expected V_c Waveform During Correct Operation.....	28
Figure 3.4.1.2. V_c Waveform at Failure.....	29
3.5. ESP32 and TFT Touchscreen.....	30
3.5.1. Operation.....	30
3.5.2. Validation.....	30
3.5.2.1. Power-On Behavior and Initialization.....	31
3.5.2.2. Touchscreen Mapping and Interface Validation.....	31
3.5.2.3. System Runtime Stability.....	31
3.5.2.4. Relay Trigger.....	32

- 3.6. Subsystem Conclusion..... 32
- 4. AI Security Camera Subsystem..... 33
 - 4.1. Subsystem Introduction..... 33
 - 4.2. Subsystem Overview..... 33
 - Figure 4.2: Functional Block Diagram of AI Security Camera Subsystem..... 33
 - 4.3. Subsystem Validation..... 34
 - 4.3.1. Frame Rate (FPS) Stability Test..... 34
 - Figure 4.3.1: Application FPS Over 1 Minute Validation Test..... 34
 - Table 4.3.1: Application FPS Over 1 Minute Validation Test Data..... 35
 - 4.3.2. AI Latency Test..... 35
 - Figure 4.3.2: Latency Test Results with Average Graph..... 35
 - Table 4.3.2: Latency Test Results Data..... 36
 - 4.3.3. Connection Robustness Test..... 37
 - Figure 4.3.3: Camera Reliability Validation Test Visual Graph..... 37
 - 4.4. Subsystem Conclusion..... 38
- 5. MCU, Sensors, and Database Subsystem..... 38
 - 5.1 Subsystem Introduction..... 38
 - 5.2 Microcontroller Unit (MCU)..... 38
 - 5.2.1 Operation..... 38
 - 5.2.2 Validation..... 39
 - Table 5.2: MCU Subsystem Validation Summary..... 39
 - 5.3 Sensors..... 40
 - 5.3.1 Operation..... 40
 - 5.3.2 Validation..... 41
 - Figure 5.3.1: PIR Detection..... 41
 - Figure 5.3.2: Digital Light Sensor Readings..... 42
 - 5.4 Database..... 42
 - 5.4.1 Operation..... 42
 - 5.4.2 Validation..... 43
 - Figure 5.4: Database Telemetry..... 43
 - 5.5 Conclusion..... 43
- Table of Contents..... 3
- List of Tables..... 4
- List of Figures..... 5
- 1. Introduction..... 6
- 2. System Overview..... 7
 - Figure 2.1: System Block and Connectivity Diagram..... 7
- 3. Integrated System..... 8
 - 3.1 Integration Overview..... 8
 - 3.2 Power System Integration..... 9
 - 3.3 Control and Interface Integration..... 10

3.4 Sensor and Automation Integration.....	10
3.5 Communication and Database Integration.....	11
3.6 Security System Integration.....	12
4. Integrated System Validation and Performance Results.....	12
4.1 Introduction.....	12
4.2 Power Delivery and Rail Stability Validation.....	13
Table 4.2.1 12 V Output Voltage Under Load.....	13
Figure 4.2.1 12 V Output Voltage vs Load Condition.....	14
4.2.2 3.3 V Rail Stability Under Full System Load.....	14
Table 4.2.3 3.3 V Output Voltage Under Full System Load.....	14
Table 4.2.4 3.3 V Rail Summary Metrics.....	14
4.2.3 Summary.....	14
4.3 Local Control Interface Performance (HMI Validation).....	15
4.3.1 Functional Control and System Reliability.....	15
Table 4.3.1 Local Control Interface Validation Results.....	15
4.4 Sensor-Based Automation Performance and Reliability.....	15
4.4.1 Motion Sensor Detection Reliability.....	16
Figure 4.4.1 Outdoor PIR Trigger Success Rate vs Distance.....	16
Figure 4.4.2 Indoor PIR Trigger Success Rate vs Distance.....	16
Table 4.4.1 Motion Sensor Performance Summary.....	17
4.5 Remote Monitoring and Battery Data Validation.....	17
4.5.1 Battery Poll Accuracy and Reliability.....	17
Figure 4.5.1 Battery Poll Success Rate Over Time.....	17
Figure 4.5.2 Mobile Application Battery Display Interface.....	18
4.6 Security Response and Notification Performance.....	18
4.6.1 Motion Detection to Application Notification Latency.....	18
Figure 4.6.1 Security Event Image Capture and Application Display.....	18
Figure 4.6.2 End-to-End Latency vs Test Number.....	19
Table 4.6.1 Latency Measurement Summary.....	19
4.7 Local Control Response and Latency.....	19
4.7.1 HMI Command Execution and System Behavior.....	19
Table 4.7.1 HMI System Validation Checks.....	20
4.7.2 HMI Response Latency.....	20
Figure 4.7.1 HMI Command Latency by Lighting Zone.....	20
4.8 Sensor-Based End-to-End System Validation.....	21
4.8.1 Sensor-Driven System Behavior.....	21
Table 4.8.1 Sensor-Based System Validation Checks.....	21
4.8.2 Sensor-to-Output Latency.....	21
Figure 4.8.1 Sensor-Based Lighting Latency by Zone.....	22
4.9 Application-Based End-to-End Control Validation.....	22
4.9.1 Remote Control System Behavior.....	22

Final Report
Solar Home Lighting

Table 4.9.1 Application Control Validation Checks.....	23
4.9.2 Application-to-Output Latency.....	23
Figure 4.9.1 Remote (Application) Latency by Zone.....	23
4.10 Solar panel regulation under load.....	23
4.10.1 Dual energy delivery (load + storage).....	23
Figure 4.10.1 Load power, battery charging power, and total useful output power over time.....	24
Figure 4.10.2 State of charge of Battery A and Battery B during operation.....	24
Figure 4.10.2 Solar panel voltage compared to battery voltages.....	25
5. Overall System Results.....	26
Figure 5.1 System Implementation on Model Residential Structure.....	26
Figure 5.2 Fully Integrated System Hardware Bench Assembly.....	27

Solar Home Lighting

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Concept of Operations

Final Report
Solar Home Lighting

CONCEPT OF OPERATIONS FOR Solar Home Lighting

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Table of Contents

Table of Contents	3
List of Tables	5
List of Figures	6
1. Executive Summary	6
2. Introduction	7
2.1. Background	7
2.2. Overview	8
2.3. Referenced Documents and Standards	9
3. Operating Concept	10
3.1. Scope	10
3.2. Operational Description and Constraints	10
3.3. System Description	10
3.4. Modes of Operations	11
3.5. Users	11
3.6. Support	11
4. Scenarios	12
4.1. Airbnb Host #1	12
4.2. Independent/Off-Grid #2	12
4.3. Nighttime Security Monitoring	12
5. Analysis	12
5.1. Summary of Proposed Improvements	12
5.2. Disadvantages and Limitations	13
5.3. Alternatives	13
5.4. Impact	14

List of Tables

No table of figures entries found.

List of Figures

Figure 1. Solar Panel Security System Block Diagram

1

1. Executive Summary

This project presents the design of a unified, solar-powered residential lighting and monitoring system intended to address growing concerns related to energy reliability, household security, and fragmented smart-home solutions. Homeowners increasingly face rising electricity costs and frequent power disturbances, while existing commercial systems often depend on multiple separate devices that operate with poor coordination relying entirely on grid power. As a result, security monitoring, lighting control, and user interaction frequently become inconsistent or unavailable during periods of power instability.

The proposed system integrates a photovoltaic charging network, battery storage, modular lighting controls, and an embedded camera with AI-based recognition into a single coordinated platform that remains operational during grid fluctuations. Lighting functions are managed through sensor-driven automation, manual controls, and mobile application inputs, while the camera subsystem provides day and night visual awareness with real-time notifications to the user. The mobile application offers remote access to system status, battery information, security alerts, and light control, enabling flexible interaction regardless of the user's location.

By combining security awareness, lighting management, and user interface capability into one platform, the system provides a reliable and self-sustaining solution suitable for modern residential settings. Its development requires interdisciplinary collaboration across hardware design, embedded firmware, renewable-energy systems, and mobile software. The result is a user-centered, energy-efficient system designed to maintain continuous operation while improving the overall safety and experience of the homeowner.

2. Introduction

This project introduces a solar-powered residential lighting and monitoring system designed to reduce dependence on the electrical grid. The system is intended to address common issues faced by homeowners, including rising energy costs, inconsistent grid reliability, and the fragmentation of current lighting and basic security solutions. By integrating sensor-based lighting control with a mobile application for system monitoring and user interaction, the design provides continuous operation and improves accessibility within a variety of residential environments. The objective of this project is to present a coordinated, renewable-energy-driven approach that enhances reliability while remaining straightforward for homeowners to operate.

2.1. Background

Residential lighting, monitoring, and security systems have traditionally relied on grid electricity and often operate as separate, uncoordinated devices. Standard lighting systems provide only basic illumination and offer limited automation, while commercial smart-security products depend on continuous electrical and network connectivity to deliver features such as live video, notifications, and remote access. During power outages or network interruptions, these systems frequently become unavailable, reducing reliability at times when consistent operation is most needed. Existing solar-powered lighting products address part of this issue but are generally limited to simple outdoor use and do not integrate with broader home functions such as indoor lighting control or coordinated monitoring.

Rising energy prices and declining grid reliability underscore the limitations of grid-dependent residential systems. According to the U.S. Energy Information Administration, the national average residential electricity price increased from 12.6 cents per kilowatt-hour in 2012 to more than 16.3 cents per kilowatt-hour in 2023—an increase of over 29%. Several states, including California, New York, and Massachusetts, have experienced increases exceeding 40% over the past decade. At the same time, the frequency and duration of power outages have grown significantly; the U.S. Department of Energy reports that weather-related outages have increased by roughly 50% since the early 2000s. These trends create growing interest in energy systems that remain functional without depending solely on the electrical grid.

Solar adoption has risen in parallel with these challenges. More than 4.3 million U.S. homes now operate with rooftop photovoltaic systems, with residential solar capacity more than doubling between 2018 and 2023. This shift reflects a broader movement toward decentralized, homeowner-controlled energy systems that offer greater resilience and cost predictability. However, most consumer-grade solar lighting products are not designed to manage multiple indoor and outdoor lighting zones, interface with security devices, or provide meaningful awareness during grid outages.

Security considerations further highlight the need for more robust, reliable residential systems. According to FBI reporting, the average financial loss per residential burglary exceeded \$97,000 in 2022, and multiple studies show that visible lighting and monitoring systems serve as effective deterrents. For example, research from the University of North

Final Report

Solar Home Lighting

Carolina at Charlotte found that approximately 60% of surveyed offenders would avoid homes with clear signs of active alarms or surveillance equipment. Maintaining consistent lighting and monitoring capability—especially during power disruptions—therefore plays a critical role in basic home safety.

Recent advances in low-cost microcontrollers, embedded artificial intelligence, and low-power wireless communication now allow residential lighting and security functions to be integrated into a single, coordinated platform. Modern controllers can perform local sensor processing, object detection, and communication without relying on cloud services or significant power reserves. These capabilities enable a level of system autonomy and coordination not previously available in consumer-grade solar lighting systems.

Collectively, these trends demonstrate the need for a unified residential platform that reduces dependence on the electrical grid, remains operational during outages, and integrates lighting and basic monitoring functions into a coordinated, reliable system.

2.2. Overview

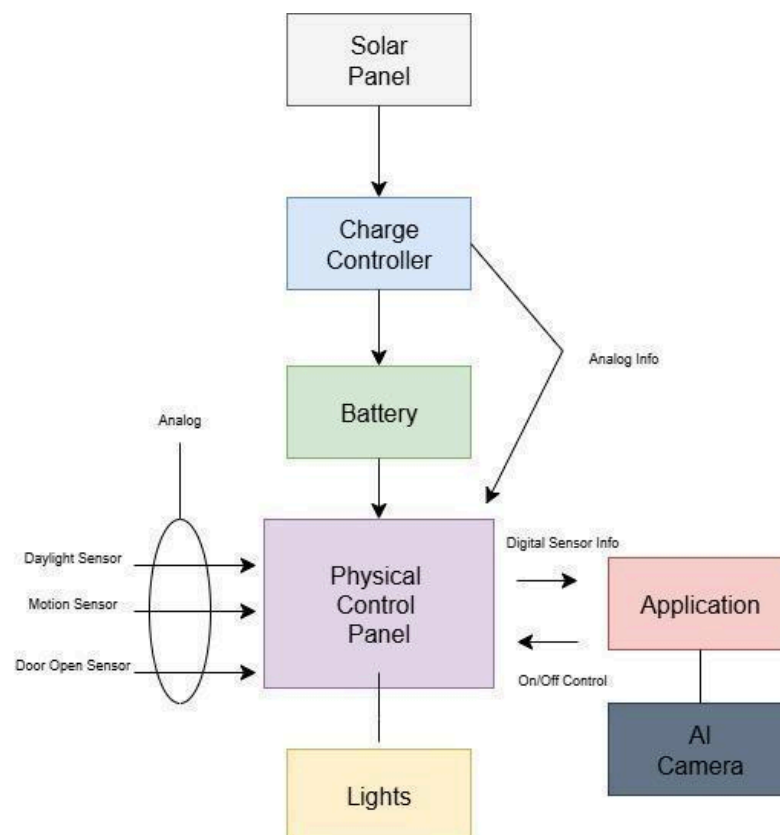


Figure 1: Solar Panel Security System Block Diagram

Our system integrates renewable energy and camera security systems by using a solar panel and charge controller to supply the solar power to a battery while the other provides power to the load. This then works by powering a physical control panel built around

multiple ESP32 microcontrollers. The control panel will then take in analog signals from daylight, motion, and door opening sensors. In contrast, an ESP32 and motion sensors will also handle person detection, allowing the camera to capture human presence and send security notification signals to the mobile application. In the application, the user will be able to receive camera gallery, battery percentage, light switch controls, and security notifications. Conclusively, this design provides a robust, low-cost, energy-efficient, and user-friendly solution for a security and light system in a residential home.

2.3. Referenced Documents and Standards

- [1] IEEE Standards Association, IEEE Standard for Information Technology—Telecommunications and Information Exchange Between Systems Local and Metropolitan Area Networks—Specific Requirements Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications, IEEE Std 802.11™, 2020. Accessed at: https://standards.ieee.org/standard/802_11-2020.html

- [2] Apple Inc., Human Interface Guidelines for iOS, Apple Developer Documentation, 2023. Accessed at: <https://developer.apple.com/design/human-interface-guidelines/ios/overview/introduction/>

- [3] International Electrotechnical Commission (IEC), IEC 61724: Photovoltaic system Performance monitoring - Guidelines for measurement, data exchange and analysis, IEC Standard, 2017. Accessed at: <https://webstore.iec.ch/publication/33622>

3. Operating Concept

3.1. Scope

The project encompasses the development of a solar-powered residential lighting and monitoring system intended for use in both indoor and outdoor environments. The system is designed to operate independently of the electrical grid through the use of renewable energy storage and to provide coordinated lighting and basic security awareness. User interaction is supported through both local controls and a mobile application. For demonstration purposes, a representative model will be constructed to illustrate core functionality, while the overall design remains applicable to full-scale residential deployment.

3.2. Operational Description and Constraints

The system is intended to provide a modular lighting and security camera network for residential homes that is functioning entirely on solar power, independent from the power grid. The system will be controlled off of a local physical interface as well as a Wi-Fi-based app. The lighting aspect of the system will function with a dual purpose of recreational lighting for the interior and exterior of the home, as well as exterior security flood lights. The security camera portion will operate within the app to notify the user of detected motion outside of the home. There will be several motion sensors outside the house to detect possible trespassers and trigger both the camera functionality and security lights. For recreational lighting purposes, there will be motion sensors, door sensors, and a programmable timer to automatically engage or disengage the lights.

The following are constraints relating to the operational description:

- The solar panels need to be placed in an area that provides consistent direct sunlight during the day.
- The battery bank needs to be sized according to the number of lights to ensure adequate operating time.
- The physical control panel needs to be mounted within the range of the home's wifi to allow app functionality.
- Users are required to have a smartphone to use the security feature of this system.

3.3. System Description

The smart solar home lighting system integrates a renewable energy source, control electronics, and user-friendly software to provide a reliable and efficient home lighting and security system. The system consists of 4 subsystems: the solar panel network, microcontroller, sensors and lighting, and the application. This interconnected design ensures a sustainable and user-accessible solution for a modern home.

- **Solar Panel Network:** The solar panel array will be a roof-mounted unit fed through a charge controller into deep-cycle batteries. One battery will collect a charge throughout the day while the other provides power to the load. The charge controller will be designed to implement MPPT charging to account for inclement weather that may interfere with charging conditions for maximum efficiency.

- **Microcontroller:** The microcontroller will interface with the mobile app, sensors, solar/battery network, and lighting. It reads inputs and applies the logic accordingly. It also manages comms, low power modes, enforcing limits when needed.
- **Sensors and lighting:** Motion sensors and cameras will communicate with the MCU when motion is detected. Once sensors are triggered, home lighting for designated areas will turn on, and users will be notified through the mobile app. Areas with cameras will turn on once motion is detected.
- **Application:** The Application will act as a tool that stores important system data (Camera, battery percentage, light status, and database logs) while also acting as a wireless control panel for basic operation, such as turning lights on and off. The application will communicate through Wifi via Android device and an ESP32. The ESP32 will be mounted and connected to our main control panel, where signals will be able to receive and send out vital data.

3.4. Modes of Operations

The smart Solar-powered lighting and security system supports three modes of operation - Automatic, Manual, and Security. Each is designed to adapt system behavior based on environmental conditions, user interaction, and security needs. These modes determine how the lighting subsystem, sensors, camera, and user interface cooperate to achieve the efficient and reliable operation under various scenarios.

3.4.1 Automatic Mode

Automatic mode enables the system to make lighting decisions based on real time sensor data and preconfigured user settings. It is designed to maximize energy while providing convenient adaptable control without requiring constant user input.

Operational Behavior:

- Lights turn on automatically when motion is detected under low light conditions defined by the user.
- Lighting remains active for a configurable timeout period; if no motion is detected after a set interval the lights shut off.
- Daylight conditions suppress outdoor lighting activation to preserve battery power.
- Users may adjust the motion threshold, timeout duration, and light level thresholds through the application.

3.4.2 Manual Mode

Manual mode provides full user control of the system, overriding any automated decision making. Users operate lights directly through local controls and the mobile application, enabling predictable control when automation is unwanted. This is intended to put user intent first, disabling automated behavior like sensor driven response.

Operational Behavior:

- Lights respond exclusively to user commands issued through one of two control options, physical or application based.

- Sensor inputs are ignored for any light zone placed into manual operation.
- The physical and application controls display real time system feedback, including the current light state and battery level for user consideration.

3.4.3 Security Mode

Security mode supports outdoor awareness by coordinating the lighting, camera, and AI recognition system when activity occurs around the home. This mode enhances situational awareness by providing illumination and processing visual information when motion is detected. The integrated AI model analysis video frames to determine and notify the user of situations occurring outside the home.

Operational Behavior:

- When movement is detected outside the security flood lights are triggered alongside the camera.
- The camera begins automatically capturing video that can be viewed real time through the application.
- The AI model analyzes the camera feed to identify the source of motion, and when necessary alerts the user through descriptive push notifications.
- During Daylight hours, the camera systems continue operation without the addition of flood lights to conserve system power.

3.5. Users

The smart solar home lighting system is intended to be used by any homeowner. The system is simple and user-friendly, requiring minimal technical training for standard operation. To interact with the system, the users will be provided with an in-app manual that walks them through the use of the system.

For installation, basic home improvement skills are needed to mount the solar panels and route cables to the lights. The design will contain modular sensors and light plugs that allow for ease of installation.

This setup will benefit typical homeowners who want to lower their electricity bill and desire a more vigilant mindset on home security. Additionally, this system will benefit homes in more remote areas with residences separate from the local power grid.

3.6. Support

A user manual will be provided with the system that gives details on use, installation, and troubleshooting. Additionally, the app will have a walkthrough mode available to teach users the process for controlling and monitoring the system within the application.

4. Scenarios

4.1. *Airbnb Host #1*

- Homeowners use the app to schedule porch lights to turn on at 7 PM for expected visitors.
- The system overrides motion detection during scheduled hours.

4.2. *Independent/Off-Grid #2*

- The system operates independently of grid power, ideal for cabins or remote homes.
- Reliable lighting and security without utility infrastructure.

4.3. *Nighttime Security Monitoring*

- The system is utilized when outside monitoring is needed for safety reasons
- Smart lighting and motion detectors can store camera data in the app in a safe location.
- Motion-detected lights can deter intruders.

5. Analysis

5.1. *Summary of Proposed Improvements*

The proposed solar home lighting system offers several important advantages over traditional home lighting systems, such as:

- **Eco-friendly/ Reduced Emissions:** By utilizing solar energy as the primary power source, the system reduces reliance on natural gas and lowers gas emissions.
- **Off-Grid Back-Up Use:** The integration of battery storage allows the lighting system to function independently of the electrical grid, ensuring reliable operations during outages.
- **Energy Efficiency:** The use of LED lighting and power management works to minimize energy consumption while handling high output, contributing to extending the life of the system.
- **Increased Security:** Motion sensors and app-based controls enhance home security by providing responsive lighting to movement in the area.
- **Mobile App Control:** The System can be controlled remotely via smartphone app that offers users convenience to monitor energy use, set timers, and adjust lighting preferences from any location.
- **Scalability:** The modular design allows for the expansion of a porch or foyer lighting to cover both interior and exterior household lighting, adapting to changing user needs.
- **Lower Energy Cost:** By reducing dependence on grid electricity, households experience lower monthly utility costs.

5.2. Disadvantages and Limitations

While the proposed solar home lighting system provides several advantages, there are some limitations that need to be considered, which include:

- **High Upfront Cost:** The initial investment for solar panels, batteries, and smart controls is higher than conventional lighting systems. Long-term energy savings will offset the cost, but the high upfront cost may still be a barrier for some households.
- **Weather Dependency:** Solar power generation relies on sunlight. Cloudy days, storms, or extended periods of low sunlight, such as winter, can reduce system performance, making it less reliable without sufficient battery storage.
- **Space Requirements:** Installation requires adequate roof and/or ground space for solar panels, which may not be feasible for all homes, especially in densely populated areas.
- **Energy Storage Limits:** Battery capacity restricts how much energy can be stored and used during non-sunlight hours. Once the battery is depleted, the system cannot provide power and must be supplemented with grid electricity.
- **Scalability vs Cost:** While the system can be expanded to provide lighting to more areas, expanding will incur additional cost, which could become a limitation.
- **Efficiency Loss:** Energy losses occur during solar conversion, battery charging/discharging, as well as system operations, which will reduce overall efficiency.

5.3. Alternatives

In evaluating the proposed solar home lighting system, there are several alternatives to consider, each with its own set of tradeoffs compared to the proposed system:

- **Conventional Grid-Based Lighting:** Traditional and inexpensive, but offers no sustainability or backup during outages.
- **Solar Lighting Without App Control:** A solar home system with panels, battery, and LED setup with only physical switches. This alternative is more affordable but lacks monitoring, scalability, and smart functionality.
- **Hybrid Solar System:** Combines solar with a generator or grid connection for backup. This increases reliability but adds cost, complexity.
- **Commercial Smart Home Lighting:** Uses grid electricity to power smart LEDs with wifi control. These, however, don't reduce energy cost or environmental impact.

5.4. *Impact*

- Environmental Impact: Reduces carbon emissions by using solar energy and energy-efficient LEDs. There will still be some environmental footprint through battery and panel production.
- Societal Impact: Provides reliable lighting, improves home security, and enhances user convenience.
- Ethics: May raise concerns regarding data privacy on the mobile app. Affordability for lower-income households is also a factor, as well as the sourcing of materials used in the system.

Solar Home Lighting 3

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INTERFACE CONTROL DOCUMENT

REVISION – 1
25 January 2018

REVISION - 1
04 April 2026

INTERFACE CONTROL DOCUMENT FOR Solar Home Lighting

PREPARED BY:

Author Date

APPROVED BY:

Jack Holdridge Date

John Lusher II, P.E. Date

T/A Date

Change Record

Rev	Date	Originator	Approvals	Description
1	10/4/2025	Photon Phools		Draft Release
2	12/6/2025	Photon Phools		Updated content and formatting.
3	4/27/26	Photon Phools		Final update for ECEN 404

Table of Contents

No table of figures entries found.	5
1. Overview	6
2. References and Definitions	6
2.1. References	6
2.2. Definitions	6
3. Physical Interface	7
3.1. Weight	7
3.2. Dimensions	8
3.3. Mounting Locations	9
4. Thermal Interface	10
4.1. Physical Controls Subsystem	10
4.2. ESP32 Camera and Application Subsystem	11
4.3. MPPT Charge Controller Subsystem	11
5. Electrical Interface	11
5.1. Primary Input Power	11
5.2. Voltage and Current Ratings	12
5.3. Polarity Reversal	13
5.4. Signal Interfaces	14
6. Communications / Device Interface Protocols	16
6.1. Physical Controls Subsystem Communication Interface	16
6.1. Wireless Communications (WiFi)	16
6.2. Host Device	16
6.3. Video Interface	16
6.4. Device Peripheral Interface	17

List of Tables

3.1.1. Physical Controls Subsystem Designed Weight	7
3.1.2. ESP32 Camera and Application Subsystem Designed Weight	8
3.1.3 MPPT Charge Controller Designed Weight	8
3.2.1. Physical Controls Subsystem Dimensions	8
3.2.2. ESP32 Camera and Application Subsystem Dimensions	9
3.2.3 MPPT Charge Controller Dimensions	9
4.1.1. Physical Controls Thermal Parameters	10
4.2.1. Camera and Application Thermal Parameters	11
4.3.1. MPPT Charge Controller Thermal Parameters	11
5.2.1. Physical Controls Subsystem Internal Power Consumption	12
5.2.2. Physical Controls Subsystem Lighting Output Load	13
5.2.3. Camera Subsystem and Application	13
5.2.4 MPPT Charge Controller	13
5.4.1. Physical Controls Subsystem Analog and Digital Signals	14
5.4.2. MPPT Charge Controller Analog and Digital Signals	15
5.4.3. MPPT Charge Controller Communication Interface	15
6.1.1. Physical Controls Communication Parameters	16

List of Figures

No table of figures entries found.

1. Overview

This Interface Control Document (ICD) defines how the subsystems of the Smart Solar Home Lighting System will interface with each other. It covers the connections between the solar power, batteries, sensors, microcontroller, lighting, camera module, and mobile applications. The ICD outlines electrical, data, and communication interfaces to ensure reliable system integration and operation.

2. References and Definitions

2.1. References

Lighting Global SHS Quality Standards v2.5
Dec 2018

IEEE Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces (IEEE 1547a-2020)
2020

National Electrical Code for Solar Photovoltaic Systems (NEC Article 690)
2023

Standard for Inverters, Converters, Controllers, and Interconnection System Equipment for Use With Distributed Energy Resources (UL 1741)
2021

ANSI Electric Connectors (ANSI C119.6)
2024

Storage Batteries (NEC Article 480)
2023

2.2. Definitions

MPPT	Maximum Power Point Tracking
PWM	Pulse Width Modulation
Wi-Fi	Wireless Fidelity
AC	Alternating Current
DC	Direct Current
RMS	Root Mean Square
HMI	Human Machine Interface
ICD	Interface Control Document
MCU	Microcontroller Unit
kWh	Kilowatts-hour
LED	Light Emitting Diode
USB	Universal Serial Bus
UART	Universal Asynchronous Receiver-Transmitter
BLE	Bluetooth Low Energy
LCD	Liquid Crystal Display
PIR	Passive Infrared Sensor
HTTP	Hypertext Transfer Protocol
MJPEG	Motion JPEG
VGA	Video Graphics Array

3. Physical Interface

This section defines the physical characteristics, dimensions, weights, and mounting requirements for all system subsystems. These physical attributes ensure mechanical compatibility during system installation and integration.

3.1. Weight

3.1.1. Physical Controls Subsystem Designed Weight

Component	Weight (g)	Number of Items	Total Weight (g)
ESP32-S3	31	2	62
2.8" TFT LCD	40	1	40
5 Channel Relay Board	90	1	90
PIR Motion Sensor	10	1	10
Daylight Sensor	17	1	17
LED Recessed Ceiling Light	9.5	3	28.5
LED Flood Security Light	296	2	592
Total Weight (Approx.)			~818 grams

3.1.2. ESP32 Camera and Application Subsystem Designed Weight

Component	Weight (g)	Number of items	Total Weight (g)
Speed Studio XIAO ESP32 S3 Sense	6.6	1	6.6

Total Weight (Approx.)			~6.6 grams
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3.1.3 MPPT Charge Controller Designed Weight

Component	Weight (g)	Number of Items	Total Weight (g)
200W Solar Panel	7520	1	7520
12V Lead Acid Battery	10433	2	20866
Charge controller PCB	1550	1	1550
Total Weight (Approx.)			~29,936 grams

3.2. Dimensions

3.2.1. Physical Controls Subsystem Dimensions

Component	Length (in)	Width (in)	Height (in)
ESP32	2.05	1.18	0.59
2.8" TFT LCD	2.8	2	0.3
5 Channel Relay Board	7.17	4.49	0.75
PIR Motion Sensor	1.5	1.5	0.75
Daylight Sensor	2.06	3.0	0.25
LED Recessed Ceiling Light	2.75	2.75	0.62
LED Flood Security Light	4.53	2.76	3.35

3.2.2. ESP32 Camera and Application Subsystem Dimensions

Component	Length (in)	Width (in)	Height (in)
------------------	--------------------	-------------------	--------------------

Speed Studio XIAO ESP32 S3 Sense	0.83	0.70	0.59
-------------------------------------	------	------	------

3.2.3 MPPT Charge Controller Dimensions

Component	Length (in)	Width (in)	Height(in)
200W Solar Panel	42.12	0.38	25.24
12V Lead Acid Battery	6.5	5	7.06
ESP32	2.05	1.18	0.59

3.3. Mounting Locations

3.3.1. Physical Controls Subsystem Panel

The subsystem is intended for indoor mounting near the main residential access point (foyer, hallway, or near a utility closet). Mounting option include:

- Wall mounting: using pre-drilled holes and drywall anchors
- Flush mounting: Inside low-voltage wall box or structured wiring panel

Mounting location must allow access to DC voltage lines and Wi-Fi coverage, and should be within visible and reachable distance for user interaction.

3.3.2. Physical Controls Subsystem Sensors

- **Motion Sensor**
 - The motion sensor will be mounted below the ESP32-EYE camera, allowing the 30 feet of motion detection range to be utilized for camera triggering purposes.
- **Reed Switch Door Sensor**
 - This will be mounted at the top of the door frame to indicate door status.
- **Daylight Sensor**
 - This will be mounted to the roof alongside the solar panel to get an accurate light level readout.

3.3.3. ESP32 Camera and Application Subsystem

- **ESP32-Camera**
 - Indoor facing outwards in wall mounted casing. In outdoor settings, the esp32-camera will be placed on a wall mount in weather-resistant housing with transparent lens cover.

3.3.4 MPPT Charge Controller

- **12V batteries and charge controller**
 - Must be in proximity to one another, preferably in the same location.
 - Indoors or safe from harsh weather elements, like a cabinet or storage house.
- **Solar Panel**
 - Must be on an elevated outdoor structure with full sun exposure.
 - Securely mounted for windy conditions.

4. Thermal Interface

This section defines the thermal characteristics, expected heat dissipation, and required cooling provisions for each subsystem. All thermal values are design estimates based on component datasheets and intended operating conditions.

4.1. Physical Controls Subsystem

The Physical Controls subsystem generates minimal thermal load during operation. Passive cooling through ventilation slots in the enclosure is sufficient under standard indoor ambient temperatures.

- Cooling method: passive ventilation
- A copper thermal pad beneath LTC3119 and LT8610AC is required (datasheet guidance).

The enclosure material is ABS plastic, suitable for residential installations. The operating temperature range is between 0-70°C.

4.1.1. Physical Controls Thermal Parameters

Component	Estimated Power Dissipation (W)	θ_{JA} (°C/W)	Expected Junction Rise (°C)
LTC3119 Buck-Boost (12 V Conditioner)	1.0–1.2 W	40 °C/W	40–48 °C
LT8610AC 3.3 V Regulator	0.8–1.0 W	45 °C/W	36–45 °C
ESP32-S3 (Relay MCU)	~0.8 W	55 °C/W	~44 °C
ESP32-S3 (Touchscreen HMI)	~0.6 W	55 °C/W	~33 °C
Relay Drivers (2N2222)	<0.2 W each	70 °C/W	<14 °C
TFT LCD Backlight	~0.5 W	-	<10 °C

Sensors (PIR, reed, daylight)	<0.05 W	-	negligible
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4.2. ESP32 Camera and Application Subsystem

Typical operating temperature for a Speed Studio XIAO ESP32 S3 Sense is between 32°F – 122°F (0°C – 50°C). The heat dissipation is minimal (<1 W power draw) but airflow will be utilized for enclosed environments

4.2.1. Camera and Application Thermal Parameters

4.3. MPPT Charge Controller Subsystem

No active cooling is required as passive cooling is sufficient under standard indoor ambient conditions, with heatsinking applied to MOSFETs and inductors. Slotted ventilation can assist in cooling. Thermal isolation is maintained between heat-generating components and temperature-sensitive sensors.

4.3.1. MPPT Charge Controller Thermal Parameters

Component	Estimated Power Dissipation (W)	θ_{JA} (°C/W)	Expected Junction Rise (°C)
LM5176 Buck-Boost Controller	~1.2–1.5 W	~40 °C/W	~48–60 °C
LM3151MHX-3.3 Buck Regulator	~0.8–1.0 W	~45 °C/W	~36–45 °C
ESP32-WROOM-32D MCU	~0.6–0.8 W	~55 °C/W	~33–44 °C
MOSFETs (CSD17310Q5A or similar)	~1.0–2.0 W	~35 °C/W	~35–70 °C
ACS781LL Hall Sensors	~0.2-0.3 W	~60 °C/W	~12–18 °C
NTC Thermistor (10k B3950)	<0.05 W	-	Negligible
Resistor Networks (Feedback/Dividers)	<0.1 W	-	Negligible
Terminal Block / Connectors (MC4, PCB)	<0.2 W	-	Negligible

5. Electrical Interface

This section defines the electrical power interfaces, connector pinouts, voltage levels, current limits, signal characteristics, and communication protocols between all subsystems.

5.1. Primary Input Power

5.1.1. Physical Controls Subsystem

The physical control will be powered by the DC battery bank from the charge controller subsystem.

- Input Voltage Range: +10.8 VDC to +14.2 VDC (nominal)
- Input Type: 2-pin terminal block (fused type)

5.1.2. ESP32 Camera and Application Subsystem

Supply voltage will be 3.3 V regulated via USB-C or battery pack. Typical current draw will be 160-200 mA during streaming, with peak current up to 500 mA.

5.1.3. MPPT Charge Controller Subsystem

The primary input power for the charge controller IC will be directly from the Solar Panel. From the IC, power will be fed to one 12V Lead acid battery, while the other discharges to the load. This is controlled via Bidirectional switches.

5.2. Voltage and Current Ratings

5.2.1. Physical Controls Subsystem Internal Power Consumption

Component	Input Voltage (VDC)	Current (mA)	Power (W)
ESP32 Microcontrollers	3.3	500	1.65
TFT LCD Touch Controller	3.3	150	0.495

Relay Control Board	3.3	150	0.495
Voltage Regulator (3.3V)	12 (nominal)	75	0.9
Voltage Regulator (12V)	12 (nominal)	90	1.1
PIR Motion Sensor	3.3	3	0.009
Daylight Sensor	3.3	2	0.007
Internal Subtotal			~4.7 W

5.2.2. Physical Controls Subsystem Lighting Output Load

Load Type	Quantity	Power per Unit (W)	Total Load (W)
LED Recessed Ceiling Light	3	2.5	7.5
LED Flood Security Light	2	10	20
Total Peak Load			27.5 W

5.2.3. Camera Subsystem and Application

Component	Input Voltage	Typical Current	Peak Current	Power
XIAO ESP32-S3	3.3 V	180 mA	300 mA	0.59 W

5.2.4 MPPT Charge Controller

Load Type	Input Voltage (VDC)	Current (mA)	Power (W)
200W Solar Panel	18.8	10.6	200
12V Lead Acid Battery	~14V	100 mAh	

ESP32	3.3	250	0.825
Total Peak Load			~201 W

5.3. Polarity Reversal

To protect against reverse polarity connection on the 12VDC input, the physical control subsystem includes an in-line Schottky diode for polarity protections. This diode will be rated for a current exceeding the max peak load continuous forward current, providing reliable protection for the subsystems peak load. Under normal load conditions the diode permits current flow with a typical voltage drop of approximately 0.5V, resulting in minimal power loss.

5.4. Signal Interfaces

This section defines the signal-level interfaces used by each subsystem, including voltage levels, expected current, and direction. All unused ESP32 pins are intentionally omitted.

5.4.1. Physical Controls Subsystem Analog and Digital Signals

Signal Name	ESP32 Pin	Direction	Voltage	Max Current	Protocol / Function
UART2_RX	IO17	Input	3.3 V	—	UART receive, 115200 baud, 8-N-1
UART2_TX	IO16	Output	3.3 V	—	UART transmit, 115200 baud, 8-N-1
PIR_IN	IO14	Input	3.3 V	<5 mA	Digital motion detection
REED_IN	IO36	Input	3.3 V	<1 mA	Door open/closed detection
IR_SENSOR_IN	IO10	Input	3.3 V	<5 mA	Auxiliary IR sensor
SPARE_SENSOR_IN	IO11	Input	3.3 V	<5 mA	Reserved future digital sensor
DAYLIGHT_ADC	IO21	Input	0–3.3 V	—	12-bit ADC daylight measurement
RELAY1_CTRL	IO12	Output	3.3 V	Logic only	Controls Relay 1 (via 2N2222 driver)
RELAY2_CTRL	IO19	Output	3.3 V	Logic only	Controls Relay 2

RELAY3_CTRL	IO9	Output	3.3 V	Logic only	Controls Relay 3
RELAY4_CTRL	IO13	Output	3.3 V	Logic only	Controls Relay 4
RELAY5_CTRL	IO35	Output	3.3 V	Logic only	Controls Relay 5
POWER_LED	IO2	Output	3.3 V	<20 mA	Status/power indication LED
TOUCH_SDA	IO21	Bidirectional	3.3 V	—	I ² C data, 400 kHz
TOUCH_SCL	IO11	Output	3.3 V	—	I ² C clock, 400 kHz
TOUCH_IRQ	IO9	Input	3.3 V	<5 mA	Touch interrupt (active-low)
TFT_SCK	IO18	Output	3.3 V	—	SPI clock
TFT_MOSI	IO20	Output	3.3 V	—	SPI MOSI
TFT_MISO	IO19	Input	3.3 V	—	SPI MISO
TFT_CS	IO5	Output	3.3 V	—	SPI chip select
TFT_DC	IO2	Output	3.3 V	—	Data/command select
TFT_RST	IO4	Output	3.3 V	—	Reset line
TFT_BL_PWM	IO14	Output	3.3 V	—	Backlight PWM, 1 kHz

The Physical Controls Subsystem exchanges digital, analog, and communication signals required for sensing, relay activation, and interaction with the touchscreen interface. This section defines each electrical signal used by the subsystem, including direction, voltage level, expected current, and protocol characteristics. Only active signals are listed; all unused ESP32 pins have been intentionally omitted. The interface enables coordination between sensors, user controls, and lighting outputs while ensuring compatibility with the system's power and communication requirements.

5.4.2. MPPT Charge Controller Analog and Digital Signals

Signal Name	ESP32 Pin	Direction	Voltage	Max Current	Protocol / Function
RTS# to EN	EN	Input	3.3V	-	Reset via UART RTS line
Thermistor	GPIO2	Input	0–3.3 V	-	NTC thermistor measures battery temperature

Load Current	GPIO4	Input	0–3.3 V	-	ACS781LL output for load current
Battery Voltage	GPIO34	Input	0–3.3 V	-	Voltage divider from solar panel to ADC
Battery Current	GPIO35	Input	0–3.3 V	-	ACS781LL output for battery current
UART_TXD	TXD0	Input	3.3V	-	UART transmit to charge controller
UART_RXD	RXD0	Output	3.3V	-	UART receive from charge controller
3.3 VIN	3V3	Input	3.3V	-	ESP32 supply voltage
DTR# to IO0	GPIO0	Input	3.3V	-	Boot mode via UART DTR line
MPPT_DAC	GPIO25	Output	0–3.3 V	~12mA	DAC output to LM5176 FB pin for MPPT control
Load Switch	GPIO26	Output	3.3V	~12mA	Disconnects the load from the battery
Panel Voltage	GPIO32	Input	0–3.3 V	-	Voltage divider from solar panel to ADC
Panel Current	GPIO33	Input	0–3.3 V	-	ACS781LL output for panel current

5.4.3. MPPT Charge Controller Communication Interface

The communication interface of the charge controller is centered on the ESP32. Using UART, DAC/PWM, and ADC, to monitor and supply an efficient charge to the battery. UART provides a wired interface for debugging and inter-microcontroller messaging, with TXD and RXD lines operating at 115200 baud and 3.3 V logic levels. The ESP32 generates either a DAC output or PWM signal to the LM5176 feedback pin, enabling real-time duty cycle adjustment for maximum power point tracking. Multiple ADC channels measure solar panel voltage and current, battery voltage and current, load current, and battery temperature, all scaled to 0–3.3 V inputs. All communication signals are referenced to the 3.3 V supply rail, ensuring consistent logic levels across analog and digital interfaces.

6. Communications / Device Interface Protocols

6.1. Physical Controls Subsystem Communication Interface

The Physical Controls Subsystem uses UART, I²C, SPI, and PWM signaling standards for communication and device control. UART is used for inter-microcontroller messaging between the HMI and relay controller. The capacitive touch controller communicates using an I²C Fast-Mode interface at 400 kHz. The TFT display operates over a dedicated SPI bus. A PWM signal is used to regulate the backlight intensity of the display. All communication signaling operates at 3.3 V logic levels.

6.1.1. Physical Controls Communication Parameters

Interface	ESP32 Pins (Relay MCU)	Voltage	Direction	Protocol Parameters	Description
UART2 (Inter-ESP32 Link)	TX: IO16, RX: IO17	3.3 V	Bidirectional	115200 baud, 8-N-1	Command and status messaging between Relay MCU and HMI MCU
I ² C (Touchscreen FT6206)	SDA: IO21, SCL: IO11	3.3 V	Bidirectional	400 kHz Fast Mode	Capacitive touch event and coordinate reporting
PWM (Backlight Control)	IO14	3.3 V	Output	1 kHz, 0–100% duty	TFT LCD backlight control (if physically integrated)

6.2. Wireless Communications (WiFi)

Standard: IEEE 802.11 b/g/n at 2.4 GHz

Function: Provides primary communication path between the ESP32 camera subsystem and Android mobile application

Bandwidth: MJPEG video stream operates at up to 2 Mbps depending on resolution. MQTT sensor/control messages operate under 10 kbps, which ensures minimal overhead

6.3. Host Device

Mobile Device: Android smartphone or tablet running on Flutter/SwiftUI based application.

6.4. Video Interface

Type: MJPEG over HTTP streaming from an onboard ESP32 server

Protocol: HTTP GET request handled by ESP32 web server library

Resolution: VGA (640x480) 15 fps

Solar Home Lighting 3

Jack Holdridge

Kyle Walcher

Matthew Greer

Adam Garcia

FUNCTIONAL SYSTEM REQUIREMENTS

REVISION – 1
2 October 2025

REVISION - 1
04 April 2026

FUNCTIONAL SYSTEM REQUIREMENTS FOR Solar Home Lighting

PREPARED BY:

Author Date

APPROVED BY:

Jack Holdridge Date

John Lusher, P.E. Date

Hailey Ward Date

Change Record

Rev	Date	Originator	Approvals	Description
1	10/4/2025	Solar Home Lighting System		Draft Release
2	12/6/2025	Photon Phools		Updated content and formatting.
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Table of Contents

List of Tables	4
List of Figures	5
1. Introduction	6
1.1. Purpose and Scope	6
1.2. Responsibility and Change Authority	7
2. Applicable and Reference Documents	8
2.1. Applicable Documents	8
2.2. Reference Documents	8
2.3. Order of Precedence	9
3. Requirements	9
3.1. System Definition	9
3.2. Characteristics	10
3.2.1. Functional / Performance Requirements	10
3.2.2. Physical Characteristics	12
3.2.3. Electrical Characteristics	12
3.2.4. Environmental Requirements	15
3.2.5. Failure Propagation	16
4. Support Requirements	17
Appendix A: Acronyms and Abbreviations	18
Appendix B: Definition of Terms	19

List of Tables

2.1.1. Applicable Documents Table	8
2.2.1. Reference Documents Table	9
3.2.3.1.1. Table of Inputs	13
B. Definitions of Terms	19

List of Figures

Figure 1.1.1. Conceptual Product Layout	7
Figure 3.1.1. Block Diagram of System	10

1. Introduction

1.1. Purpose and Scope

The purpose of this project is to design, develop, and demonstrate a fully integrated solar-powered residential lighting and security system capable of operating independently from the electrical grid. The system shall support both interior and exterior applications and

Final Report
Solar Home Lighting

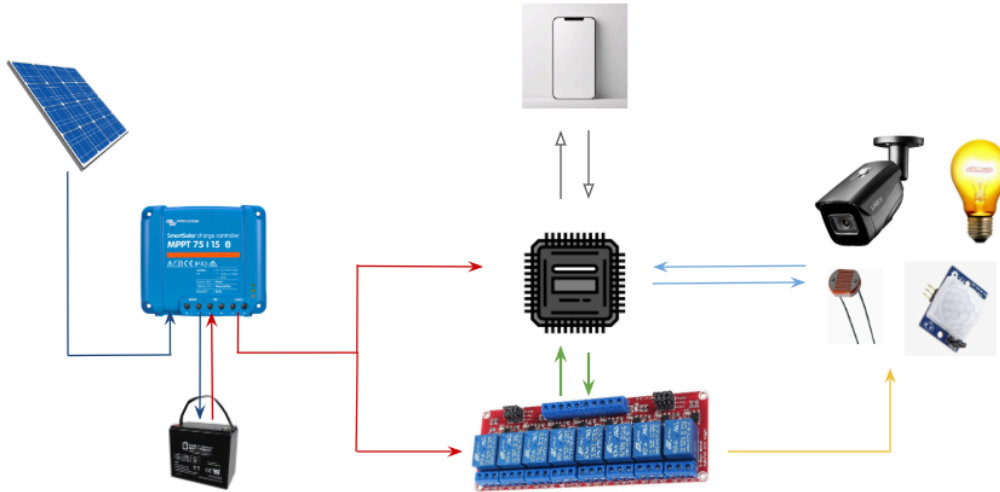
shall combine the functionality of traditional lighting control with modern smart security features into a unified, renewable energy-powered platform.

This system encompasses the complete development of a modular system architecture, including solar energy generation, battery storage, sensor driven automation, real time monitoring, and user control via both a local interface and mobile application. The system is intended to enhance residential security while reducing energy consumption and utility costs by leveraging solar energy as the primary power source.

For demonstration purposes, a function prototype will be built, simulating a typical porch and foyer installation. However, the system is designed to be scalable and expandable to support multiple rooms and outdoor zones across a full residential property. The final implementation shall include motion detection, door sensors, environmental light sensing, programmable timers, and a network of lighting and surveillance devices. All of which will be managed through a central microcontroller and ESP32 based wireless communication.

The scope of this work includes hardware and software integration, embedded system development, sensor interfacing, application design, and functional validation of system performance under typical residential operating conditions.

Figure 1.1.1. Conceptual Product Layout



The Solar Home Lighting System uses sunlight to power lights and small devices in a home. During the day, the solar panel will collect energy from the sun and send it to a battery for storage via the charge controller. A separate battery will be used to feed the load, power distribution will be controlled in accordance to user preference and power availability.

Users can connect to the system using a mobile app, which shows how much energy is being made, how full the battery is, and lets them turn lights on or off remotely. All the system's data, like how much power was used or if there were any problems, is saved in a database.

1.2. Responsibility and Change Authority

All project requirements shall be verified under the authority of the team captain, Jack Holdridge, who shall ensure each requirement is met in accordance with the project specification. No requirement may be altered without prior approval from both the team captain and the project sponsor, Professor Won Jang.

Subsystem Ownership is as follows:

- MPPT Charge Controller – Adam Garcia
- Physical Controls – Jack Holdridge
- Android Application and Camera – Kyle Walcher
- MCU/Sensors & Database – Matthew Greer

2. Applicable and Reference Documents

2.1. Applicable Documents

The following documents, of the exact issue and revision shown, form a part of this specification to the extent specified herein:

2.1.1. Applicable Documents Table

Document Number / Title	Revision / Release Date	Description
Lighting Global SHS Quality Standards v2.5	Dec 2018	<i>Defines minimum quality, durability, and safety for solar home system kits up to 350W. Includes testing protocols and eligibility criteria.</i>
IEEE 1547a	2020	<i>IEEE Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces--Amendment 1: To Provide More Flexibility for Adoption of Abnormal Operating Performance Category III</i>
NEC Article 690	2023	<i>National Electrical Code for Solar Photovoltaic Systems. Covers wiring, grounding, and safety for PV installations.</i>
UL 1741	2021	<i>Standard for Inverters, Converters, Controllers, and Interconnection System Equipment for Use With Distributed Energy Resources.</i>
ANSI C119.6	2024	<i>Electric Connectors—Non-Sealed, Multiport Connector Systems Rated 600 V or Less for Aluminum and Copper Conductors.</i>
IEC 62852	2014	<i>Connectors for DC-application in photovoltaic systems - Safety requirements and tests</i>
NEC Article 480	2023	<i>National Electrical Code for Storage Batteries</i>

2.2. Reference Documents

The following documents are reference documents utilized in the development of this specification. These documents do not form a part of this specification and are not controlled by their reference herein.

2.2.1. Reference Documents Table

Document Number / Title	Revision / Release Date	Description
ESP32 TRM (Version 5.5)	2025.08	<i>ESP32 - Technical Reference Manual Version 5.5</i>
ISO 9241-210	2019	<i>Ergonomics of human-system interaction</i>
ISO/IEC 14882	2024	<i>Programming languages — C++</i>
IEEE 802.11	2020	<i>IEEE Standard for Information Technology--Telecommunications and Information Exchange between Systems</i>
ISO/IEC 9075-1	2023	<i>Information technology — Database languages SQL</i>

2.3. Order of Precedence

In the event of a conflict between the text of this specification and an applicable document cited herein, the text of this specification takes precedence without any exceptions.

All specifications, standards, exhibits, drawings or other documents that are invoked as “applicable” in this specification are incorporated as cited. All documents that are referred to within an applicable report are considered to be for guidance and information only, except ICDs that have their relevant documents considered to be incorporated as cited.

3. Requirements

3.1. System Definition

The Smart Solar Residential Lighting and Security System integrates four main subsystems, each fulfilling a critical role in delivering a reliable, energy-efficient solution for residential applications. The solar power network centers on an MPPT charge controller that optimizes energy harvesting by regulating the variable DC output from the solar panels to efficiently charge the battery. This controller communicates with the battery management systems and central microcontroller to ensure continuous power availability.

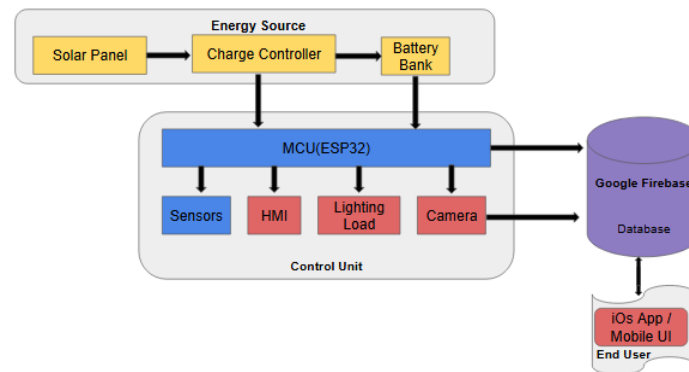
The physical control subsystems provide a user-friendly touch screen interface for lighting control and essential system status monitoring. It also hosts DC voltage conversions and serves as the central connection point for all system components, ensuring efficient power distribution and safe operation compliant with residential electric codes.

The lighting and security monitoring functions are supported by an android application paired with an ESP32-EYE camera module. This subsystem streams live video feeds and uses AI-powered motion detection to alert users of security events, while securely logging data to a cloud database for later review.

At the core of the system, the microcontroller and sensor subsystem collect environmental and motion data, execute control logic, and maintain real-time communication with the mobile app and cloud database via Wi-Fi. Sensor inputs trigger lighting adjustment and security alerts, ensuring a responsive and adaptive system behavior.

Together, these interconnected subsystems create a cohesive, scalable solution that provides solar-powered, automated lighting and security, addressing the need for sustainable energy use and enhancing residential awareness.

Figure 3.1.1. Block Diagram of System



3.2. Characteristics

3.2.1. Functional / Performance Requirements

3.2.1.1. Lighting Duration

The system shall provide continuous home/residential operation of exterior and interior lighting for a minimum of eight hours during night house conditions, powered solely from solar energy.

Rationale: This ensures that residential homes can receive reliable illumination throughout the night that is not dependent on grid electricity. This is critical to both the usability of the system and ability to reduce monthly costs.

3.2.1.2. Motion Detection Accuracy

The system will be able to detect and log human motion events with an accuracy of at least 90% under typical residential operating conditions, including varying lighting level and regular environmental noise

Rationale: Reliable detection of motion is essential for the systems role as a resource for security. A reliable, High detection accuracy will minimize false negatives and ensures timely push notifications that are generated for the homeowner

3.2.1.3. Camera Streaming Latency

The mobile android application developed in Kotlin shall stream live video from the ESP32-EYE camera with no more than two seconds when operating over standard WiFi

Rationale: The requirement will ensure that the homeowner experiences video feedback that is as close to real time as possible for responsive decision making for the user

3.2.1.4. Database Storage

The system shall maintain at least thirty days of stored event snapshot data in its cloud (firebase) for later review and analysis

Rationale: A month of storage strikes between useful historical/security data and practical storage constraints, allowing homeowners to review the past without over the top resource costs.

3.2.1.5. User Control and Responsiveness

The Android application will allow the user to toggle lighting states (ON / OFF) and received push notifications of change within 2 to 3 seconds of command execution

Rationale: Ensures the mobile will feel seamless and responsive to the user, which is essential for modern usability

3.2.1.6. Motion Triggered Lighting

The system shall activate exterior lighting once triggered by motion detection for at least 30 seconds unless reset by further activity

Rationale: Automatic light response provides both security and deterrence. This will provide users more safety and controllability within the application.

3.2.1.7. System Start-Up Sequence

The system shall complete its full start-up sequence within 10 seconds from the application of power, including initialization of the microcontroller, sensors, wireless communications, and user interface.

Rationale: Defines user expectations for system responsiveness and ensures the system becomes operational in a predictable amount of time. The PIR sensor shall stabilize within 30 seconds of power-on to allow accurate motion detection. PIR sensors require a warm-up period to avoid false triggers. The camera module shall begin streaming and AI-based recognition within 3 seconds after the system boots. Allows camera-based alerts to function shortly after startup without delaying system security.

3.2.1.8. System Start-Up Sequence

The system shall enter a safe shutdown mode within 5 seconds of power removal or command shutdown, ensuring lighting loads are disabled and data logging is halted without corruption.

Rationale: Prevents unintended lighting activation and protects data integrity under unexpected power loss.

3.2.1.9. Hardware and Mobile Interface Interaction

The system shall maintain a bidirectional communication interface between the hardware control units (MCU, sensors, lighting modules, and ESP-32 camera) and the android application. All hardware status signals should be updated and transmitted to the application in real time with a less than 5 seconds delay.

Rationale: This requirement clarifies the necessary interaction pathways between physical hardware components and the mobile application, ensuring that users receive accurate system status and can remotely control all major functions in a timely manner.

3.2.2. Physical Characteristics

3.2.2.1. Subsystem Size

The physical control unit shall not exceed 15 x 20 cm footprint, while individual sensors and camera shall fit within a 10x10x10 cm volume.

Rationale: Ensures compact sizing for ease of installation in tight residential spaces while maintaining discreet placement for cameras and sensors

3.2.2.2. Battery Housing

The battery housing shall be fully enclosed in a weather resistant casing that is also ventilated

Rationale: Protects deep cycle battery from weather exposure while allowing airflow

3.2.2.3. Cable Management and Mounting

All interconnections between the subsystems shall utilize standard cable runs and shall be designed for mounting with basic tools

Rationale: Simplifies installation and ensures compliance with safe residential practices

3.2.3. Electrical Characteristics

3.2.3.1. Inputs

- a. The Solar Home Lighting System shall tolerate any combination and sequence of input signals as defined in the Interface Control Document (ICD) without sustaining damage, degradation of life expectancy, or operational malfunction. This resilience applies equally whether the system is energized or de-energized.
- b. No permissible sequence of input commands shall induce system failure, reduce expected service life, or impair functionality.

Rationale: The system is engineered to minimize the risk of damage or malfunction arising from user or technician actions, ensuring reliability in real-world operation.

3.2.3.1.1. Table of Inputs

Parameter	Requirement	Notes
Input Voltage Range	12–24 VDC	Compatible with solar batteries and MPPT output
System Peak Power	$\leq 50 \text{ W}$	From requirement 3.2.3.1.1
Converter Efficiency	$\geq 85\%$	Applies to all DC/DC converters
Input Ripple Tolerance	0–200 mV	Based on converter datasheet limits
Output Regulation (Logic Rail)	$3.3 \text{ V} \pm 3\%$	ESP32 logic supply
Lighting Output Regulation	12 VDC	Supports LED loads
Overcurrent Protection	Required	Per requirement 3.2.3.1.4
Short-Circuit Protection	Required	Must survive without damage
Reverse Polarity Protection	Required	Diode or active method
Thermal Derating	Components shall operate within -20°C to $+55^{\circ}\text{C}$	Matches environmental requirement

3.2.3.1.1. Power Consumption

The maximum peak power of the system, including operation of lights, camera, sensors, wireless communication, shall not exceed 50 watts

Rationale: Limiting peak power draw ensures easy compatibility with mid level solar generation capacity and affordable battery storage. This will enable the system to remain cost effective while still meeting performance requirements

3.2.3.1.2. Input Voltage Range

The system shall accept DC input from 12 to 24 volts, compatible with standard solar batteries

Rationale: Flexibility will allow for seamless integration with commercially available solar panels and storage systems without custom hardware

3.2.3.1.3 DC/DC Conversion Efficiency

All DC/DC converters (bucks and boost) shall maintain a minimum conversion efficiency of 85% under load conditions

Rationale: High efficiency is required to minimize energy loss and maximize usable solar energy

3.2.3.1.4. Overcurrent and Short Circuit Protection

The system shall include overcurrent protection on all major supply rails and shall withstand short circuit episodes without permanent damage to the system

Rationale: Protects the system from wiring faults, load failures, and user error. This will ensure safety and reliability

3.2.3.1.4. Data and Control Signals

All MCU and sensor interfaces should conform to logic levels of 3.3 V with proper protection for I/O pins

Rationale: Aligns with ESP32 electrical limits, ensuring proper and safe communication between subsystems

3.2.3.2. Outputs

3.2.3.2.1. Energy Output

The Solar Home Lighting System shall provide a regulated DC output compatible with 12V appliances.

Rationale: Ensures LED lighting can be used off-grid.

3.2.3.2.2. Battery Status Output

The Solar Home Lighting System shall include a digital interface to provide the battery charge level for both batteries.

Rationale: The user and technician can monitor battery performance and plan accordingly.

3.2.3.2.3. Raw Video Output

The Solar Home Lighting System application shall include a raw video interface to support external recording.

Rationale: Too much data to store videos; only photos are stored internally. It would be used for security.

3.2.3.2.4. Connectors

The Solar Home Lighting System shall use weatherproof DC connectors compliant with IEC 62852.

Rationale: Ensures safe connections in outdoor environments.

3.2.3.2.5. Wiring

The Search and Rescue System shall follow the guidelines outlined in NEC Article 690 and use UL-listed cables.

Rationale: Ensures electrical safety and minimizes fire risk.

3.2.4. Environmental Requirements

The Solar Home Lighting System shall be designed to withstand and operate in the environmental conditions and laboratory tests specified in the following subsections.

Rationale: These requirements are based on scenarios where the system must maintain reliable operation under varying and harsh weather.

3.2.4.1. Thermal

The Solar Home Lighting System shall operate between -20°C to $+55^{\circ}\text{C}$.

Rationale: This range accommodates extreme hot and cold conditions in the U.S.

3.2.4.2. Rain

The Solar Home Lighting System shall withstand heavy rainfall and storm conditions, including exposure to high precipitation rates.

Rationale: Ensures waterproofing and short circuits, and corrosion.

3.2.4.3. Solar Radiation

All components exposed to outdoor elements shall be UV-resistant and capable of withstanding high solar irradiance.

Rationale: Prolonged exposure to sunlight can degrade plastics and coatings of components.

3.2.4.2. Wind

The Solar Home Lighting System shall withstand 150 km/h wind speeds without structural failure and detachment.

Rationale: High winds pose a risk of damage to rooftop installations.

3.2.5. Failure Propagation

The Solar Home Lighting System shall not allow propagation of faults beyond its system interface, ensuring that failures within the system do not affect connected external devices or the grid interface.

3.2.5.1. Failure Detection, Isolation, and Recovery (FDIR)

3.2.5.1.1 Built-In Test (BIT)

The Solar Home Lighting System shall include an internal subsystem that generates diagnostic signals and evaluates system responses to determine the presence of faults.

3.2.5.1.1.1. BIT Critical Fault Detection

The BIT shall detect critical faults in the Solar Home Lighting System with a minimum detection rate of 95 percent.

Rationale: This requirement ensures reliable fault detection in remote or off-grid environments where manual diagnostics are impractical.

3.2.5.1.1.2. BIT False Alarms

The BIT shall have a false alarm rate of less than 5 percent.

Rationale: Minimizing false alarms reduces unnecessary maintenance interventions and improves user trust in system diagnostics.

3.2.5.1.1.3. BIT Log

The BIT shall record the results of each diagnostic test in a log stored within the Solar Home Lighting System. This log shall be accessible for retrieval and clearing by maintenance personnel.

Rationale: Persistent logging supports fault analysis and long-term reliability tracking.

3.2.5.1.2 Isolation and Recovery

The Solar Home Lighting System shall support fault isolation and recovery by enabling subsystems to be reset, disabled, or reconfigured based on BIT results.

Rationale: This capability ensures that faults can be contained and mitigated without requiring full system shutdown, preserving partial functionality in critical lighting scenarios.

4. Support Requirements

The Smart Solar Residential Lighting and Security System requires stable Wi-Fi connection within the home to enable communication between the control panel and mobile applications. Users must provide access to Wi-Fi and maintain the solar panel installation in an area with sufficient sunlight for optimal performance. A compatible Android device is additionally required to operate the mobile application and receive security notifications.

Final Report
Solar Home Lighting

The system package includes all necessary hardware components, such as solar panels, batteries, control boards, sensors, and lighting units. Users will use the Android mobile application for wireless control and monitoring. A comprehensive user manual and an in-app walkthrough guide are provided to assist with installation, operation, and troubleshooting.

Technical support is available via email and phone during standard business hours. In the event of hardware malfunction or operational issues in the field, users can contact support for remote diagnostic and troubleshooting. Field issues related to solar panel position or battery performance may require an on-site inspection or user intervention as guided by support. Periodic software updates will be pushed automatically to enhance system functionality and security.

Appendix A: Acronyms and Abbreviations

MPPT	Maximum Power Point Tracking
PWM	Pulse Width Modulation
Wi-Fi	Wireless Fidelity
AC	Alternating Current
DC	Direct Current
RMS	Root Mean Square
HMI	Human Machine Interface
ICD	Interface Control Document
MCU	Microcontroller Unit
km/h	Kilowatts per hour
LED	Light Emitting Diode

Appendix B: Definition of Terms

This appendix provides definitions for specialized technical terms and acronyms used throughout the Functional System Requirements document. These definitions ensure consistency and clarity when interpreting the system description, requirements, and validation criteria.

B. Definitions of Terms

Solar Irradiance	The amount of solar power received per unit area, measured in watts per square meter (W/m^2).
MPPT (Maximum Power Point Tracking)	A control method used by the charge controller to continuously adjust voltage and current for maximum solar-panel power output under varying sunlight conditions.
Lux	A unit of illuminance equal to one lumen per square meter, used to quantify visible light intensity detected by the system's ambient-light sensor.
HMI (Human-Machine Interface)	The touchscreen display and control interface that allows users to view system data and issue lighting or configuration commands.
Firebase	A cloud-based database service used to store sensor data, camera images, and system status information for mobile-app access.
ESP32	A low-power microcontroller with integrated Wi-Fi and Bluetooth, used as the system's primary processing and communication unit.

Solar Home Lighting 3

Jack Holdridge

Kyle Walcher

Matthew Greer

Adam Garcia

ECEN 403 EXECUTION PLAN

Final Report

Solar Home Lighting

	9/17/2025	9/24/2025	10/1/2025	10/8/2025	10/15/2025	10/22/2025	10/29/2025	11/5/2025	11/12/2025	11/19/2025	11/26/2025	12/1/2025	12/6/2025	
Established firmware framework & Wi-Fi connection test														Kyle W.
Test connection between ESP32-EYE and Android application														Kyle W.
Test functionality with OpenAI														Kyle W.
Camera fully embedded with polished UI														Kyle W.
Validate application communicates seamlessly with camera and database														Kyle W.
Functional System Requirements														Jack H.
Design DC converters and select components														Jack H.
Design and verify touch screen wiring														Jack H.
Design PCB and order alognside components.														Jack H.
Program touch screen interface														Jack H.
Receive Physical Control circuit components														Jack H.
Assemble PCB and connect sensors, load, and HMI														Jack H.
Finish coding, implement initial control logic, perform calibration test														Jack H.
Validate assembled HMI functions as expected														Jack H.
Design DC converters and select components														Adam G.
Designed and simulated MPPT circuit														Adam G.
Design Charge controller PCB and components on Altium														Adam G.
Order PCB and components														Adam G.
Validate battery & loads are safely powered														Adam G.
Test converters & MCU program														Adam G.
Installed ESP-IDF, created Firebase database														Matthew G.
Implement MCU-to-SQL communication & validate sensors														Matthew G.
Validate sensor accuracy (lux, PIR, temp)														Matthew G.
Circuit PCB, MCU, and physical controls integration														Matthew G.

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Jack Holdridge

Kyle Walcher

Matthew Greer

Adam Garcia

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Test Name	Success Criteria	Methodology	Status	Responsible Engineer(s)
Verify MPPT charge controller performance	Charge controller maintains >90% efficiency under varying solar input	Measure voltage/current with simulated solar load	UNTESTED	Adam Garcia
Battery/load safety validation	No over-current or over-voltage during charging/discharging	Measure with multimeter + load test	UNTESTED	Adam Garcia
Physical controls DC converters	Consistent designed output voltage levels at peak operation	Measure with multimeter	COMPLETE	Jack Holdridge
Physical controls lighting relays	100% correct actuation across 50+ command cycles	50 button press trials	COMPLETE	Jack Holdridge
Physical controls HMI functionality	Touchscreen responds <500 ms, <5% error	50 button press trials	COMPLETE	Jack Holdridge
Verify data update on HMI screen	System updates light ON/OFF and sensor status <1 s	50 test updates with sensor parameters	COMPLETE	Jack Holdridge
Sensor detection (PIR, lux, temperature)	PIR detects motion ≥6 ft, lux sensor ±5%, temp ±2 °C	Log sensor readings vs. reference instruments	COMPLETE	Matthew Greer
Control logic (auto dusk/dawn)	Lights turn ON ≤20 lux, OFF ≥40 lux (5 lux hysteresis)	Dimmable light sweeps	COMPLETE	Matthew Greer
MCU-to-Database communication	Write latency <1 s (85% <2 s), ≥99% success	200 test messages logged	COMPLETE	Matthew Greer
App-to-Database refresh	UI updates ≤5 s after DB change	Stopwatch across 50 updates	COMPLETE	Kyle Walcher
Camera integration	ESP32-EYE streams 90%+ reliable	100 test captures, record drop rate	COMPLETE	Kyle Walcher
Power consumption check	System avg. current supports ≥12 h operation @ avg. load draw	12 h load run logged with probe	UNTESTED	Adam Garcia + Jack Holdridge
Integration test (end-to-end)	Sensor → MCU → DB → App works as expected	20 functional test cycles	UNTESTED	Matthew Greer + Jack Holdridge
System reliability	Continuous 7 days operation ≤1 reset/failure	7 day uptime log	UNTESTED	Jack Holdridge + Adam Garcia
Environmental tolerance	Operates at 0–40 °C, 20–90% RH	Lab stress tests	UNTESTED	Jack Holdridge + Adam Garcia

Solar Home Lighting

Jack Holdridge

Kyle Walcher

Matthew Greer

Adam Garcia

SUBSYSTEM REPORTS

SUBSYSTEM REPORT
FOR
Solar Home Lighting

TEAM 22

APPROVED BY:

Jack Holdridge Date

Prof. Kalafatis Date

Hailey Ward Date

Change Record

Rev	Date	Originator	Approvals	Description
1	12/6/25	Jack Holdridge		Original Release
2	4/27/26	Jack Holdridge		

Table of Contents

Table of Contents	3
List of Tables	4
List of Figures	5
1. Introduction	6
2. Charge Controller Subsystem	6
2.1. Subsystem Introduction	6
2.2. Subsystem Overview	6
2.3. Solar Panel to Battery Buck Converter (18.8V to 14.4V - 13.6V)	7
2.3.1. Operation	7
2.3.2. Simulation Validation	7
2.4. 3.3V Buck Converter	9
2.4.1. Operation	9
2.4.2. Simulation Validation	9
2.5. ESP32 Perturb & Observe (P&O) Program	9
2.5.1. Operation	9
2.5.2. Program Validation	10
2.6. Subsystem Conclusion	11
3. Physical Controls Subsystem	11
3.1. Subsystem Introduction	11
3.2. Subsystem Overview	12
3.3. 3.3V Buck Converter	13
3.3.1. Operation	13
3.3.2. Validation	14
3.4. 12V Buck-Boost Converter	20
3.4.1. Operation	20
3.4.2. Partial Validation	20
3.4.1. Failure Root Cause Analysis	27
3.5. ESP32 and TFT Touchscreen	29
3.5.1. Operation	29
3.5.2. Validation	29
3.6. Subsystem Conclusion	31
4. AI Security Camera Subsystem	32
4.1. Subsystem Introduction	32
4.2. Subsystem Overview	32
4.3. Subsystem Validation	33
4.4. Subsystem Conclusion	37
5. MCU, Sensors, and Database Subsystem	37
5.1 Subsystem Introduction	37
5.2 Microcontroller Unit (MCU)	37

Final Report
Solar Home Lighting

5.2.1 Operation	37
5.2.2 Validation	38
5.3 Sensors	39
5.3.1 Operation	39
5.3.2 Validation	40
5.4 Database	41
5.4.1 Operation	
5.4.2 Validation	41

List of Tables

Table 3.3.2.6. Percent Error Results	21
Table 3.4.2.6. Percent Error Results	27
Table 4.3.1: Application FPS Over 1 Minute Validation Test Data	35
Table 4.3.2: Latency Test Results Data	36
Table 5.2: MCU Subsystem Validation Summary	39

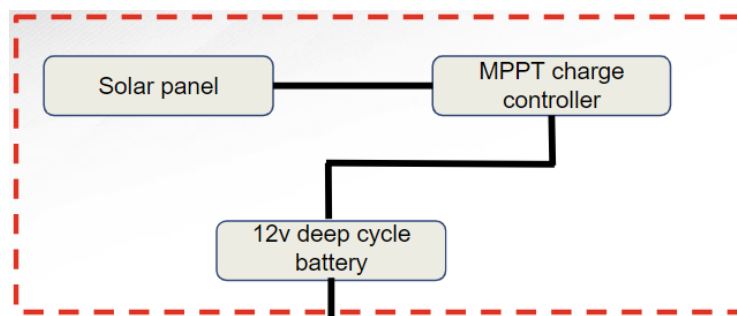
List of Figures

Figure 2.3.3 14.4V Buck conversion from OC Voltage	8
Figure 2.3.4 14.4V Buck conversion from Nominal 18.8V	9
Figure 2.3.5 14.4V Buck conversion from low voltage input	9
Figure 2.3.2.2 Verification of ESP32 DAC implementation	10
Figure 2.1.3 3.3V LM3150 Buck converter during Bulk/Absorption Stage	11
Figure 2.5.3 ESP32 adjusting DAC according to Battery SOC	12
Figure 2.5.4 ESP32 P&O according to efficiency of system	13
Figure 3.2.1. Physical Controls System Overview	14
Figure 3.3.2.1. Line Regulation $I_{out} = 0A$	16
Figure 3.3.2.2. Load Regulation $I_{out} = 1.75A$	16
Figure 3.3.2.3. Load Regulation	17
Figure 3.3.2.4. Efficiency at 3.3V	18
Figure 3.3.2.5.1. No load	19
Figure 3.3.2.5.2. Nominal Load $I_{out} = 0.8$ Amps	20
Figure 3.4.2.1. Line Regulation $I_{out} = 0A$	23
Figure 3.4.2.2. Load Regulation $I_{out} = 1.75A$	23
Figure 3.4.2.3. Load Regulation	24
Figure 3.4.2.4. Efficiency at 3.3V	25
Figure 3.4.2.5.1. No load	26
Figure 3.4.2.5.2. Nominal Load $I_{out} = 0.8$ Amps	26
Figure 3.4.1.1. Expected V_c Waveform During Correct Operation	28
Figure 3.4.1.2. V_c Waveform at Failure	29
Figure 4.2: Functional Block Diagram of AI Security Camera Subsystem	33
Figure 4.3.1: Application FPS Over 1 Minute Validation Test	34
Figure 4.3.2: Latency Test Results with Average Graph	36
Figure 4.3.3: Camera Reliability Validation Test Visual Graph	37

1. Introduction

The Solar Home Lighting System is composed of several coordinated subsystems that work together to deliver reliable off-grid lighting, security monitoring, and user control. These subsystems together include the solar charge controller and battery power system, the physical control panel and power distribution hardware, the microcontroller based sensing and automation logic, and AI enabled camera with mobile application interface. Each subsystem was designed to satisfy its individual functional requirements while maintaining compatibility with the overall system as defined in the ConOps, FSR, and ICD. By validating each subsystem, it demonstrated correct operation under expected conditions

2. Charge Controller Subsystem



2.1. Subsystem Introduction

The Maximum Power Point Tracking (MPPT) charge controller is a solar energy management system designed to maximize power transfer from the solar panel to a lead-acid battery. The controller uses a microcontroller to adjust duty cycles and charging stages, ensuring safe and efficient battery charging under varying sunlight conditions. Integrated sensors measure panel voltage and current, battery voltage and current, and load current, providing real-time data that enables precise monitoring of system performance.

2.2. Subsystem Overview

The MPPT charge controller is built to maximize energy transfer from a 200 W solar panel into a 12 V, 100 Ah sealed lead-acid battery while ensuring safe charging across bulk, absorption, and float stages. At the heart of the system is the LM5176 synchronous buck-boost controller, which steps down the solar panel's operating voltage (18–22 V) to the battery's required charging levels. The ESP32 microcontroller manages system intelligence: it runs the MPPT algorithm, adjusts duty cycles through DAC/PWM outputs, and switches feedback resistor networks to set the correct charging voltage for each stage.

A suite of sensors provides real-time monitoring: ACS781LL hall-effect sensors measure panel, battery, and load currents; voltage dividers track solar and battery voltages. This data enables the ESP32 to calculate battery state of charge and charging efficiency. Safety features include over-voltage, under-voltage, and over-current, as well as a load disconnect switch to prevent battery over-discharge.

2.3. Solar Panel to Battery Buck Converter (18.8V to 14.4V - 13.6V)

2.3.1. Operation

The solar panel buck converter will regulate the input voltage from the solar panel so that the 12V lead-acid battery can safely charge. The LM5176 Buck-Boost converter is used in this design to step down the panel's nominal 18.8 V input to the battery's required charging voltages of 14.4 V (Bulk/Absorb) and 13.6 V (Float). The ESP32 microcontroller supervises the LM5176 by adjusting its feedback network and DAC/PWM control signals, allowing dynamic regulation of duty cycle to track the maximum power point under varying irradiance conditions.

Input and output filtering, compensation components, and an appropriately sized inductor are selected to support stable operation and minimize voltage deviation. Reliable performance of the LM5176 stage is essential, as excessive ripple or poor transient response can reduce charging efficiency, stress the battery, or disrupt system monitoring. The converter, therefore, serves as the fundamental power stage enabling safe, efficient energy storage and stable operation of the overall MPPT charge controller.

2.3.2. Simulation Validation

To validate the LM5176 buck converter, the charge controller circuit was designed using PSpice for TI. These simulations replicate the behavior of the LM5176 as it steps down the solar panel's input voltage to match the battery's charging requirements. The converter maintains regulation across varying input voltages and responds appropriately to control signals from the ESP32's DAC output. This confirms that the output voltage can be dynamically adjusted to support bulk, absorption, and float charging stages, even in the absence of physical hardware.

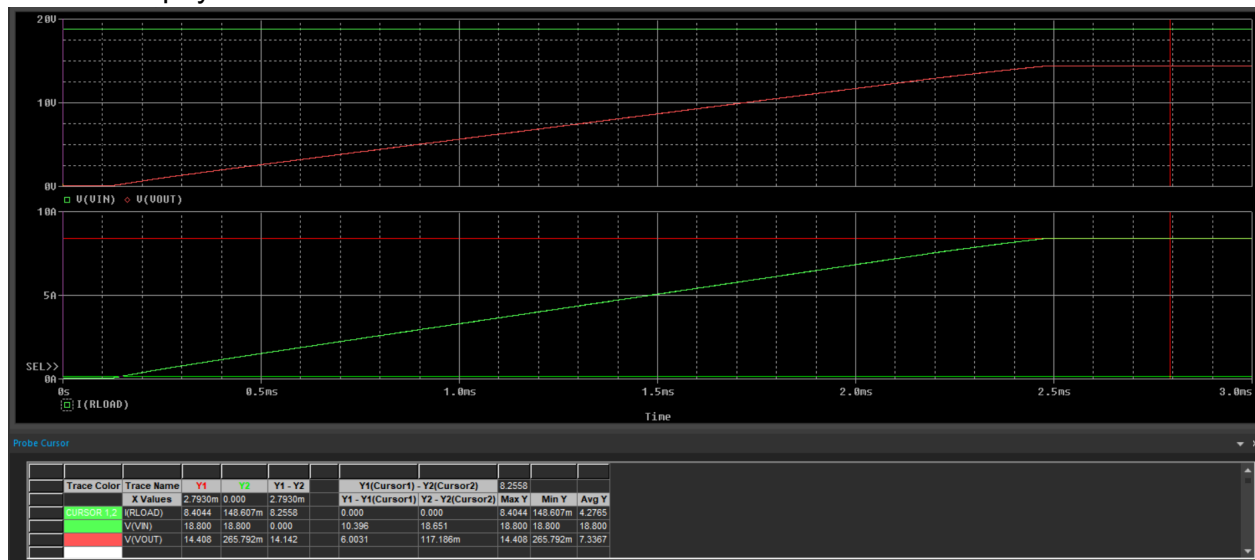


Figure 2.3.3 14.4V Buck conversion from OC Voltage

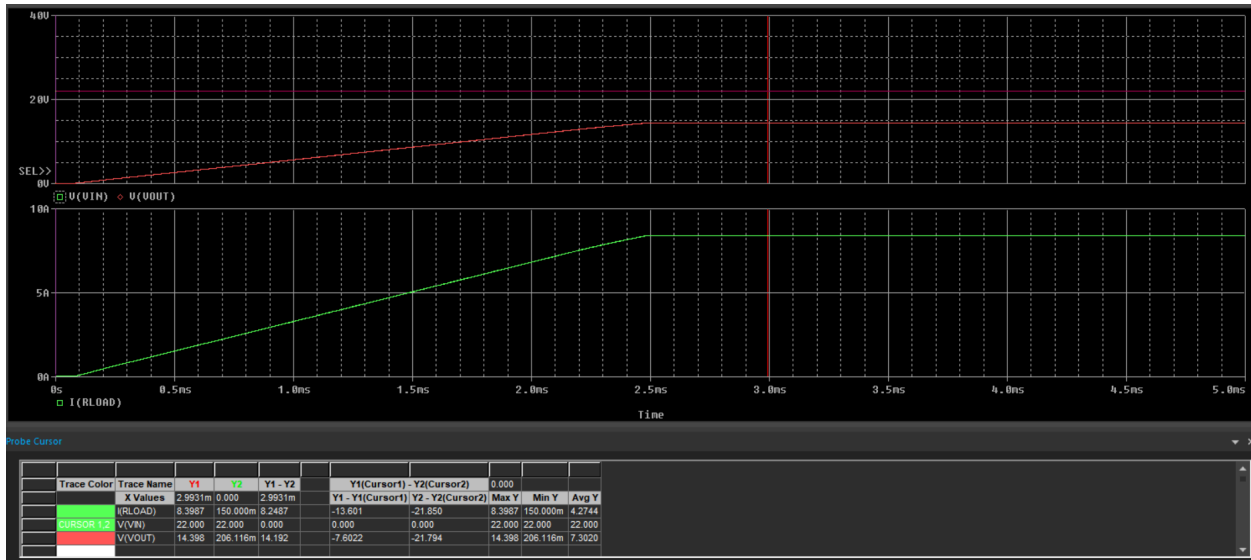


Figure 2.3.4 14.4V Buck conversion from Nominal 18.8V

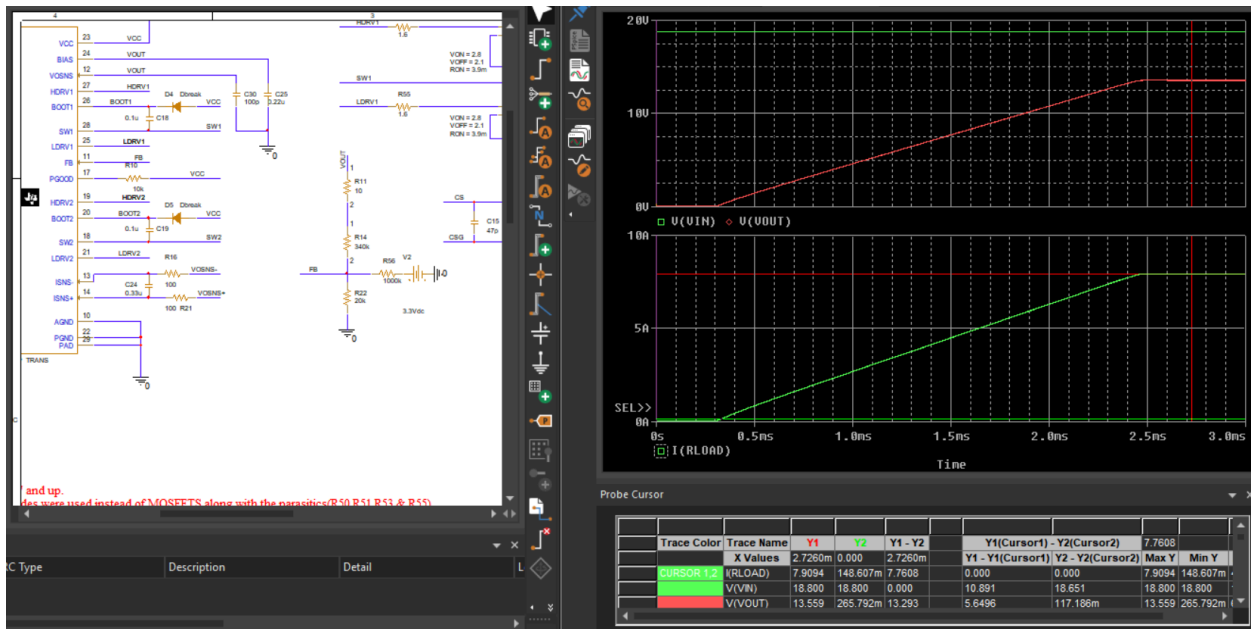


Figure 2.3.5 14.4V Buck conversion from low voltage input

2.3.2.1. 18.8V to 13.6V Step down (ESP32 DAC implementation)

A critical verification for MPPT implementation is demonstrating how the ESP32's DAC output interacts with the LM5176 feedback network to control converter duty cycle. In Figure 2.3.2.2, the DAC is modeled as a 3.3 V source connected through a 1 MΩ resistor to the LM5176 feedback pin. This configuration ensures that when the DAC reaches its maximum output, the LM5176 regulates the converter to approximately 13.6 V, corresponding to the battery's float charging stage. By scaling the DAC voltage through the resistor network, the ESP32 can dynamically bias the feedback loop, enabling precise adjustment of the converter's output voltage across different charging stages.

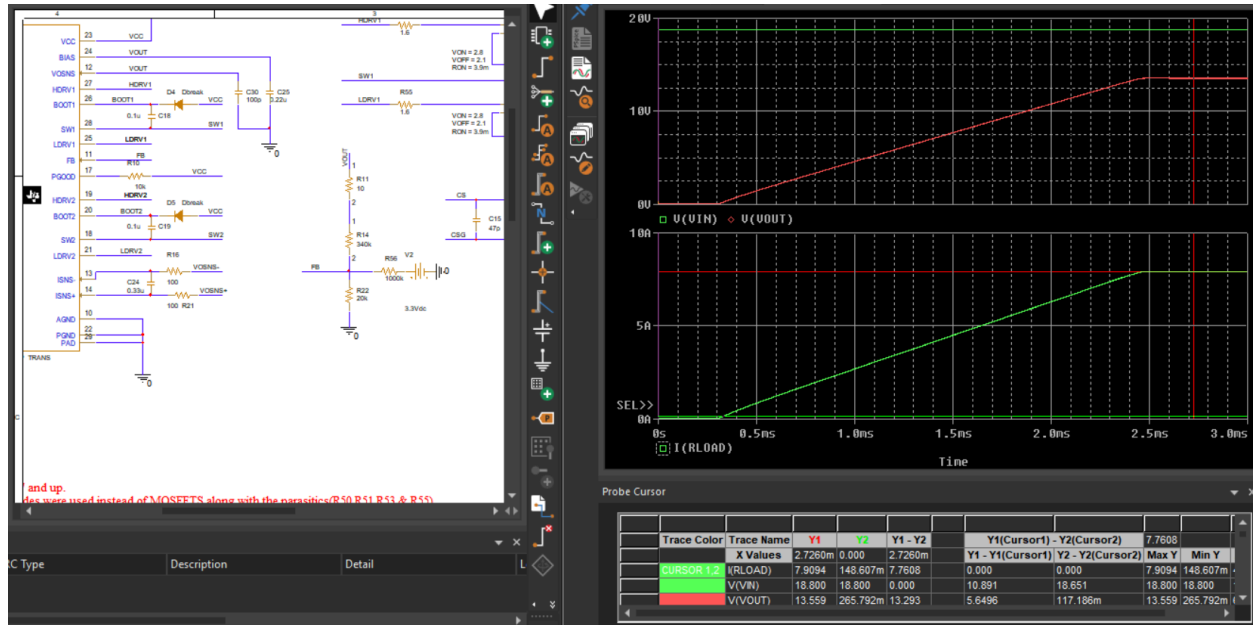


Figure 2.3.2.2 Verification of ESP32 DAC implementation

2.4. 3.3V Buck Converter

2.4.1. Operation

The 3.3 V buck converter provides the regulated logic supply required for all digital elements of the charge controller. It is implemented using the LM3150 synchronous step-down controller, which converts the 12–14.4 V from the battery bus into a stable 3.3 V output. This rail powers the ESP32 microcontroller and ACS781LL making reliable operation of the converter essential for system control and communication.

The LM3150 operates at a high switching frequency, allowing the use of compact inductors and capacitors while maintaining low output ripple and fast transient response. Input filtering ensures stable operation across the battery's voltage range, while the output filter and compensation network are tuned to support continuous conduction mode under dynamic digital loads. As the ESP32 and communication peripherals draw varying current, particularly during Wi-Fi transmission, UART logging, and Bluetooth activity, the converter must maintain regulation across rapidly changing load conditions.

2.4.2. Simulation Validation

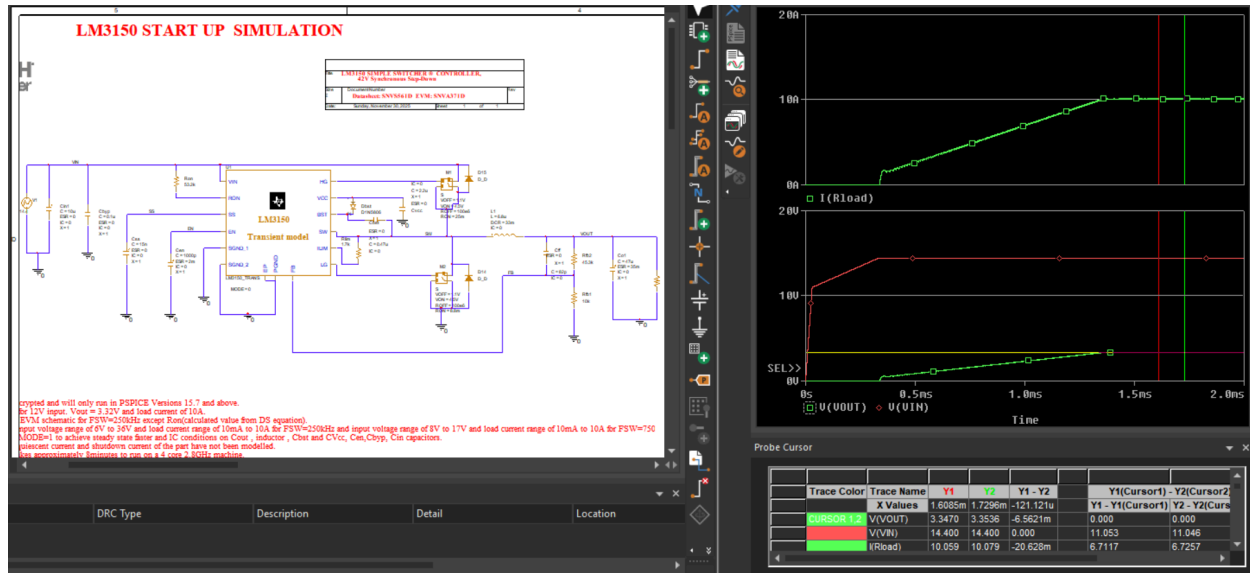


Figure 2.1.3 3.3V LM3150 Buck converter during Bulk/Absorption Stage

2.5. ESP32 Perturb & Observe (P&O) Program

2.5.1. Operation

The ESP32 microcontroller coordinates power conversion, sensing, protection, and communication. It executes the maximum power point tracking (MPPT) algorithm, dynamically adjusting the LM5176 buck converter through DAC and PWM outputs to optimize energy transfer from the solar panel. The ESP32 continuously acquires data from integrated sensors, including voltage dividers for panel and battery monitoring, and hall-current sensors for current measurement. With this information, the ESP32 will calculate the efficiency of the system and battery State of Charge (SOC). The ESP32 reads solar panel and battery voltages/currents, calculates power and efficiency, tracks battery state of charge (SOC), and drives the LM5176 buck converter via its DAC output to regulate charging. It uses a state machine to switch between charging stages

2.5.2. Program Validation

To validate program operation, test cases were developed to emulate the behavior of the solar panel and battery, ensuring that the ESP32 correctly drives the DAC output. Figure 2.5.3 illustrates the relationship between battery state of charge (SOC) and DAC value. As SOC increases, the DAC adjusts to regulate the LM5176 feedback network accordingly. During the Bulk stage, the battery draws high current, and maximum power point tracking (MPPT) is most effective for rapid charging. Once the battery approaches ~80% SOC or current demand decreases significantly, MPPT provides diminishing returns, and the converter transitions to the Absorption stage, holding the battery at 14.4 V until current tapers. At full capacity, the system enters the Float stage, where the DAC sets a lower

target voltage (13.6 V) to maintain charge while preventing overcharging and minimizing battery stress.

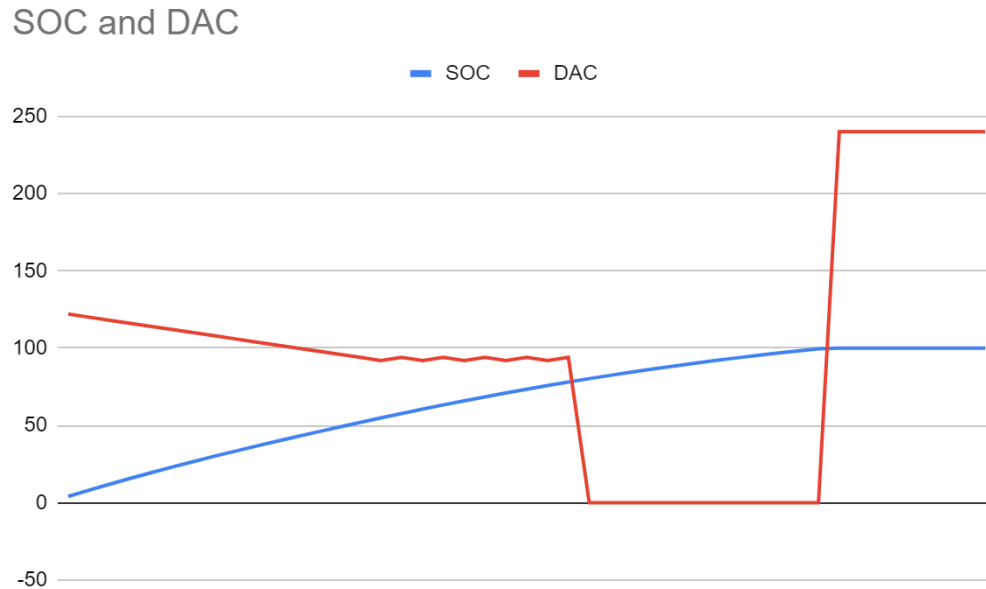


Figure 2.5.3 ESP32 adjusting DAC according to Battery SOC

Figure 2.5.4 illustrates the implementation of maximum power point tracking (MPPT) during the Bulk stage. The ESP32 monitors system efficiency while perturbing the DAC output to the LM5176 feedback pin using the Perturb and Observe (P&O) method. The DAC value is adjusted incrementally. If a change results in higher efficiency, the controller continues in the same direction, if efficiency decreases, the algorithm reverses direction and steps the DAC opposite. Through this iterative process, the ESP32 converges toward the operating point that maximizes panel power transfer to the battery.

DAC vs. efficiency for Bulk

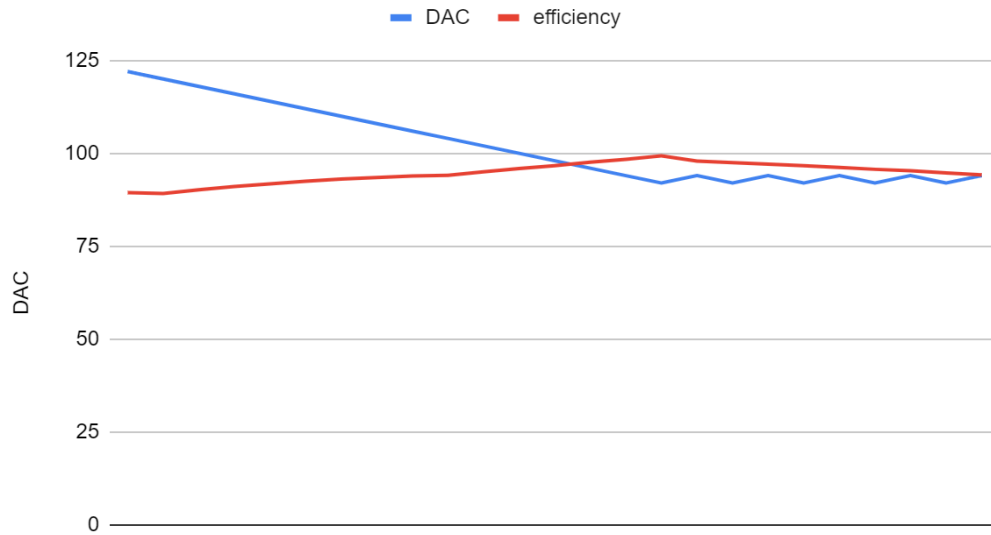


Figure 2.5.4 ESP32 P&O according to efficiency of system

2.6. Subsystem Conclusion

As the solar input and battery load fluctuate, the LM5176 converter must maintain regulation across rapidly changing operating points, ensuring continuous conduction and efficient energy transfer. Real-time information from the sensor components allows efficient regulation of charging stages while enforcing safety cutoffs for over-voltage, under-voltage, over-current, and deep discharge. By integrating control logic, sensing, protection, and communication, the ESP32 transforms the LM5176 synchronous buck-boost converter into an adaptive solar energy management system.

3. Physical Controls Subsystem

3.1. Subsystem Introduction

The Physical Controls Subsystem provides the local interface, power regulation, and lighting-control capability required for the Smart Solar Residential Lighting and Security System. It enables the homeowner to manually operate lighting zones through a touchscreen display while also supporting the microcontroller logic that executes user commands and routes control signals to the relay hardware. In addition to handling all user interaction, the subsystem delivers stable low-voltage power to the system's electronics through dedicated DC-DC conversion stages.

This subsystem forms the foundation for local operation of the overall system by supplying regulated power, processing interface inputs, and actuating the 12 V lighting loads. The following sections describe its electrical design, hardware operation, and validation testing.

3.2. Subsystem Overview

The Physical Controls Subsystem begins with a 10.8-14.2 V rail which is distributed to two internal domains: the relay hardware, which requires a stable 12 V supply for coil actuation, and a dedicated 3.3 V buck converter that generates the logic-level rail for all digital electronics. The regulated 3.3 V output powers both ESP32-S3 microcontrollers as well as the 2.8-inch ILI9341 TFT display and FT6206 capacitive touch controller. These elements make up the digital control layer of the subsystem. The touchscreen ESP32 functions as the human-machine interface, rendering the graphical control panel and interpreting user input. When the user selects a lighting function, the touchscreen microcontroller converts this interaction into a structured control command and transmits it over a dedicated UART link to the relay-control ESP32. The relay-control ESP32 receives these commands and responds by actuating the appropriate relay channels on the five-channel relay module. Each relay is driven through an isolated transistor interface, allowing the 3.3 V logic domain to safely control 12 V lighting loads. The relay controller returns acknowledgment messages back to the touchscreen ESP32 to maintain consistent system state awareness.

Through this hierarchy—input voltage distribution from 10.8 to 14.2 V, regulated power conversion, coordinated microcontroller communication, and relay-based switching—the subsystem delivers reliable and responsive lighting control under all expected operating conditions. The following sections describe the detailed operation and validation of each major component: the 3.3 V converter, the 12 V converter, and the ESP32 touchscreen and relay-control system.

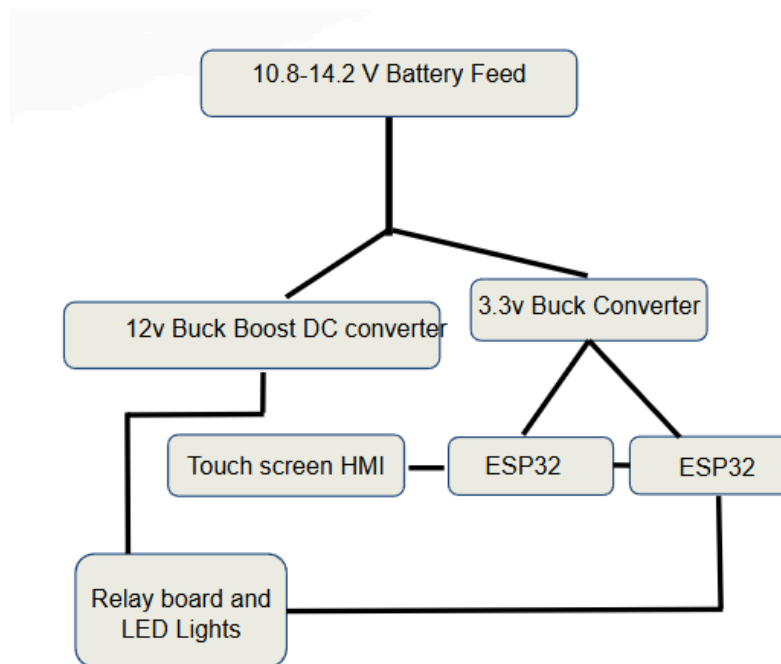


Figure 3.2.1. Physical Controls System Overview

3.3. 3.3V Buck Converter

To confirm that the 3.3 V converter met expected performance, a series of electrical tests were conducted under controlled bench conditions. These tests evaluated the converter's ability to maintain stable voltage output across varying input voltages and load currents.

3.3.1. Operation

The 3.3 V buck converter provides the regulated logic supply for all digital elements of the Physical Controls Subsystem. It is implemented using the LT8610AC synchronous step-down regulator, which converts the subsystem's incoming 10.8–14.2 V rail to a stable 3.3 V output. The converter operates at a high switching frequency, allowing the use of compact inductors and capacitors while maintaining low ripple and fast transient response. This 3.3 V rail powers both ESP32-S3 microcontrollers as well as the 2.8-inch ILI9341 TFT display and FT6206 capacitive touch controller. As these devices draw dynamically varying current—particularly during Wi-Fi transmission, display refresh operations, and UART communication—the converter must maintain regulation across rapidly changing load conditions. The LT8610AC design includes appropriate input and output filtering, compensation components, and an inductor selected to support stable continuous conduction across the subsystem's operational range. Reliable operation of the buck converter is essential, as voltage deviation or excessive ripple can produce ESP32 brownout resets or display instability. The converter therefore serves as the fundamental power stage enabling stable digital control and user interaction.

3.3.2. Validation

Validation of the 3.3 V converter was performed to ensure proper electrical behavior across expected operating conditions. Testing focused on confirming stable output regulation, acceptable ripple performance, and consistent efficiency.

3.3.2.1. Line Regulation

Line regulation was evaluated by sweeping the input voltage from 10.8 V to 14.2 V under two load conditions: no load and a high load of 1.75 A. The results for both cases are shown in Figures 3.3.2.1 and 3.3.2.2.

Under no-load conditions ($I_{out} = 0$ A), the converter output varied by less than 0.1% across the entire input range. The curve remained nearly flat, indicating that changes in input voltage had minimal effect on the 3.3 V output. This behavior confirms that the LT8610AC maintained strong closed-loop control even in a low-current state.

When tested at 1.75 A, representing a near-maximum subsystem load, the output deviation increased slightly, ranging from approximately -0.55% to -0.77% . Although the magnitude of variation was greater than in the no-load condition, the response remained smooth and well within acceptable tolerance for ESP32 operation. No signs of instability, oscillation, or dropout were observed during the sweep.

Final Report

Solar Home Lighting

Overall, the converter demonstrated stable performance across the full input voltage range. The measured deviations are consistent with expected behavior of a buck converter operating under varying duty cycles, and the results verify that the 3.3 V rail remains properly regulated for all expected operating conditions of the subsystem.

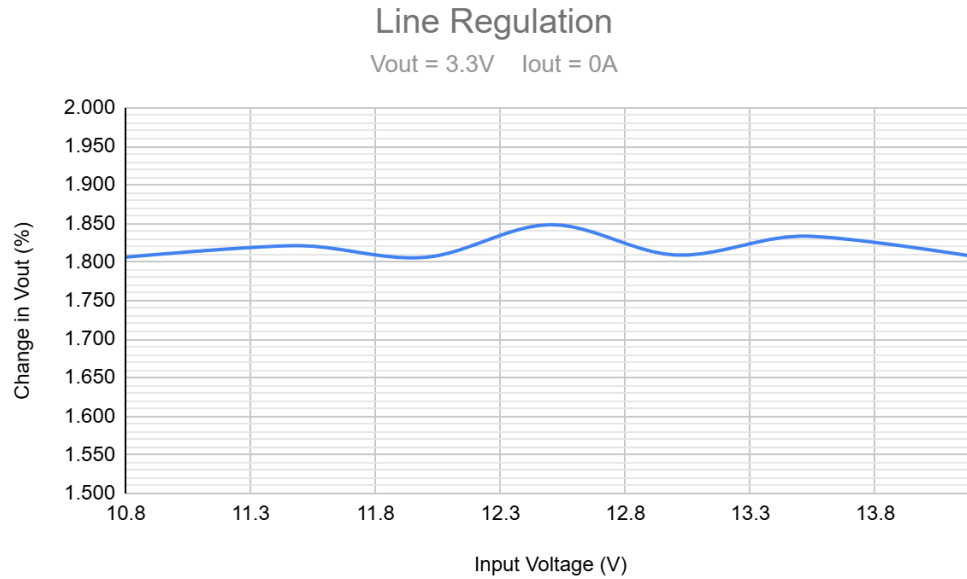


Figure 3.3.2.1. Line Regulation $I_{out} = 0A$

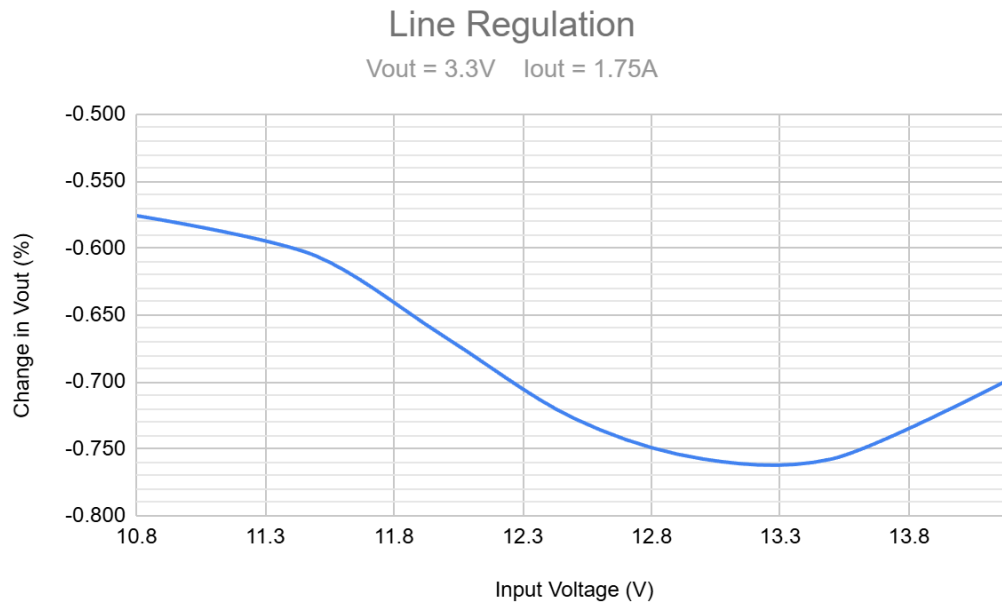


Figure 3.3.2.2. Load Regulation $I_{out} = 1.75A$

3.3.2.3. Load Regulation

Load regulation was evaluated by increasing the output current from 0 A to 2.0 A, measuring the percent change in output voltage at input voltages of 10.8 V, 12 V, and 14.2 V. The results are presented in Figure 3.3.2.3.

Across all three input voltages, the converter showed a smooth and controlled decrease in output voltage as the load increased. From 0 A up to approximately 1.7 A, the change in V_{out} remained within about -0.5% to -2.5%, and the curves for all input voltages were closely aligned. This indicates that the converter maintained stable output regulation across the majority of the load range, with performance largely unaffected by variations in V_{in} .

As the load approached the upper limit of the test, the converter exhibited a more rapid voltage drop. At the full 2.0 A load, V_{out} deviation increased to approximately -5.5% to -6.0% depending on input voltage. This behavior is expected when operating near the converter's maximum current capability, where conduction losses, switch resistance, and inductor saturation effects become more significant.

Overall, the converter maintained acceptable regulation across its entire tested range up to 2.0 A. Output degradation occurred only at the extreme high-load condition, confirming that the 3.3 V rail remains suitable for all expected subsystem operating currents.

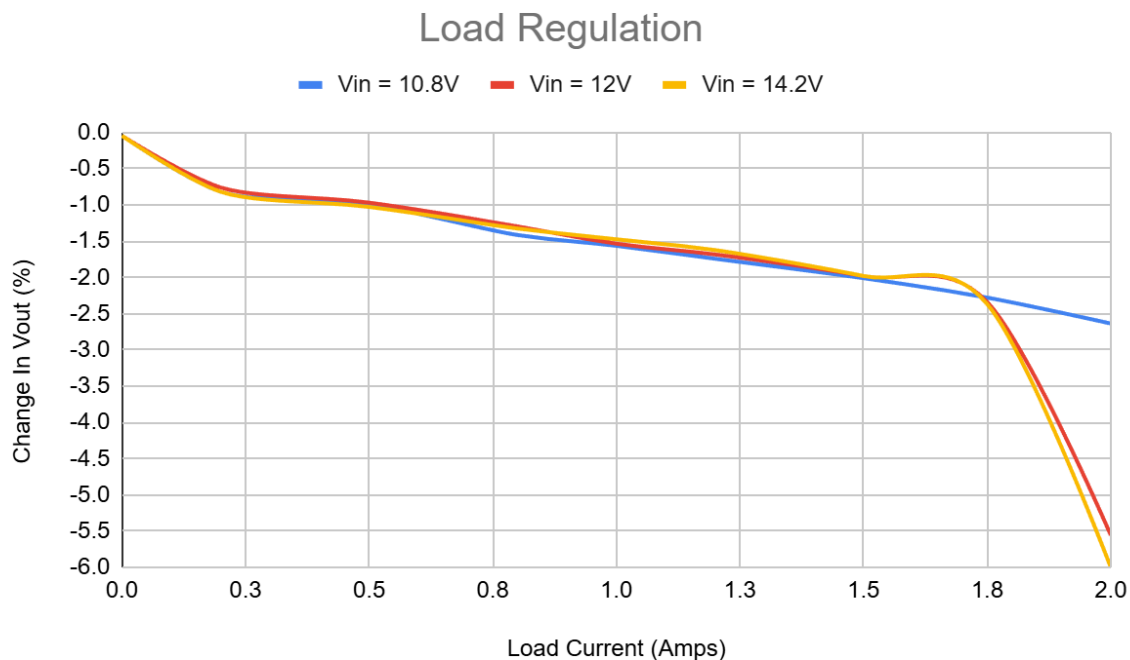


Figure 3.3.2.3. Load Regulation

3.3.2.4. Efficiency

Efficiency was measured across load currents from 0.2 A to 2.0 A for input voltages of 10.8 V, 12 V, and 14.2 V. The results are shown in Figure 3.3.2.4.

For all three input conditions, efficiency increased rapidly as load current rose from 0.2 A to approximately 0.7 A, reaching a peak of about 92-93%. This operating region corresponds to the converter's optimal balance of switching and conduction losses. Beyond 0.7 A, efficiency gradually decreased with increasing load, falling to approximately 88-89% at 1.5 A.

As the converter approached its maximum tested load of 2.0 A, efficiency dropped more noticeably. At this point, efficiency reached roughly 82-85%, depending on the input voltage, with the lowest efficiency occurring at $V_{in} = 14.2$ V. This behavior is consistent with increased conduction losses and the higher duty cycles required to maintain output regulation under heavy load.

Across the entire tested range, efficiency curves for all input voltages were closely aligned, indicating that V_{in} had only a minor effect on performance. The converter demonstrated high efficiency in the load region most representative of normal subsystem operation (0.3–1.2 A), confirming suitability for continuous powering of the ESP32 microcontrollers and display hardware.

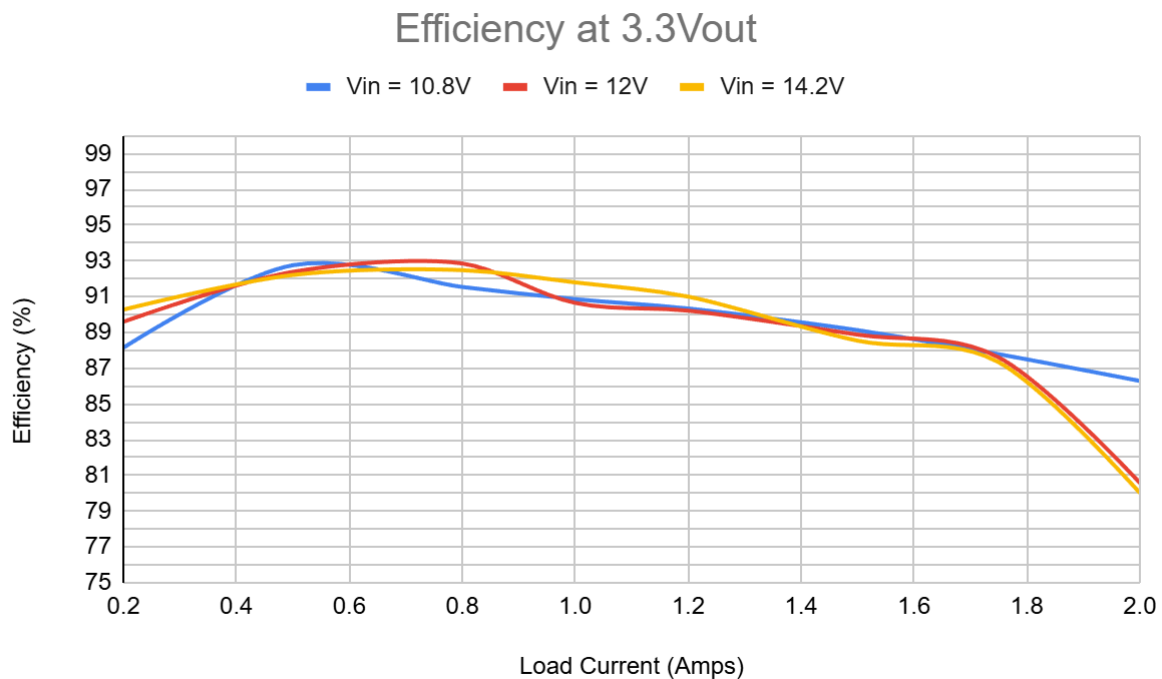


Figure 3.3.2.4. Efficiency at 3.3V

3.3.2.5. Output Voltage Ripple

Ripple measurements were taken under two conditions: no load and a nominal operating load of 0.8 A. At no load, the converter produced approximately 71 mV peak-to-peak ripple on the 3.3 V output (Figure 3.3.2.5.1). This value aligns with expectations for a high-frequency buck converter operating with minimal inductor current and reduced energy storage.

Under nominal load conditions ($I_{out} = 0.8 \text{ A}$), ripple increased slightly to approximately 90 mV peak-to-peak (Figure 3.3.2.5.2). This rise in ripple is consistent with higher inductor current ripple at increased load and the converter operating closer to its continuous conduction region. The magnitude of ripple remained stable, with no signs of oscillation, discontinuities, or switching irregularities.

Across both test conditions, the observed ripple remained well within acceptable limits for powering ESP32 microcontrollers and the TFT display, neither of which showed instability or brownout behavior during operation. The results confirm that the output filtering network is adequate for the subsystem's digital load requirements.

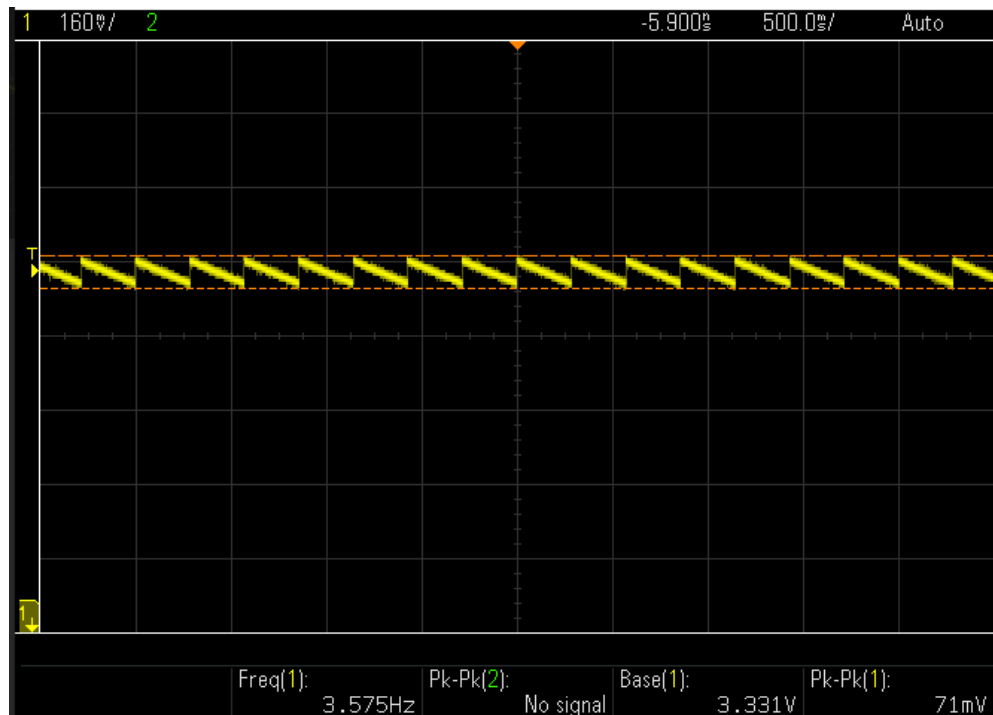


Figure 3.3.2.5.1. No load

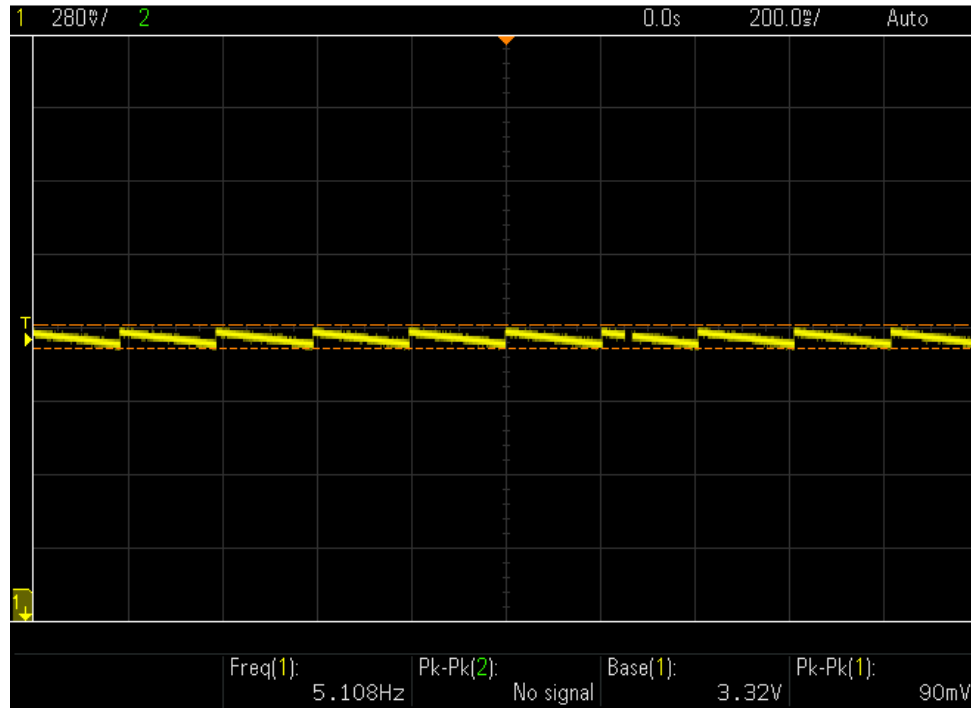


Figure 3.3.2.5.2. Nominal Load $I_{out} = 0.8$ Amps

3.3.2.6. Percent Error

Percent error was calculated across a range of input voltages and load conditions to quantify the deviation of the converter's output from the nominal 3.3 V setpoint. The results are summarized in Table 3.3.2.6.

Under no-load conditions, output voltage measured approximately 3.357-3.359 V, corresponding to a percent error of 1.73-1.79%. This slight positive deviation is expected due to minimal load current and the converter operating in a low-duty, discontinuous conduction region.

At nominal load (0.8 A), accuracy improved significantly, with percent error reduced to 0.36–0.48% across the input voltage range. This reflects the converter transitioning into a more stable conduction mode, where switching behavior and inductor current are better regulated.

At normal maximum load (1.75 A), the converter displayed small negative deviations, ranging from -0.52% to -0.61%, indicating a slight drop in output voltage under higher current demand. These values remain well within acceptable limits for the ESP32 and associated logic circuitry.

At the absolute maximum tested load (2.0 A), percent error increased to -0.88% to -4.15%, with the largest deviation occurring at $V_{in} = 14.2$ V. This increased drop corresponds to the converter operating near its upper current limit, where conduction losses and switch resistance become more significant.

Across the full range of operating conditions, the percent error remained within levels that support stable microcontroller and display operation. Only the highest load case showed notable deviation, which aligns with the design's expected maximum load boundary.

Vin	Iin	Iout	Vout	State	%error
10.8	0	0	3.358	No Load	1.76%
12	0	0	3.359	No Load	1.79%
14.2	0	0	3.357	No Load	1.73%
10.8	0.268	0.8	3.312	Nominal Load	0.36%
12	0.242	0.8	3.313	Nominal Load	0.39%
14.2	0.202	0.8	3.316	Nominal Load	0.48%
10.8	0.606	1.75	3.283	Normal Max	-0.52%
12	0.546	1.75	3.28	Normal Max	-0.61%
14.2	0.463	1.75	3.281	Normal Max	-0.58%
10.8	0.703	2	3.271	Absolute Max	-0.88%
12	0.647	2	3.182	Absolute Max	-3.58%
14.2	0.549	2	3.163	Absolute Max	-4.15%

Table 3.3.2.6. Percent Error Results

3.4. 12V Buck-Boost Converter

3.4.1. Operation

The 12 V rail is generated using the LTC3119 buck-boost converter, selected for its ability to maintain a regulated 12 V output from the system's variable input range of 10.8-14.2 V. The converter was designed to support a normal operating load of 0.7-1.0 A, with a continuous output capability of 2 A and a peak load tolerance of up to 3 A for short-duration relay actuation or transient events.

The buck-boost topology is intended to ensure uninterrupted light functionality during battery discharge, seamlessly transitioning between buck and boost operation as required. The design follows the LTC3119 datasheet recommendations for inductor current rating, compensation network, and input/output capacitor values. Under correct implementation, the converter should provide sufficient power for all lighting loads.

However, during system testing, the converter operated reliably only up to approximately 0.85 A, significantly below the intended 2-3 A capability. Beyond this point, the converter failed to maintain regulation and output voltage collapsed. This load limitation directly affected the ability to validate the converter at its full design rating and prompted further investigation into potential hardware faults.

3.4.2. Partial Validation

Validation testing assessed converter performance across line and load conditions within the functional range available on the assembled hardware.

3.4.2.1. Line Regulation

Figures 3.4.2.1 and 3.4.2.2 show the measured line regulation performance of the 12 V buck-boost converter at 0 A and 0.8 A, respectively. Testing was conducted over the full operating input range of 10.8 V to 14.2 V, reflecting the expected voltage swing of a typical residential battery system.

At 0 A load, the converter output deviated between approximately -0.30% and 0%, indicating relatively tight regulation when the converter is lightly loaded. The output remains close to nominal across all input voltages, with no significant drift or instability as V_{IN} increases. Minor variations appear between 12.3 V and 13.3 V, but remain well below 0.3%, showing good unloaded regulation behavior.

At 0.8 A, the highest current the hardware could sustain, the output deviation increases slightly, ranging from -1.0% to -0.7%, but the overall trend remains consistent across the input sweep. The voltage droop deepens at the lower end of the V_{IN} range and improves modestly around mid-range input levels before rising slightly near 14 V. No oscillatory behavior or step changes were present, confirming stable operation within the converter's reduced functional load capability.

Overall, the line regulation results demonstrate that the converter maintains predictable behavior across varying input voltages for both no-load and functional-load conditions. Although the converter could not reach its intended operating range of 2 A nominal, the partial data indicates that within the 0-0.8 A region the converter remains stable and exhibits regulation performance that follows expected buck-boost characteristics.

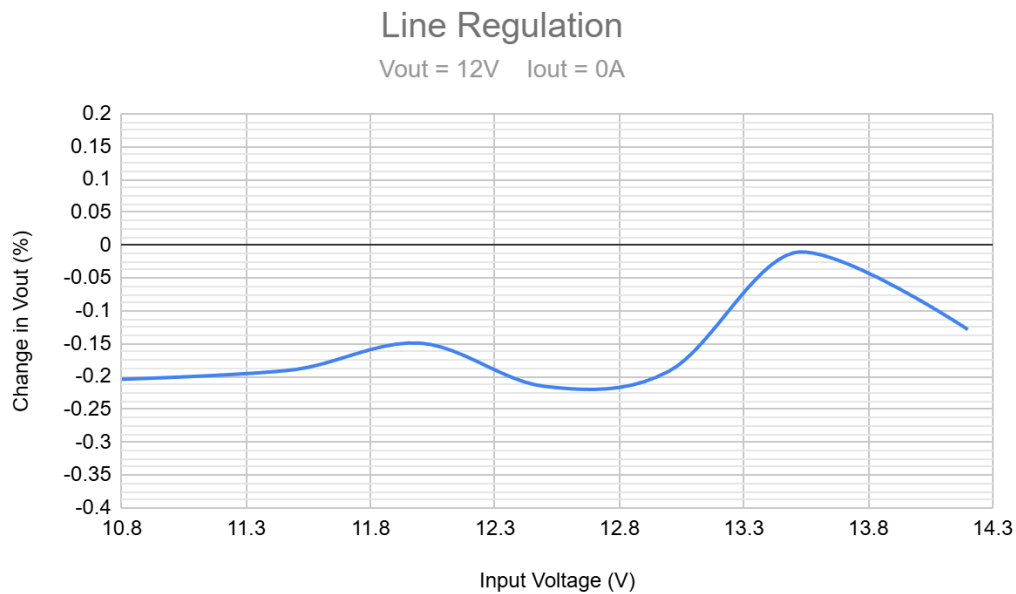


Figure 3.4.2.1. Line Regulation $I_{out} = 0A$

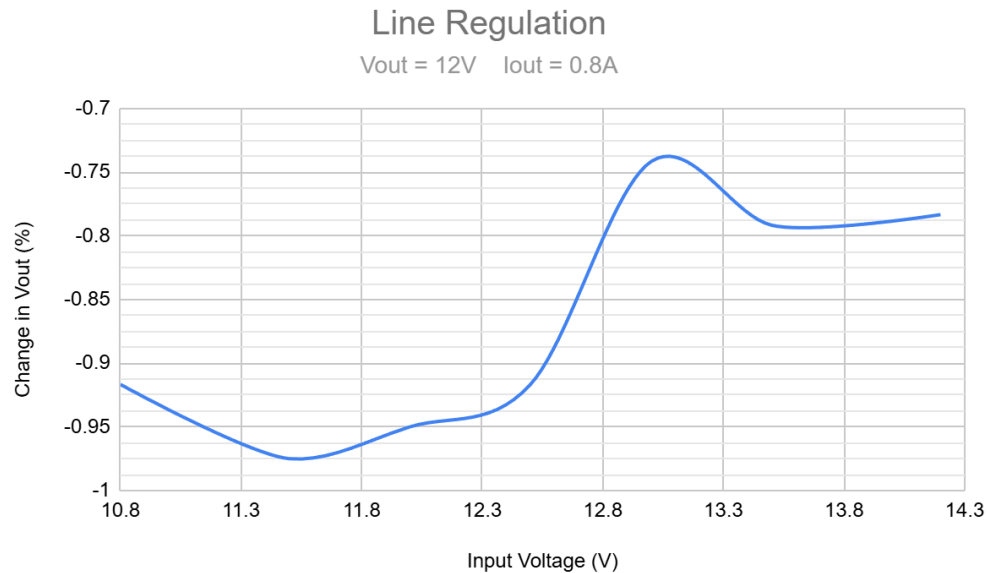


Figure 3.4.2.2. Load Regulation $I_{out} = 1.75A$

3.4.2.3. Load Regulation

Figure 3.4.2.3 illustrates the load-regulation performance of the 12 V buck-boost converter for the three tested input voltages (10.8 V, 12 V, and 14.2 V) across the converter's functional load range of 0.1 A to 0.8 A. Although the converter was designed to support up to 2A continuous output, hardware limitations restricted successful operation to approximately 0.8 A. Within this region, the converter remained stable and produced consistent output-voltage behavior.

Across all three curves, the output voltage shows a gradual negative deviation as load current increases. For $V_{IN} = 10.8V$, the converter demonstrates the best overall regulation, with deviations beginning near -0.25% at 0.1 A and decreasing to approximately -0.55% at 0.8 A. The slope is gradual, indicating that the converter maintains reasonable voltage control despite approaching the edge of its functional current capability.

For $V_{IN} = 12V$, regulation remains relatively flat at lighter loads, with deviations clustered near -0.30% up to around 0.3 A. Beyond this point, the output droop becomes more pronounced, reaching approximately -0.60% at 0.8 A. This trend indicates that the converter handles moderate load conditions adequately but begins to show increased voltage sag as the current approaches the functional limit.

For $V_{IN} = 14.2V$, the converter exhibits the greatest deviation at light load, beginning around -0.40% . The regulation briefly improves near 0.25 A before steadily declining with increasing load, ultimately reaching approximately -0.70% at 0.8 A. This behavior suggests reduced efficiency at higher input voltage, consistent with the converter's elevated switching and conduction losses at lighter load conditions.

Overall, the load-regulation results indicate that the converter maintains predictable and stable performance within its reduced operating range. Although output droop increases with load, no instability or oscillatory behavior was observed, and the trends remain consistent with expected buck-boost converter characteristics operating below their nominal current capability.

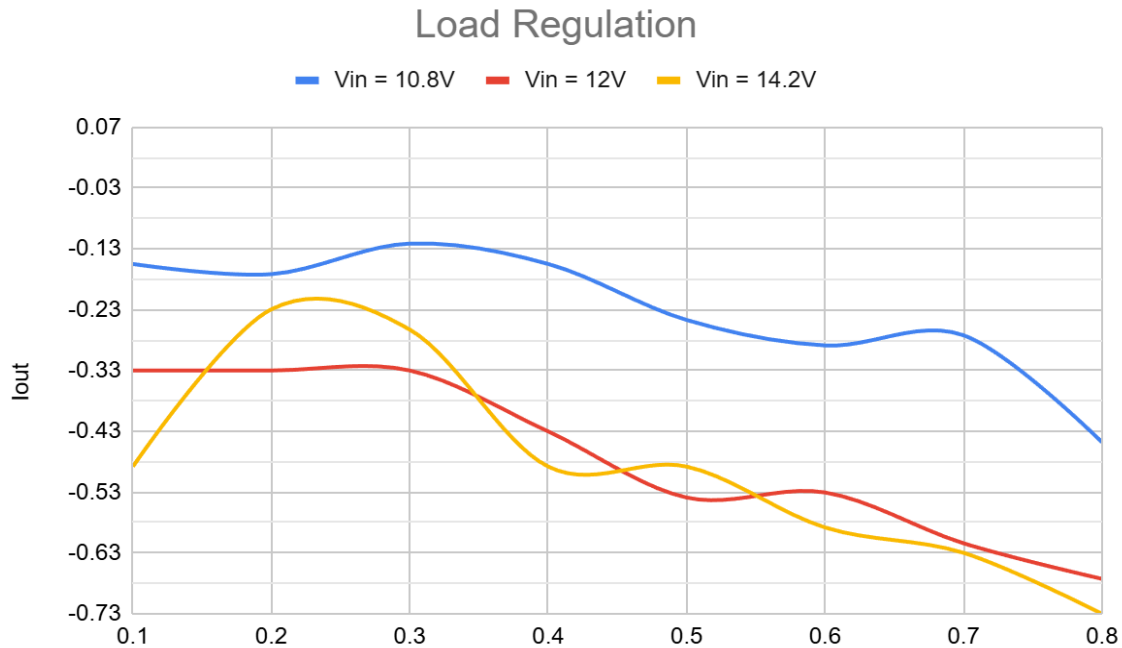


Figure 3.4.2.3. Load Regulation

3.4.2.4. Efficiency

Figure 3.4.2.4 shows the measured efficiency of the 12 V buck-boost converter across load currents from 0.1 A to 0.8 A for three input voltages (10.8 V, 12 V, and 14.2 V). Although the converter was originally designed for higher load capability, only data up to 0.8 A is shown due to the converter's functional limitation above this range. Within the operating region, efficiency follows a consistent and predictable trend.

For $V_{IN} = 10.8 V$, efficiency starts at approximately 89% at 0.1 A and increases steadily with load, reaching a peak of nearly 96% around 0.6 A, after which it tapers slightly but remains above 93% at 0.8 A. This curve represents the best overall performance, as lower input voltage typically reduces switching losses in buck-boost converters operating near unity conversion ratio. For $V_{IN} = 12 V$, the converter demonstrates a slower rise in efficiency, beginning near 85% at 0.1 A and climbing to roughly 93% at 0.5–0.6 A. Beyond this region, efficiency decreases marginally due to increasing conduction losses and the converter approaching its functional current limit. Overall, the performance remains within an acceptable range for a converter operating in buck-boost mode. For $V_{IN} = 14.2 V$, efficiency is the lowest at light load (approximately 76% at 0.1 A), likely due to increased switching and gate-drive losses at higher input voltage. Efficiency improves significantly with load, peaking around 94–95% near 0.6 A, and stabilizes above 93% as the converter approaches

0.8 A. This trend aligns with expected behavior: as load increases, the converter operates in a region where switching losses are proportionally less dominant.

Across all input conditions, the converter demonstrates strong mid-load efficiency performance, with peak efficiency values above 93% and in some cases approaching 96%. These results indicate that, despite the reduced maximum-load capability, the converter operates efficiently within its functional range and is well-suited for the system's typical current demands (0.7–1.0 A).

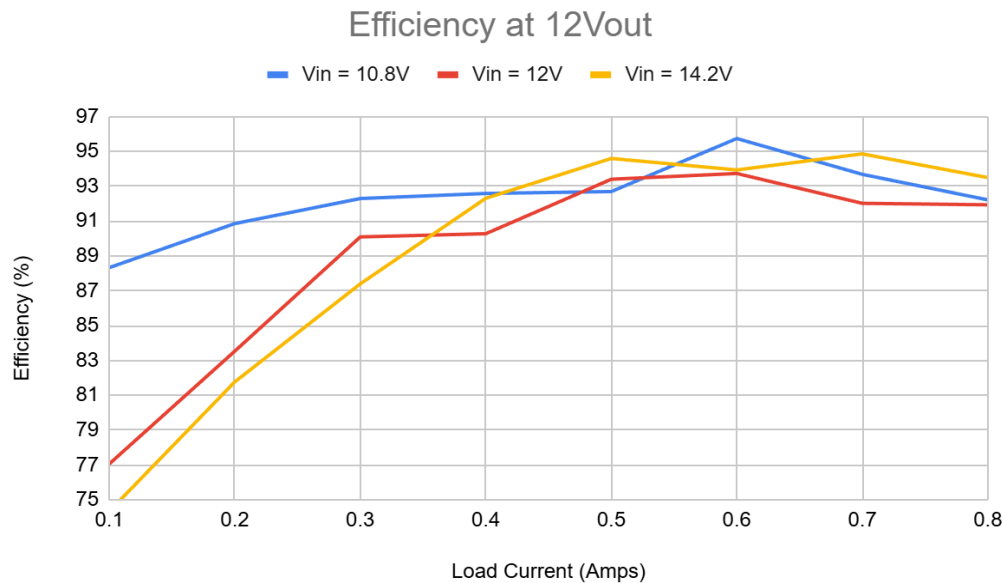


Figure 3.4.2.4. Efficiency at 3.3V

3.4.2.5. Output Voltage Ripple

Ripple measurements for the 12 V converter are shown in Figures 3.4.2.5.1 and 3.4.2.5.2. At no load, the output exhibited approximately 181 mV peak-to-peak ripple.

Under nominal load (0.8 A) the highest current the converter could sustain before failing the ripple increased to approximately 325 mV peak-to-peak.

Both values are significantly higher than expected for this converter architecture and exceed typical ripple limits for a stable buck-boost stage. The elevated ripple is consistent with the hardware fault identified during testing, in which the converter was unable to deliver current beyond ~0.85 A. This degraded operating condition likely caused improper switching behavior and reduced filtering effectiveness, leading to the unusually large ripple amplitudes.

Because the unit did not reach its intended operating region, these ripple results represent the behavior of a partially failed converter rather than the expected performance of the design.

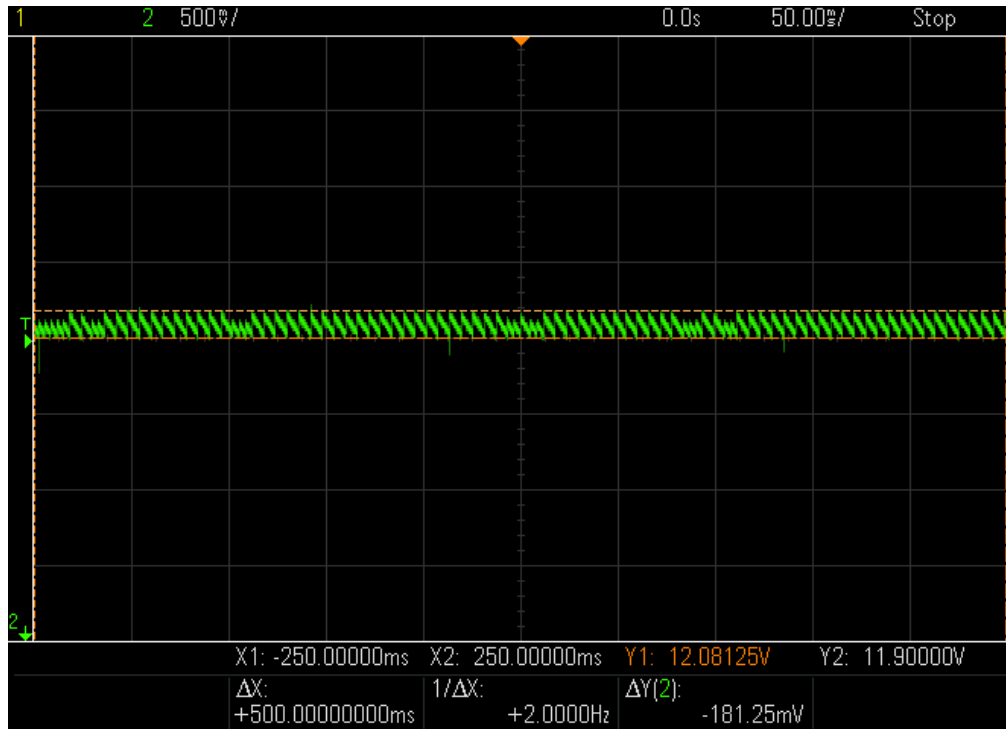


Figure 3.4.2.5.1. No load

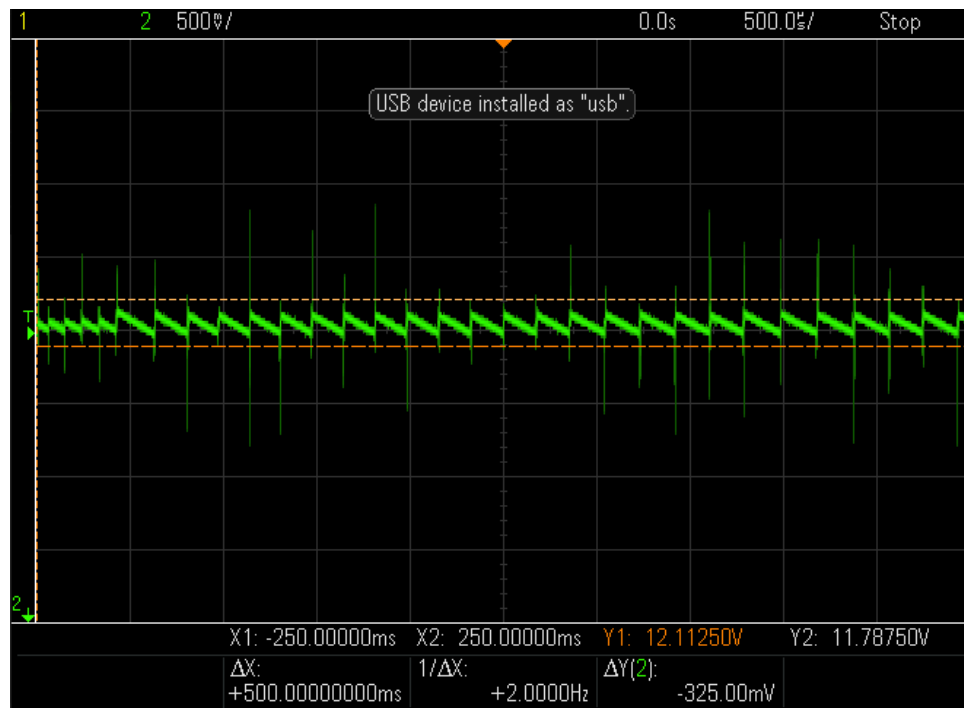


Figure 3.4.2.5.2. Nominal Load $I_{out} = 0.8$ Amps

3.4.2.6. Percent Error

Table 3.4.2.6 summarizes the percent error of the 12 V converter across all measured operating points.

Under no-load conditions, the output remained within -0.13% to -0.25%, demonstrating stable voltage regulation when the converter is lightly loaded.

At the single-light and dual-light load levels (approximately 0.45 A and 0.8 A), the percent error ranged from -0.70% to -0.96%, which is still within an acceptable tolerance for a 12 V regulation stage. The gradual increase in negative error reflects the additional voltage drop as the converter is brought closer to its functional limit.

The final operating region—labeled “Limit”—captures the highest load currents the converter was physically able to sustain: 0.855–1.0 A, depending on input voltage. At these points, the percent error increased to -1.0% to -1.10%, and the converter was unable to provide additional current. This behavior confirms the earlier observation that the hardware fault prevented the converter from reaching its intended design range (2 A nominal, 3 A peak). The percent-error values in this region represent operation at the edge of failure rather than normal performance.

Because the converter’s current ceiling was reached prematurely, these percent-error results do not reflect the expected accuracy of the full design but instead illustrate the degraded behavior of the partially failed unit.

Vin	Iin	Iout	Vout	State	%error
10.8	0	0	11.97	No Load	-0.25%
12	0	0	11.984	No Load	-0.13%
14.2	0	0	11.983	No Load	-0.14%
10.8	0.526	0.45	11.912	Single Light Load	-0.73%
12	0.481	0.45	11.907	Single Light Load	-0.78%
14.2	0.405	0.45	11.916	Single Light Load	-0.70%
10.8	0.935	0.8	11.891	Dual Light Load	-0.91%
12	0.859	0.8	11.885	Dual Light Load	-0.96%
14.2	0.708	0.8	11.903	Dual Light Load	-0.81%
10.8	0.979	0.855	11.877	Limit	-1.02%
12	0.898	0.865	11.881	Limit	-0.99%
14.2	0.884	1	11.868	Limit	-1.10%

Table 3.4.2.6. Percent Error Results

3.4.1. Failure Root Cause Analysis

During validation, the 12 V DC-DC converter operated correctly up to approximately 0.8 A of load current. Above ~0.9–0.93 A, the converter consistently entered a collapse-and-restart failure mode in which VOUT dropped to only a few millivolts, momentarily recovered, and then failed again. VIN remained stable during all events, and the measured IC temperature during testing was low (~94 °F), indicating the failure was not caused by normal operating thermal stress. Instead, the failure is attributed to a combination of PCB grounding/layout issues and possible internal degradation of the LTC3119 IC caused by overheating during assembly.

Figure 3.4.1.1 shows the VCC/VC node under correct operation at ~0.8 A. The waveform remains relatively stable, with small switching-related perturbations as expected for a functional compensation loop.

However, as shown in Figure 3.4.1.2, once the converter reaches approximately 0.93 A, the VCC/VC waveform becomes a distorted, noisy signal that resembles a corrupted version of the SW1 switching node. Instead of a clean bias supply, the IC's internal control voltage is overtaken by switching noise. When the effective VCC drops or becomes unstable, the LTC3119's internal protection circuitry disables switching. With switching halted, VCC temporarily rises again, causing the chip to attempt a restart—producing the cyclical collapse observed during testing.

This distorted control-voltage behavior directly indicates a loss of control-loop stability and instability in the device's internal bias generator, which strongly supports a grounding-related failure mechanism.

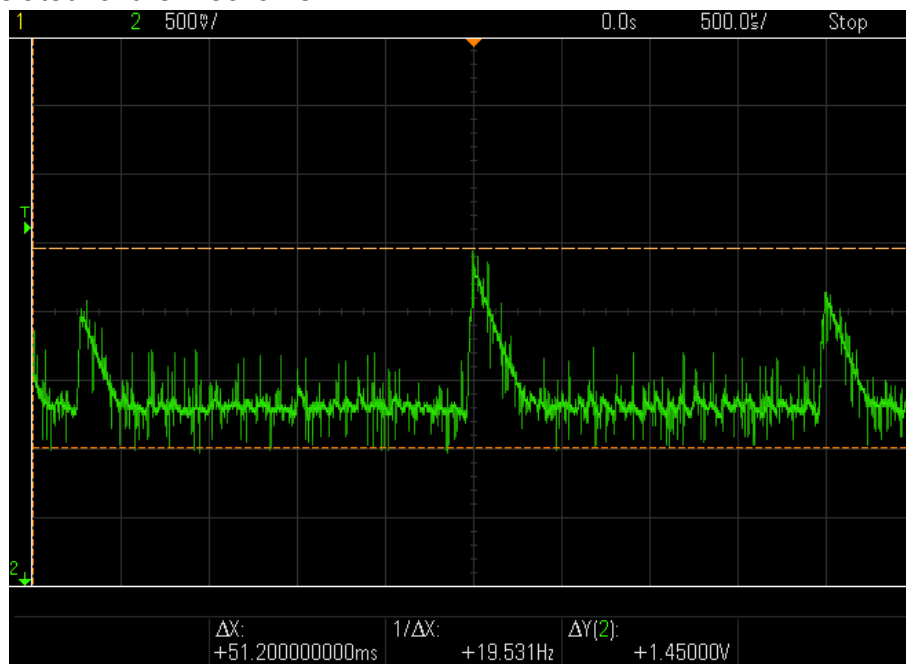


Figure 3.4.1.1. Expected Vc Waveform During Correct Operation

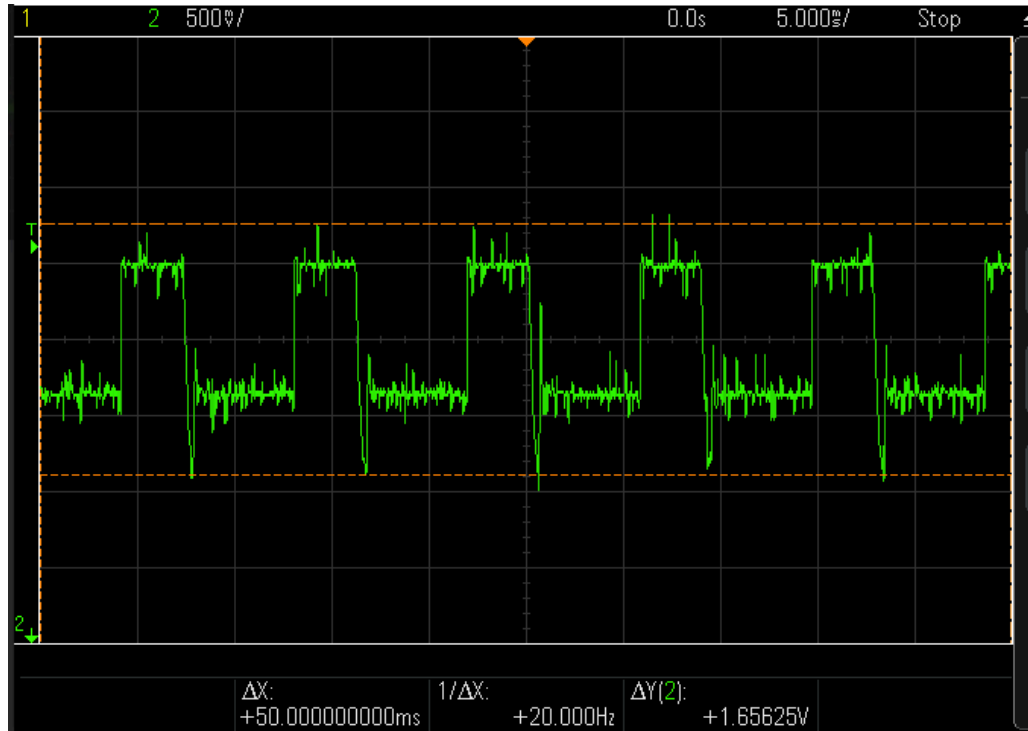


Figure 3.4.1.2. Vc Waveform at Failure

In this PCB revision, the SGND, VC, FB, and VCC/MPPC/SVCC decoupling networks were tied directly into the global ground polygon that also carries the high-di/dt switching currents of the power stage, preventing the LTC3119's analog control circuitry from having a quiet and stable ground reference. As output load increases, the corresponding rise in switching-current amplitude creates significant ground bounce across the shared plane, causing SGND and the internal VCC reference to shift relative to true ground. This noise couples into the VCC/VC node, contaminating the control voltage with switching artifacts and destabilizing the error amplifier. The distorted VCC waveform observed during failure directly confirms this behavior. In addition to the grounding issue, the LTC3119 IC was overheated during installation, and this likely weakened the internal bias regulator or error-amplifier structures, reducing their noise immunity and narrowing the device's usable control-voltage margin. While the grounding flaw alone can explain degraded current capability, the combination of ground noise and latent thermal damage makes the control loop far more susceptible to collapse as load increases, ultimately limiting the converter to approximately 0.93 A well below its intended operating range.

The failure of the 12 V converter at approximately 0.93 A is therefore attributed to the combined effects of improper grounding layout and thermal overstress of the LTC3119 IC during assembly. The shared analog/power ground return allowed switching noise to corrupt the VCC/VC control node, while the overheated IC exhibited reduced tolerance to this disturbance, causing the control loop to collapse and forcing the device into its internal protection mode at elevated load. A future hardware revision that provides a properly isolated SGND region, a short and direct return path for the VCC/MPPC/SVCC decoupling capacitor, and improved assembly temperature control is expected to eliminate this failure mode and restore the converter's full rated output-current capability.

3.5. ESP32 and TFT Touchscreen

3.5.1. Operation

The touchscreen and relay-control subsystem forms the primary user interface and digital switching layer of the Physical Controls System. It is implemented using two ESP32-S3 microcontrollers that divide responsibility between graphical user interaction and relay actuation. The subsystem is powered by the regulated 3.3 V domain generated from the main supply, which supports the logic-level electronics, while the 12 V rail provides the energy required for relay coil activation through isolated transistor drivers.

The first ESP32-S3 operates as the human-machine interface (HMI) and is responsible for rendering the graphical control panel on a 2.8-inch ILI9341 TFT display. An FT6206 capacitive touch controller detects user interaction with the displayed buttons. Each button corresponds to one of the five lighting channels in the system and is dynamically color-coded—green when the associated relay is active and red when inactive. This microcontroller translates user presses into structured UART messages, such as “R2:ON” or “R4:OFF,” which are transmitted to the relay-control ESP32 for execution.

The second ESP32-S3 performs all relay-driving functions. Upon receiving a command from the HMI controller, it decodes the message, activates or deactivates the appropriate transistor stage, and energizes the corresponding 12 V relay coil. This arrangement electrically isolates the logic circuitry from the higher-voltage load domain while ensuring robust and deterministic switching behavior. After completing an action, the relay microcontroller returns an acknowledgment message to the HMI so that the graphical interface remains synchronized with the actual system state at all times.

To enhance usability and reduce energy consumption, the subsystem incorporates automatic backlight management. The HMI includes a settings interface that allows manual adjustment of screen brightness, and inactivity triggers an automatic reduction or shutoff of the TFT backlight. Touch input immediately restores illumination, ensuring that the interface remains responsive even during extended idle periods. Both ESP32-S3 units include dedicated power-indication LEDs, providing immediate visual confirmation of subsystem power status during operation and testing.

Together, the UART communication link, touchscreen interface, relay-control logic, and coordinated power-management features create a responsive, low-power, and user-friendly subsystem capable of reliably controlling lighting loads across the full 10.8–14.2 V input range. The following subsection presents the detailed validation results demonstrating the performance and correctness of this architecture.

3.5.2. Validation

A comprehensive validation sequence was performed to confirm the functional correctness, reliability, electrical integrity, and command-response behavior of the Touchscreen and Relay-Control Subsystem. Validation included initial power-on behavior, touchscreen

mapping confirmation, long-duration reliability testing, and command-trigger validation under repeated use. All tests were conducted using direct bench instrumentation

3.5.2.1. Power-On Behavior and Initialization

When the subsystem was energized, both microcontrollers successfully booted without brownout or instability, and the TFT display initialized immediately. The capacitive touch controller and backlight system were detected and configured correctly at startup. These results confirm proper wiring, sufficient current delivery from the 3.3 V converter, and correct boot configuration for both ESP32-S3 modules.

3.5.2.2. Touchscreen Mapping and Interface Validation

The touch panel was exercised across the full display area to verify that the FT6206 controller correctly reported coordinate ranges and that all interactive elements aligned with their graphical regions.

During calibration, the screen was touched near each corner to establish the usable coordinate bounds. The measured coordinates were:

- **Top-Left:** (X = 10, Y = 320)
- **Bottom-Left:** (X = 237, Y = 318)
- **Top-Right:** (X = 20, Y = 2)
- **Bottom-Right:** (X = 237, Y = 1)

These readings confirmed that the touch controller reported stable, repeatable values and that the usable coordinate space spanned the full physical screen area, with minor offsets attributable to finger size and bezel margin. These boundary readings were then used to map the buttons precisely onto the display so that the graphical elements matched the physical touch detection regions.

After calibration was applied, all on-screen control buttons responded accurately within their intended areas, with no false triggers, edge misalignments, or missed presses observed during evaluation.

3.5.2.3. System Runtime Stability

A continuous runtime test was conducted over a two-hour period. After approximately 45 seconds of inactivity, the automatic backlight-sleep feature engaged as expected, reducing display power while keeping the system responsive. Throughout the two-hour session, the HMI and relay-control ESP32 modules remained fully operational, with no resets, freezes, communication failures, or degraded touch response. User interaction immediately restored the backlight, confirming proper performance of the sleep/wake logic.

3.5.2.4. Relay Trigger

The command-response behavior between the two ESP32-S3 modules was tested by actuating each of the five relay channels 50 times. This produced a total of 250 individual relay operations. Every press resulted in the correct UART command being transmitted to the relay controller, followed by successful relay activation and acknowledgment. Voltage measurements confirmed that each relay correctly delivered 12 V to its output terminals on every activation cycle. No missed triggers, delayed responses, or desynchronization events were observed during testing. The consistency of these results demonstrates that the UART protocol, relay drivers, transistor interfaces, and synchronization logic operate reliably under repeated use.

3.6. Subsystem Conclusion

The Physical Controls Subsystem integrates the low-voltage power stages and digital interface components necessary to operate the lighting and user-interaction elements of the Smart Solar Residential Lighting and Security System. Across multiple validation procedures, the subsystem demonstrated strong performance in the majority of its functional domains, with reliable conversion efficiency, stable user-interface behavior, and consistent relay-control operation.

The 3.3 V buck converter met its regulation and efficiency expectations, maintaining output stability across the full 10.8-14.2 V input range and up to 2 A of load. Line and load regulation remained within acceptable bounds, ripple measurements were low, and percent-error results confirmed accurate voltage control throughout all tested operating points. This converter is therefore verified as suitable for supplying all logic-level electronics within the subsystem.

The 12 V buck-boost converter exhibited stable voltage regulation and good efficiency up to approximately 0.8-0.85 A, but a hardware failure prevented testing within the intended current range of 2 A nominal and 3 A peak. Ripple levels and percent-error measurements reflect this limitation and represent behavior under degraded conditions rather than the expected performance of a properly functioning design. Despite this, the partial results provide insight into the converter's pre-failure behavior and confirm that the control architecture, layout, and test protocol remain valid for future hardware revisions.

The ESP32 touchscreen and relay-control architecture performed reliably across all validation criteria. The human-machine interface consistently detected touch inputs, mapped coordinates accurately, and communicated deterministically with the relay controller through UART. Long-duration runtime testing confirmed stable operation with no resets or lockups, and all 250 relay-activation cycles succeeded without error. The subsystem's responsiveness, synchronization, and low-power backlight management further demonstrated its suitability as the central user-interaction layer for the system.

Overall, the Physical Controls Subsystem achieved its functional objectives, with the exception of the identified fault in the 12 V converter hardware. All other components met or exceeded design expectations, validating the subsystem's readiness for integration with the solar-charging, sensor, and mobile-application subsystems. The results confirm that the architecture, interface definitions, and control logic provide a strong foundation for reliable system performance in the final Smart Solar Residential Lighting and Security platform.

4. AI Security Camera Subsystem

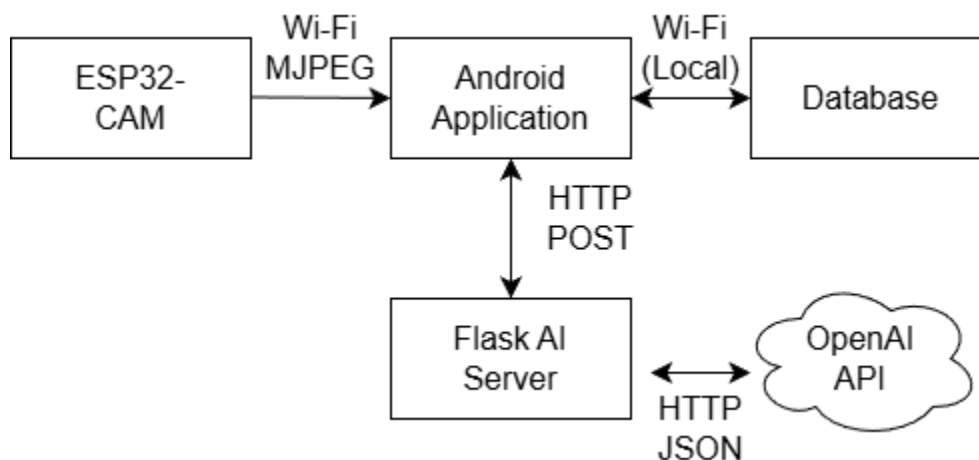
4.1. Subsystem Introduction

The AI enhanced Wireless Camera Subsystem is responsible for acquiring visual data from the ESP32-CAM module, transmitting the image stream to the android application, and enabling automated scene analysis through a cloud based AI model. This subsystem exists to provide real time visual monitoring, remote access, and intelligent feedback without requiring wired infrastructure or a local compute board. This provides easy accessibility for any user looking to implement the solar home lighting project. The subsystems integrate three major components: ESP32-CAM, Android Application, and Flask AI backend. The ESP32-CAM captures frames using an OV2640 sensor and exposes them over an MJPEG HTTP stream. The android application receives, decodes, displays, and stores frames while giving user remote control over lighting and analysis capabilities. The Flask AI backend receives short bursts of images from the app and uses OpenAI's model for automated, real time monitoring for security purposes. Together, these components form the imaging and intelligence backbone of the entire project.

4.2. Subsystem Overview

A block diagram of the subsystem is shown below.

Figure 4.2: Functional Block Diagram of AI Security Camera Subsystem



The primary focus of the AI Camera Subsystem was ensuring that all components would communicate properly and in a timely manner over local Wi-Fi and ensuring easy accessibility within the user application. Each component was successfully integrated over local Wi-Fi and displayed in a reliable manner as validated by the validation tests below.

The frame rate stability, connection verification, and AI Analysis latency were validated after the subsystem was mostly completed, and preformed within expectation laid out in the validation plan

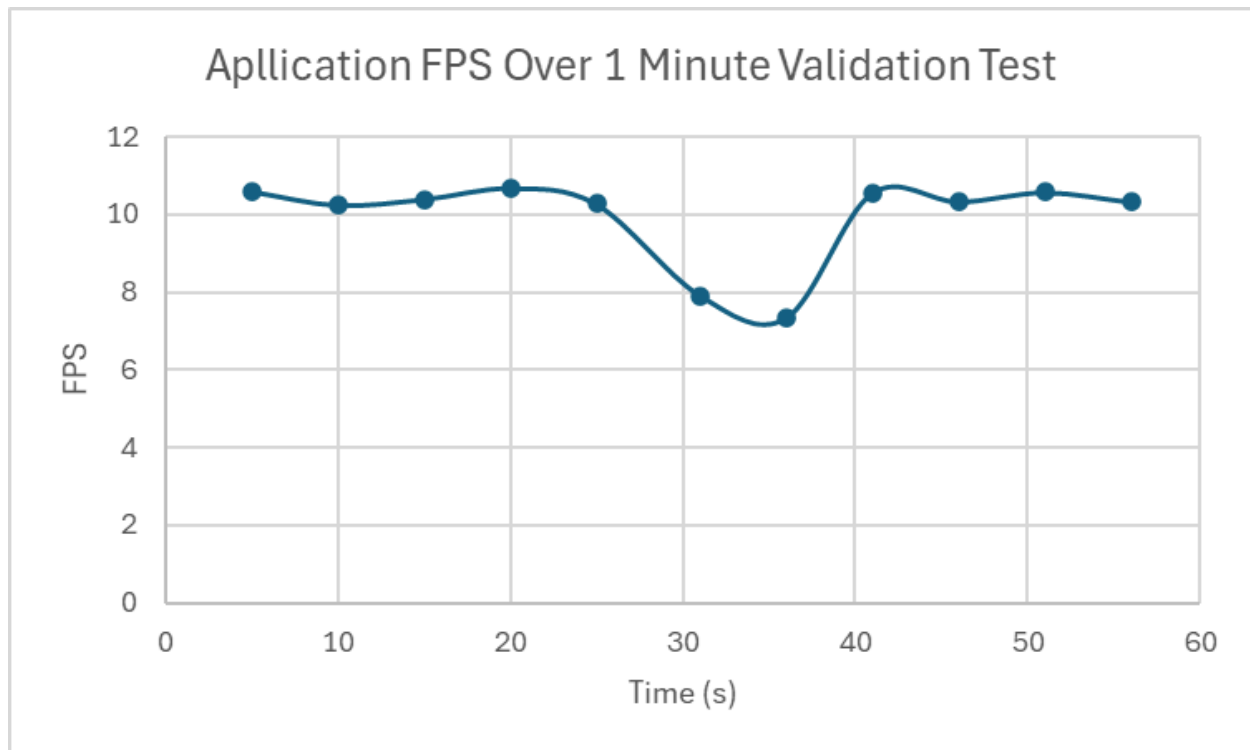
4.3. Subsystem Validation

To validate the reliability of the camera subsystem, quantitative tests were conducted on stream performance, AI pipeline latency, and network stability. Testing focused on real world conditions that the system would face during normal operation.

4.3.1. Frame Rate (FPS) Stability Test

A 60 second FPS sampling test was implemented inside the camera fragment portion of the application. The application logged average FPS every 5 seconds producing a time series of frame throughout under normal Wi-Fi conditions

Figure 4.3.1: Application FPS Over 1 Minute Validation Test



The purpose of this validation test was to evaluate the consistency of the MJPEG decoding pipeline and verify that the ESP32-CAM can maintain real time performance while streaming locally within the application. A table detailing exact frames is shown below.

Table 4.3.1: Application FPS Over 1 Minute Validation Test Data

Time (sec)	FPS
5	10.61
10	10.25
15	10.4
20	10.69
25	10.28
31	7.91
36	7.32
41	10.56
46	10.34
51	10.58
56	10.33
Average FPS	9.93

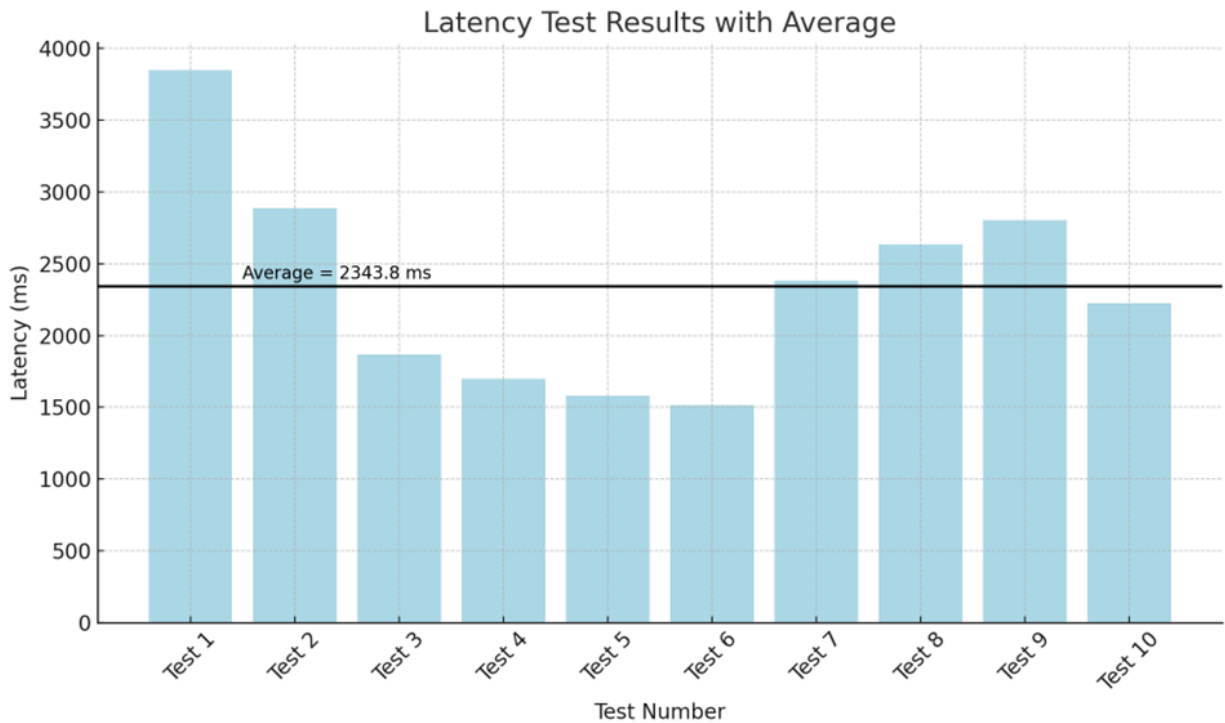
The subsystem demonstrated stable performance with minimal drift, confirming a reliable streaming pipeline.

4.3.2. AI Latency Test

To quantify AI processing responsiveness, the app recorded the elapsed time from image upload to summary reception. This was done Time stamping each process from the start of the AI analysis, adding the total and running the test 10 times.

Figure 4.3.2: Latency Test Results with Average Graph

Final Report
Solar Home Lighting



There was an average of 2.3s Latency for the whole system to return back to the application and display, which is well under the projected hypothesis of less than 5 seconds. An exact measurement of the test data is shown below.

Table 4.3.2: Latency Test Results Data

Test #	Time (ms)
1	3849
2	2885
3	1868
4	1696
5	1582
6	1513
7	2383
8	2633
9	2803

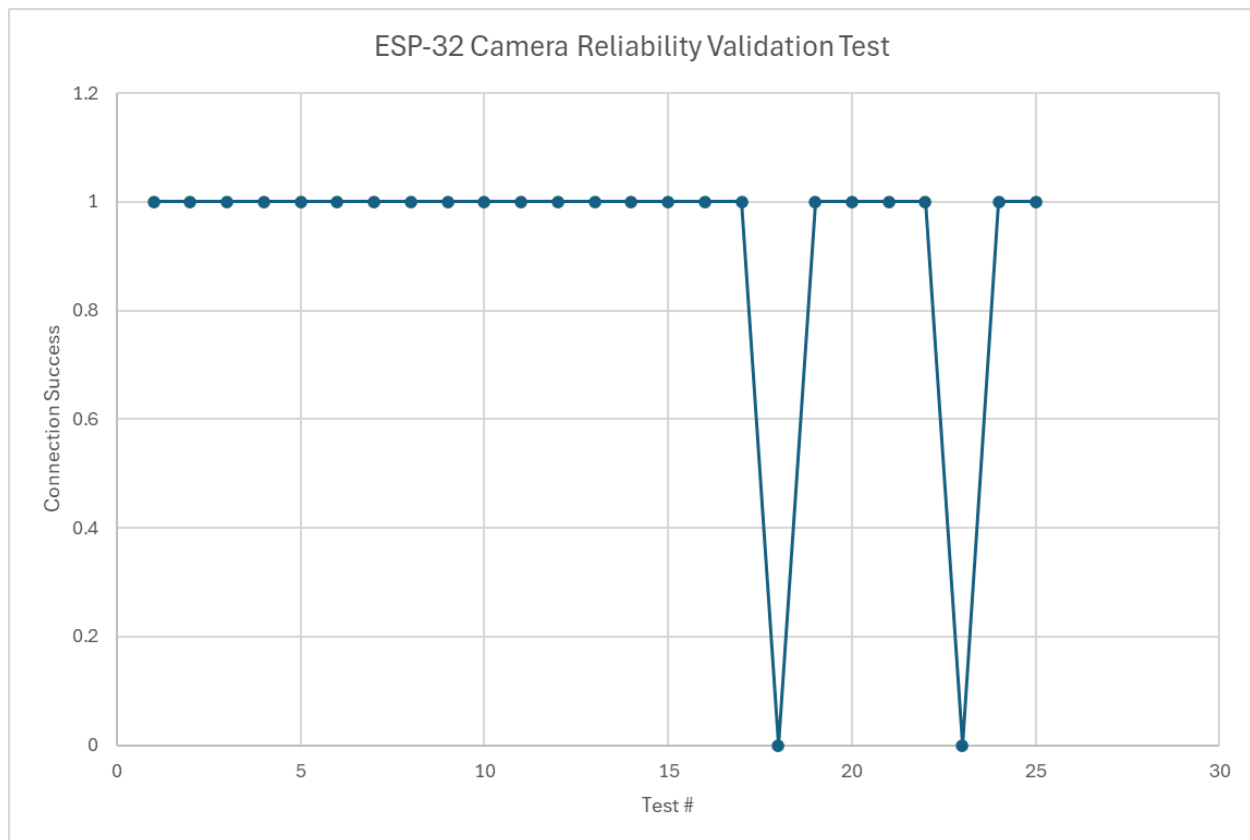
10	2226
Average	2343.8

This conclusively validates that the AI backend is fast enough for real time environmental feedback.

4.3.3. Connection Robustness Test

This validation test was performed to check the connectivity from the Application to the ESP-32 Camera over 25 tries to ensure solid performance and communication between the two components

Figure 4.3.3: Camera Reliability Validation Test Visual Graph



The camera connected successfully 23 times with 2 failures due to connection timeout. This gives a 92% reliability within the camera which is above projected standards of 92% conclusively validating the robustness of the connection to the camera and application.

4.4. Subsystem Conclusion

Each part of the subsystem was shown to communicate and work successfully at a rate higher than projected initially. The Camera AI subsystem effectively streams MJPEG streams over local Wi-Fi, streams within the application, and streams and stores data at a reliable rate.

5. MCU, Sensors, and Database Subsystem

5.1 Subsystem Introduction

The MCU, Sensors, and Database subsystem serves as the central control layer of the Solar Home Lighting System, coordinating real-time sensing, decision-making, and two-way communication across the entire system. This subsystem interprets data from the charge controller as well as sensors including daylight levels and motion activity and uses these inputs to execute all system logic, such as determining night conditions, activating the porch light, and generating daily performance metrics.

In addition to sensor processing, the MCU maintains continuous communication with the cloud database, enabling both data upload and remote command retrieval. This bidirectional link allows the user to monitor system status and adjust configuration parameters through the application, which the MCU interprets and applies in real time. Through this architecture, the subsystem ensures synchronized behavior between the physical hardware and the user-facing app.

Through its combined sensing, control, and communication capabilities, the MCU, Sensors, and Database subsystem forms the decision-making backbone of the Solar Home Lighting System, ensuring coordinated and reliable operation under both real and simulated conditions.

5.2 Microcontroller Unit (MCU)

5.2.1 Operation

The Microcontroller Unit (ESP32) serves as the primary controller for the Solar Home Lighting System. The MCU periodically polls the charge controller to retrieve processed telemetry, including battery status, panel status, load behavior, and state of charge (SOC). In the current design, the MCU is configured to request this telemetry at an interval of 30 seconds, which is sufficient to capture system trends without overloading the communication link or database.

In addition to receiving telemetry from the charge controller, the MCU directly reads the daylight (LDR) and motion (PIR) sensors. The LDR is used to determine whether the system is in Day Mode or Night Mode, while the PIR sensor is used to detect motion events near the light. When the LDR indicates low light and the system enters Night Mode, any

valid motion detection triggers the MCU to energize the porch light relay for a configured duration and log the event.

The MCU takes the telemetry with local sensor readings to turn data into telemetry packets and daily performance metrics and uploads them to the cloud database at scheduled intervals. Over the same link, the MCU also retrieves configuration parameters and control flags from the user-facing application, allowing remote adjustment of thresholds, timers, and operating modes. In addition the MCU is responsible for controlling the relay logic balancing the response between the physical controls and application controls.

5.2.2 Validation

Validation of the Microcontroller Unit (MCU) focused on confirming correct telemetry polling, sensor-driven logic execution, relay activation, and database communication.

Table 5.2: MCU Subsystem Validation Summary

Validation Item	Procedure	Expected Result	Result
Charge-Controller Polling	Observe MCU requests at 30-second intervals	Telemetry received every cycle with no dropouts	Pass
Day/Night Detection State	Cover/uncover LDR	System switches between Day and Night modes	Pass
PIR Motion Detection	Trigger PIR during Night Mode	Motion event logged and relay activated	Pass
Relay Output	Measure relay ON duration	Relay deactivates when signaled	Pass
Database Upload	Inspect Firebase entries	Telemetry uploaded with correct timestamps	Pass
Remote Commands	Change app-control flags	MCU updates behavior after small latency	Pass
Simulation Mode	Generate bench telemetry	Simulated values appear in database as real data	Pass
Connectivity Loss Handling	Disable Wi-Fi during upload	MCU retries and uploads once reconnected	Pass

The MCU successfully polled the charge controller at the intended 30-second interval, with all telemetry received and logged correctly. Day/night transitions were validated using controlled changes in the LDR sensor, and the system reliably entered Night Mode at low light levels. PIR testing confirmed immediate relay activation and accurate timeout behavior. Remote override commands issued through the application updated relay and system behavior in real time.

Simulation mode produced predictable SOC and panel-status values that matched expected system behavior, confirming the MCU's ability to operate without live solar hardware. Connectivity loss testing showed that the MCU safely buffered data and retransmitted it once Wi-Fi was restored.

Overall, the MCU demonstrated reliable performance across all operational scenarios, confirming proper logic execution, sensor integration, relay control, and database synchronization.

5.3 Sensors

5.3.1 Operation

The Sensor Interface Layer consists of two primary environmental sensors: the Passive Infrared (PIR) motion sensor and the digital ambient light sensor used for daylight detection. Together, these sensors provide the MCU with the necessary information to determine lighting conditions and activate the porch light based on motion and environmental brightness.

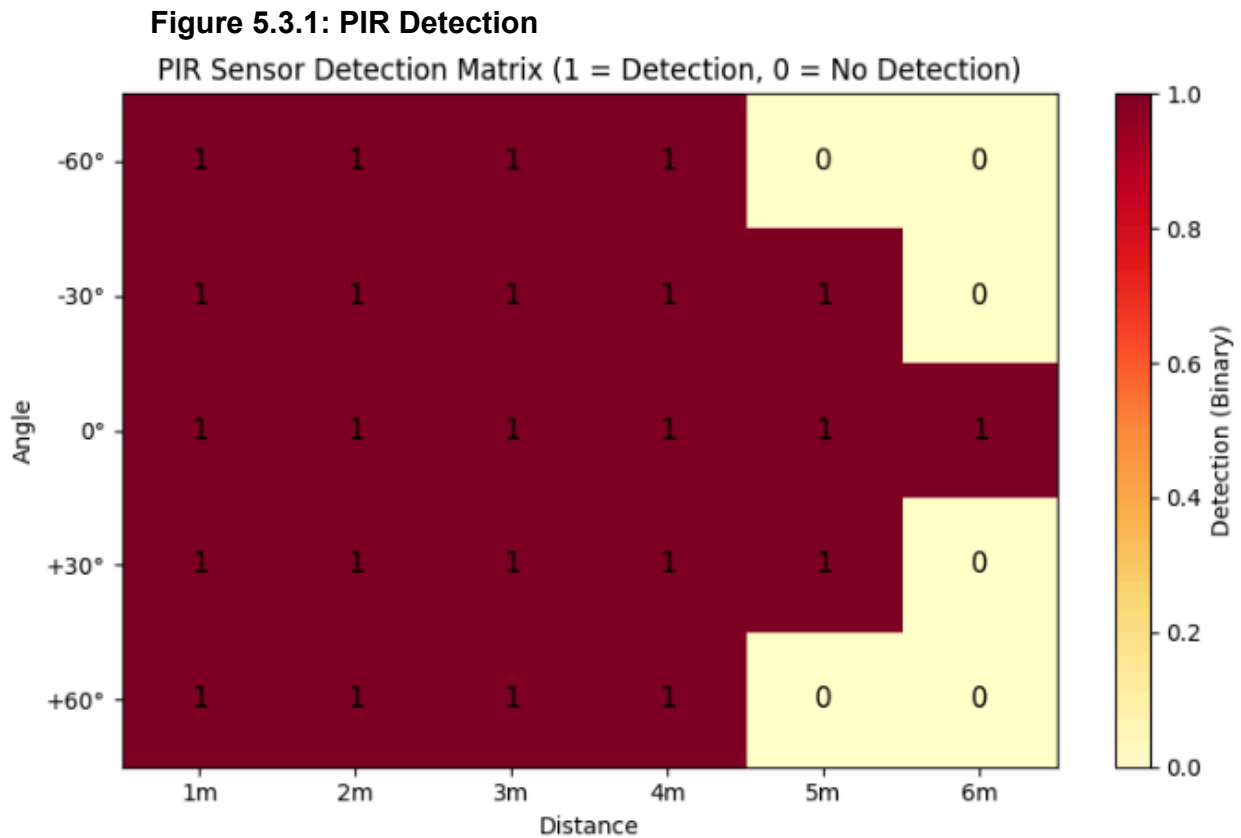
The PIR sensor detects infrared energy changes caused by human movement. When motion is detected, the sensor outputs a HIGH digital signal to the MCU. During Night Mode, this immediately triggers the security-light relay for the configured duration. The PIR sensor offers a practical detection range of approximately 3–5 meters and a horizontal field of view of roughly $\pm 60^\circ$, which is consistent with typical PIR hardware. This allows the system to reliably capture motion events in front-facing areas such as sidewalks or entry paths.

Instead of a traditional analog LDR, the system uses a digital ambient light sensor, which provides a raw brightness value through its digital output interface. The MCU periodically reads this brightness value and compares it to a predefined threshold to determine day and night conditions. When the sensor reading falls below the threshold, the MCU sets the internal state to dark=yes and transitions the system into Night Mode. When the reading rises above the threshold, the MCU returns to Day Mode (dark=no). This digital sensor provides stable and noise-resistant measurements, enabling consistent and reliable day/night classification for the lighting system.

5.3.2 Validation

Validation of the Sensor Interface Layer focused on verifying reliable PIR motion detection across practical angles and distances, as well as ensuring accurate day/night classification using the digital ambient light sensor. Both sensors were evaluated under controlled test conditions that reflect expected real-world operation.

The PIR sensor was tested by recording binary detection results at multiple distances (1–6 meters) and angles (-60° to $+60^\circ$). Each test position was marked as either detected (1) or not detected (0). The resulting detection matrix was visualized to illustrate the sensor's effective coverage.



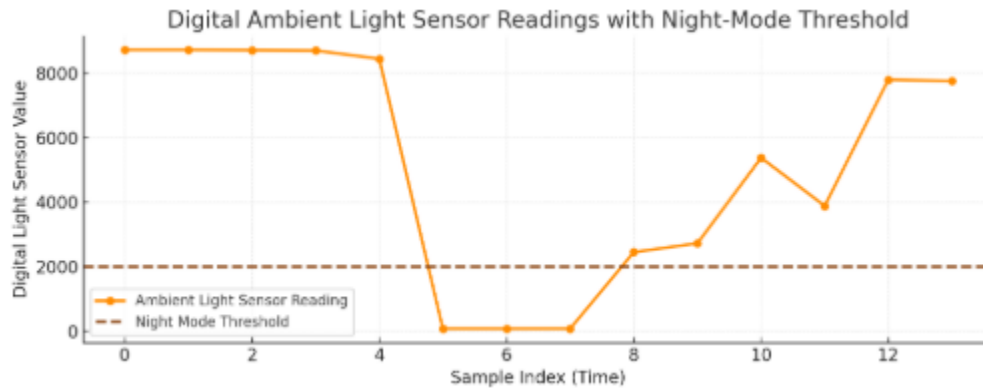
As shown in Figure 5.3 the PIR sensor reliably detected motion within 3–4 meters across $\pm 60^\circ$, with diminishing sensitivity beyond 5–6 meters and at extreme angles. This behavior is consistent and confirms the sensor's suitability for night-time motion-based lighting activation.

Final Report

Solar Home Lighting

The digital light sensor was validated by measuring brightness readings during controlled lighting changes. The sensor output was logged while transitioning from bright indoor conditions to partial shade and complete darkness. A threshold value (≈ 2000 units) was used to determine when the system should enter Night Mode.

Figure 5.3.2: Digital Light Sensor Readings



The digital ambient light sensor outputs high values (approximately 8400–8800) under bright conditions. At sample index 5, the sensor is covered or exposed to low light, causing the reading to drop sharply below 100. This is well below the night-mode threshold of ~ 2000 units, prompting the MCU to classify the environment as dark (dark=yes). As light increases again across samples 8–13, the readings rise above the threshold and the system correctly returns to Day Mode (dark=no). This behavior confirms that the sensor accurately distinguishes between bright and dark conditions and the MCU interprets the measurements to control the porch light.

5.4 Database

5.4.1 Operation

The Database Subsystem serves as the communication bridge between the Solar Home Lighting System and the user-facing application. The subsystem is implemented using a real-time cloud database, which stores system telemetry, daily performance metrics, configuration parameters, and command flags. The MCU communicates with the database over Wi-Fi, enabling synchronization between the physical hardware and the mobile application.

The MCU periodically uploads charge-controller telemetry (battery status, SOC trend, panel state) along with local sensor readings and event logs. These values are timestamped and stored as structured entries within the database. In addition to raw telemetry, the MCU generates daily summary metrics such as energy-related trends which are also uploaded for visualization in the user interface.

Communication is bidirectional with the database receiving commands and set thresholds from the database. The MCU regularly looks at the database to see if the values have been updated to change these fields real time in the system

5.4.2 Validation

Validation of the Database Subsystem focused on confirming if the data uploaded reliably with consistent timestamping, successful retrieval of remote command and proper system behavior during connection loss. Validation was performed using test bench telemetry data, intentional command changes from the app and simulated wifi interruptions.

Figure 5.4: Database Telemetry



The validation of telemetry uploads and daily summary generation was validated by inspecting entries in the cloud database. As shown in figure 5.4 each daily record is timestamped and contains the relevant performance metric computed by the MCU. These results verify that the database subsystem stores structured, complete, and timely metrics during normal operation.

5.5 Conclusion

The MCU, Sensor and Database subsystem across all validation procedures demonstrated reliable sensing, decision making and data management. The PIR sensor consistently

Final Report
Solar Home Lighting

detected motion within its expected range and field of view, while the digital ambient light sensor provided stable day/night classification.

The MCU correctly integrated sensor inputs with charge-controller telemetry, executed system logic in real time, and validated proper relay activation during Night Mode. The database subsystem consistently stored data received and logged events correctly while communicating with the MCU. Overall, the subsystem met all functional requirements for real-time monitoring. These results confirm that the MCU, Sensor and Database subsystem is synchronized and works reliably, automating the lighting behavior and tracking system performance for the Solar Home Lighting System.

Solar Home Lighting

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ECEN 404 FINAL SYSTEM REPORT

GITHUB REPOSITORY:

<https://github.com/matthewmgreer-design/Senior-Design-Solar-Home-Lighting>

ECEN 404 FINAL SYSTEM REPORT FOR Solar Home Lighting

TEAM 22

APPROVED BY:

Jack Holdridge Date

Prof. Kalafatis Date

Hailey Ward Date

Change Record

Rev	Date	Originator	Approvals	Description
1	04/24/26	Photon Phools		Final ECEN 404 Release

Table of Contents

Table of Contents	3
List of Tables	4
List of Figures	5
1. Introduction	6
2. System Overview	7
3. Integrated System	8
3.1 Integration Overview	8
3.2 Power System Integration	9
3.3 Control and Interface Integration	9
3.4 Sensor and Automation Integration	9
3.5 Communication and Database Integration	9
3.6 Security System Integration	9
4. Integrated System Validation and Performance Results	9
4.1 Introduction	9
4.2 Power Delivery and Rail Stability Validation	10
4.2.2 3.3 V Rail Stability Under Full System Load	11
4.2.3 Summary	11
4.3 Local Control Interface Performance (HMI Validation)	11
4.3.1 Functional Control and System Reliability	12
4.4 Sensor-Based Automation Performance and Reliability	12
4.4.1 Motion Sensor Detection Reliability	13
4.5 Remote Monitoring and Battery Data Validation	14
4.5.1 Battery Poll Accuracy and Reliability	14
4.6 Security Response and Notification Performance	15
4.6.1 Motion Detection to Application Notification Latency	15
4.7 Local Control Response and Latency	16
4.7.1 HMI Command Execution and System Behavior	16
4.7.2 HMI Response Latency	17
4.8 Sensor-Based End-to-End System Validation	18
4.8.1 Sensor-Driven System Behavior	18
4.8.2 Sensor-to-Output Latency	18
4.9 Application-Based End-to-End Control Validation	19
4.9.1 Remote Control System Behavior	19
4.9.2 Application-to-Output Latency	20
4.10 Solar panel regulation under load	20
4.10.1 Dual energy delivery (load + storage)	20
5. Overall System Results	22

List of Tables

Table 4.2.1	12 V Output Voltage Under Load	9
Table 4.2.3	3.3 V Output Voltage Under Full System Load	10
Table 4.2.4	3.3 V Rail Summary Metrics	10
Table 4.3.1	Local Control Interface Validation Results	11
Table 4.4.1	Motion Sensor Performance Summary	13
Table 4.6.1	Latency Measurement Summary	15
Table 4.7.1	HMI System Validation Checks	16
Table 4.8.1	Sensor-Based System Validation Checks	17
Table 4.9.1	Application Control Validation Checks	19

List of Figures

Figure 2.1: System Block and Connectivity Diagram	7
Figure 4.2.1 12 V Output Voltage vs Load Condition	9
Figure 4.4.1 Outdoor PIR Trigger Success Rate vs Distance	12
Figure 4.4.2 Indoor PIR Trigger Success Rate vs Distance	12
Figure 4.5.1 Battery Poll Success Rate Over Time	13
Figure 4.5.2 Mobile Application Battery Display Interface	14
Figure 4.6.1 Security Event Image Capture and Application Display	14
Figure 4.6.2 End-to-End Latency vs Test Number	15
Figure 4.7.1 HMI Command Latency by Lighting Zone	16
Figure 4.8.1 Sensor-Based Lighting Latency by Zone	18
Figure 4.9.1 Remote (Application) Latency by Zone	19

1. Introduction

The Solar Home Lighting System is a fully integrated, solar-powered residential lighting and security platform designed to operate independently of the electrical grid. The system combines renewable energy generation, battery storage, sensor-driven automation, and both local and remote user interfaces into a unified solution for residential environments. By integrating lighting control, environmental sensing, and security monitoring into a single coordinated architecture, the system addresses the need for reliable operation during power outages while improving overall energy efficiency and user accessibility.

The purpose of this System Report is to evaluate the performance of the complete integrated system following subsystem development and validation. While individual subsystems—including the charge controller, physical controls, microcontroller and sensing logic, and application/camera interface—were designed and verified independently (see Subsystem Report), this document focuses on how these components function together as a cohesive system. Emphasis is placed on system-level operation, interface integration, and validation of key performance requirements.

The system architecture is centered around a solar energy generation and storage network, where energy collected from a photovoltaic panel is regulated through an MPPT charge controller and stored in a battery for later use. This stored energy powers the control electronics, sensors, and lighting loads through regulated voltage rails. A microcontroller-based control system processes user inputs and sensor data to determine system behavior, enabling both manual and automated operation. Users interact with the system through a local touchscreen human-machine interface (HMI) and a mobile application, providing real-time control, system monitoring, and security notifications.

This report demonstrates how the final system satisfies its intended operational objectives by analyzing integrated functionality, validating system-level requirements, and evaluating overall performance under realistic operating conditions. The results presented highlight the effectiveness of the system in delivering reliable solar-powered lighting, responsive control, and coordinated security functionality within a single platform.

2. System Overview

The system integrates energy harvesting, energy storage, intelligent control, environmental sensing, lighting actuation, and remote user interaction into a single coordinated platform. All system functionality is managed through a centralized control structure that processes sensor data, user inputs, and remote commands to control lighting behavior, monitor system status, and support security-related features.

A high-level representation of system connectivity, power flow, and communication pathways is shown in Figure 2.1.

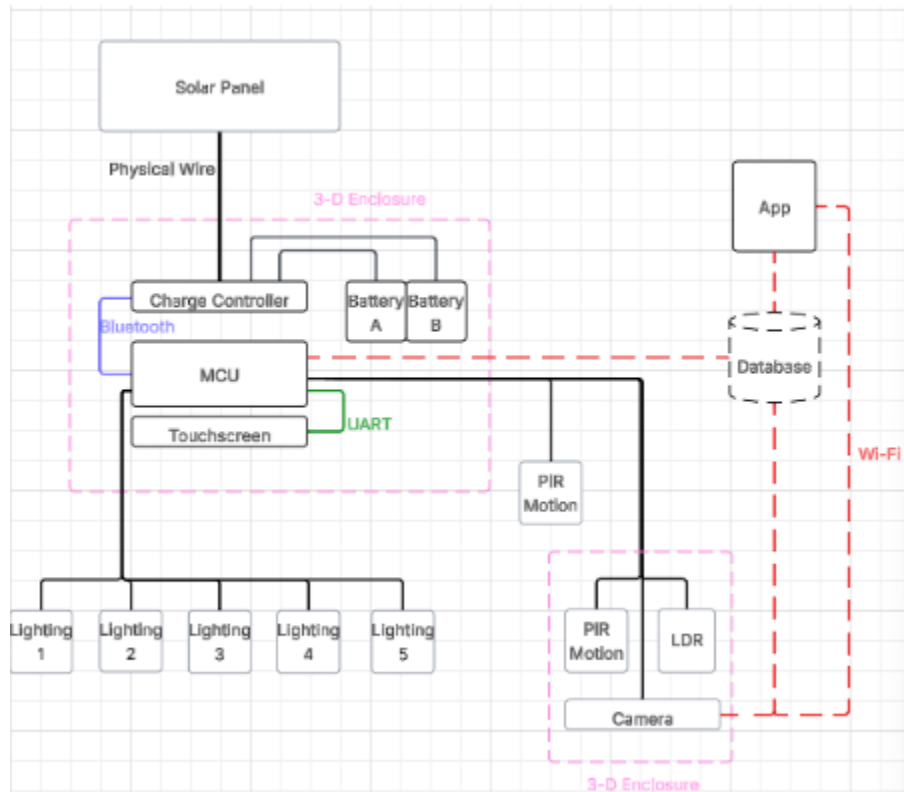


Figure 2.1: System Block and Connectivity Diagram

Solar energy is collected by the photovoltaic panel and delivered through a physical electrical connection to the internal charge regulation and storage system. Energy is stored across two battery sources, allowing the system to maintain operation during periods without solar input while supporting load demands from lighting and control electronics. Power is distributed internally to support both control logic and lighting outputs, enabling continuous system operation under varying environmental conditions.

A central microcontroller unit (MCU) governs all system behavior. The MCU processes inputs from environmental sensors, including passive infrared (PIR) motion detectors and a light-dependent resistor (LDR), to determine when automated lighting responses should occur. Based on these inputs, the system can activate lighting outputs, manage operating

modes, and initiate security-related actions such as camera triggering. Lighting loads are distributed across multiple channels, allowing independent control of different zones within the system.

User interaction is supported through two primary interfaces: a local touchscreen and a remote mobile application. The touchscreen provides direct control of lighting and system visibility, while the mobile application enables remote operation and monitoring through a wireless connection. Communication between the system and the application is handled through a cloud-based database, allowing real-time updates, command execution, and system state synchronization over Wi-Fi. Additional local communication between internal components is handled through wired protocols, ensuring reliable and low-latency control.

The integrated security functionality is driven by sensor-based event detection. When motion is detected in monitored areas, the system activates both lighting and camera responses, enabling real-time visual monitoring and user notification through the application. This coordination between sensing, control, and communication allows the system to provide both lighting automation and basic security awareness within a single device.

3. Integrated System

3.1 Integration Overview

The objective of system integration was to combine the power system, control interfaces, sensing elements, sensing elements, communication infrastructure, and security components into a unified operational platform. Integration focused on establishing reliable electrical connections, structured communication pathways, and control logic to enable consistent behavior for all modes of operation.

The system is powered from a shared battery source regulated by the charge controller, which supplies stable voltage to the lightning loads and controllers. A central microcontroller (ESP32) serves as the system core, interfacing with the touchscreen Human Machine Interface(HMI), environmental sensors, relay outputs, and wireless communication modules. Local communication is handled through wired interfaces, while remote communication is achieved through a database-backed wireless connection to the mobile application.

System integration is centered around a control architecture in which all inputs are processed through the microcontroller. User commands from both the HMI and mobile application are interpreted and executed through the same control logic to ensure consistent behavior regardless of the input source. Sensor data is continuously monitored and used to trigger automated responses, including lighting control and security related actions.

To maintain real time response, control and sensing operations were prioritized locally within the microcontroller while the network communication was handled asynchronously. This separation ensures that delays in wireless communication do not impact normal system functions. System state including lighting status, and battery information is continuously updated and synchronized across both local and remote interfaces.

The result of this integration is a system capable of simultaneous power management, user interaction, automated control, and remote monitoring within a single coordinated architecture.

3.2 Power System Integration

The power system is designed to provide a stable and continuous energy source for both the lighting loads and controllers. Energy is supplied through a solar input and regulated by a charge controller, which manages charging and discharging across two battery packs. These battery packs act as the primary energy storage and supply power to the system loads, including the microcontroller and lighting.

Integration between the power system and control architecture is achieved through telemetry communication between the charge controller and the microcontroller via UART communication. This allows the system to monitor parameters such as battery voltage, state of charge, and active battery pack. These values are used for system monitoring and are displayed on both the local touchscreen interface and the mobile application.

During operation, transitions between battery packs introduce brief power disturbances due to switching behavior within the charge controller. These disturbances were observed to cause momentary voltage instability, resulting in microcontroller resets and visible flickering of the lighting outputs. This behavior demonstrates the sensitivity of embedded control systems to power interruptions and highlights the importance of maintaining a stable supply during switching events.

To ensure reliable system performance, the power system must balance energy management with electrical stability. While the current implementation supports system operation, future improvements may include additional filtering, energy buffering, or seamless switching to reduce transient effects and improve the overall system.

3.3 Control and Interface Integration

The control and interface subsystem integrates both local and remote interaction into a unified control architecture. The system supports two interfaces: a touchscreen human-machine interface (HMI) for local control and a mobile application for remote operation. Both interfaces allow users to control lighting states.

Local control is achieved through the touchscreen HMI, which communicates directly with the microcontroller via a UART interface. User inputs are transmitted as command signals and processed immediately by the microcontroller, allowing for low-latency control of the lighting system. Remote control is implemented through a mobile application connected to a cloud-based database. User commands from the application are written to the database and periodically polled by the microcontroller, where they are interpreted and executed.

To ensure consistent system behavior, all user inputs regardless of source are processed through the same control logic within the microcontroller. This unified approach prevents conflicts between local and remote commands and ensures that the system state remains synchronized across both interfaces. The microcontroller acts as the central authority, updating lighting outputs and system states while reflecting these changes back to both the HMI and mobile application.

The system is designed to prioritize responsiveness and reliability by separating control logic from network communication. Local inputs are handled in real time, while remote commands are processed asynchronously to prevent delays caused by network latency from impacting system performance. This architecture enables seamless interaction between multiple control interfaces while maintaining stable and predictable system operation.

3.4 Sensor and Automation Integration

The sensor and automation subsystem enables the system to respond to environmental conditions without requiring direct user input. This functionality is achieved through the integration of motion sensors and an ambient light sensor, which provide real-time data to the microcontroller for decision-making.

The system uses passive infrared (PIR) sensors to detect motion in both indoor and outdoor environments, as well as an ambient light sensor to determine surrounding light levels. These sensor inputs are continuously monitored by the microcontroller and used to drive automated control logic when the system is operating in auto mode.

Automation behavior is governed by predefined control conditions. For example, the foyer lighting is activated when indoor motion is detected and remains on for a fixed duration after motion ceases. Similarly, the security lighting system responds to outdoor motion, activating the light only when ambient light levels indicate nighttime conditions and motion is detected outside. The ambient light sensor is used to differentiate between day and night operation, ensuring that lighting responses are appropriate for the environment.

Sensor inputs do not directly control system outputs; instead, they are processed through control logic within the microcontroller. This approach ensures that all automated actions are consistent with the current system state, including user-defined settings such as auto mode enablement. Additionally, security-related actions, such as triggering the camera system, are initiated based on sensor events, allowing the system to respond to activity even when lighting conditions do not require illumination.

By integrating environmental sensing with centralized decision-making, the system achieves reliable automation. This design enhances usability, reduces the need for manual interaction, and ensures that system responses remain consistent and predictable under various conditions.

3.5 Communication and Database Integration

The communication and data subsystem enables remote monitoring and control of the system through integration with a cloud-based database and mobile application. The system utilizes a wireless connection to exchange data between the microcontroller and the database, which serves as an intermediary between the embedded system and the user interface.

System state information, including lighting status, sensor readings, and battery data, is periodically transmitted from the microcontroller to the database. This allows the mobile application to provide users with real-time system visibility. In addition, user commands from the mobile application are written to the database and retrieved by the microcontroller through periodic polling. These commands are then processed and applied through the system's centralized control logic.

To maintain reliable operation, communication tasks are handled independently from the core control logic within the microcontroller. This separation ensures that delays in network communication, such as latency introduced by the cloud database, do not interfere with real-time system functions. As a result, local control and automated responses remain responsive even when remote communication is delayed or temporarily unavailable.

While the cloud-based architecture enables remote accessibility and data synchronization, it introduces inherent latency compared to local control methods. This latency is observed in the delay between issuing a command from the mobile application and the resulting system response. However, by prioritizing local processing and decoupling network operations, the system maintains stable performance and avoids disruptions caused by communication delays.

Overall, the integration of communication and data systems provides a balance between accessibility and reliability, enabling remote interaction without compromising the responsiveness of critical system operations.

3.6 Security System Integration

The security subsystem integrates motion detection, lighting control, and camera activation to provide automated monitoring and response to external activity. This functionality is centered around the outdoor passive infrared (PIR) sensor, which continuously monitors for motion in the surrounding environment.

When motion is detected, the sensor signal is processed by the microcontroller, which determines the appropriate system response based on current operating conditions. The camera system is triggered upon motion detection to capture visual data, enabling event monitoring and user notification through the mobile application. This ensures that potential activity is recorded regardless of lighting conditions.

Lighting behavior is conditionally controlled based on ambient light levels and system mode. During nighttime conditions, detected motion activates the security lighting to illuminate the area, enhancing visibility and deterrence. During daytime operation, lighting remains off to conserve energy, while the camera system continues to operate and capture events as needed.

All security-related actions are managed through the centralized control logic of the microcontroller, ensuring that responses are consistent with system settings such as auto mode enablement. By coordinating motion detection, camera activation, and lighting control, the system provides a comprehensive and context-aware security response.

This integrated approach allows the system to monitor, respond, and inform the user in real time, enhancing both functionality and user awareness without requiring continuous manual supervision

4. Integrated System Validation and Performance Results

4.1 Introduction

The final phase of development focused on validating the performance of the fully integrated Solar Home Lighting System under realistic operating conditions. While individual components and functions were verified during subsystem development, the objective of this section is to evaluate how the complete system performs as a unified device.

System-level testing was conducted to assess core functionality, including power delivery, local and remote control, sensor-driven automation, data communication, and security response. These tests were designed to verify that the system operates correctly when all features are active and interacting simultaneously, rather than in isolation.

In addition to functional validation, performance metrics such as response time, latency, reliability, and electrical stability were measured to quantify system behavior. These results provide a direct evaluation of how effectively the system meets its intended operational goals and highlight any limitations observed during integrated operation.

The following subsections present the results of these tests, organized by major system function, with each section providing measured data, analysis, and conclusions regarding overall system performance.

4.2 Power Delivery and Rail Stability Validation

Power rail stability was evaluated under representative system operating conditions. The system was powered directly from the charge controller battery, with the input voltage at three approximate levels: 10.8 V (low), 12.0 V (nominal), and 14.2 V (fully charged).

Testing was conducted while the system supplied lighting loads and supported normal operation of control electronics, sensors, camera, and wireless communication. The objective was to verify that the regulated voltage rails remain stable across the full battery range and under varying load conditions.

4.2.1 12 V Rail Regulation Under Lighting Load

The 12 V rail was tested by increasing lighting load (0, 1, 3, and 5 lights) at each battery voltage level. Output voltage was measured under each condition.

Load Condition	Vout @ 10.8 V	Vout @ 12.0 V	Vout @ 14.2 V
No Load	12.3284 V	12.3283 V	12.3287 V
1 Light	12.3259 V	12.3256 V	12.3269 V
3 Lights	12.3230 V	12.3232 V	12.3239 V
5 Lights	12.3192 V	12.3207 V	12.3207 V

Table 4.2.1 12 V Output Voltage Under Load

12 V Converter Output vs Load Condition

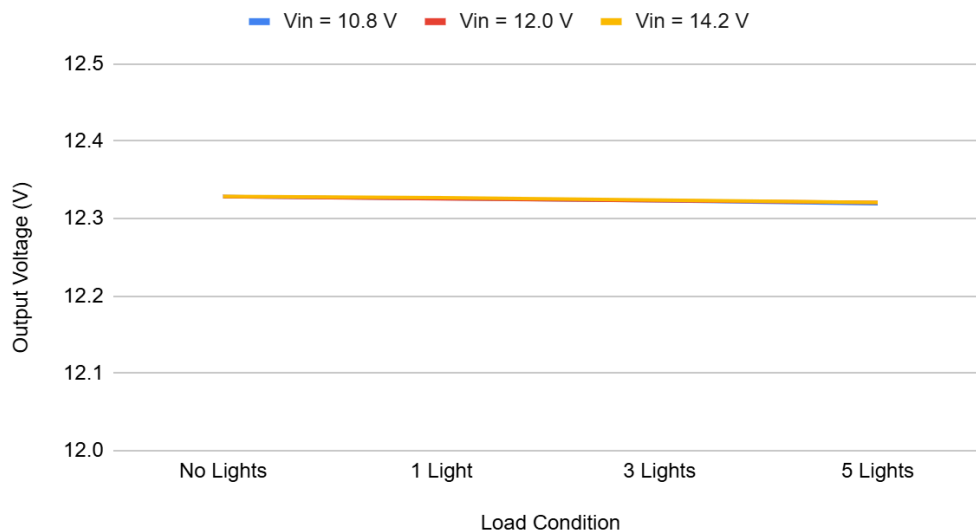


Figure 4.2.1 12 V Output Voltage vs Load Condition

The output voltage decreases slightly with increasing load but remains within 10 mV across all conditions.

4.2.2 3.3 V Rail Stability Under Full System Load

The 3.3 V rail was evaluated under full system operation, including:

- ESP32 active with Wi-Fi communication
- All lighting outputs enabled
- Camera transmitting images and notifications
- All sensors operating

Vin (V)	Vout (V)	Error from Ideal 3.300 V (V)	Percent Error (%)
10.8	3.333	0.033	1.00%
12	3.334	0.034	1.03%
14.2	3.334	0.034	1.03%

Table 4.2.3 3.3 V Output Voltage Under Full System Load

Metric	Value
Minimum Vout	3.333 V
Maximum Vout	3.334 V
Total Variation	0.001 V (1 mV)
Max Percent Variation	~0.03% across input range

Table 4.2.4 3.3 V Rail Summary Metrics

The 3.3 V output remains effectively constant across the full battery range under maximum system load.

4.2.3 Summary

Both the 12 V and 3.3 V rails maintain stable output across the full battery voltage range and under representative system loading conditions, indicating that the power architecture supports reliable system operation.

4.3 Local Control Interface Performance (HMI Validation)

The local control interface was validated by testing lighting control functionality, system startup behavior, and continuous operation under normal use conditions. Commands were issued through the touchscreen interface while monitoring lighting outputs and system state to verify correct operation and synchronization.

4.3.1 Functional Control and System Reliability

Test Scenario	Test Condition	Expected Result	Measured Result	Status
Local lighting control operation	100 manual commands issued through touchscreen (ON/OFF for Porch, Foyer, Security)	Correct lighting output activated and HMI state synchronized	100 / 100 commands executed correctly (100% success, 0 cross-trigger events)	Complete
System startup reliability	System power-cycled 30 times	HMI and lighting control initialize correctly on startup	30 / 30 successful system startups (100% success)	Complete
Continuous runtime stability	System operated continuously for 2 hours including periodic use of power-save display mode, and monitoring Battery Percentage	System remains responsive with no control failures or metric discrepancies	2 hours continuous operation with no resets or control failures; HMI recovered correctly from power-save state	Complete

Table 4.3.1 Local Control Interface Validation Results

The results show consistent and reliable operation of the local control interface, with no observed command errors, startup failures, or runtime instability.

4.4 Sensor-Based Automation Performance and Reliability

Motion sensor performance was evaluated to verify detection reliability across the operating range of both indoor and outdoor PIR sensors. Testing was conducted by triggering each sensor at increasing distances and recording detection success rate.

4.4.1 Motion Sensor Detection Reliability

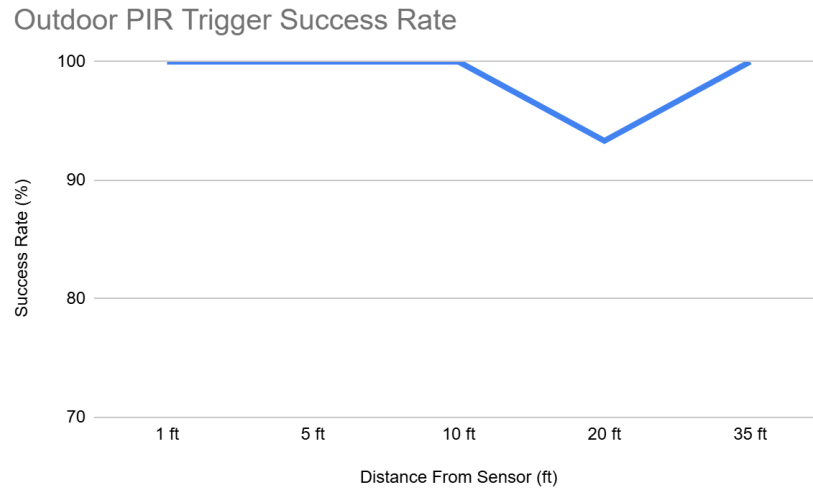


Figure 4.4.1 Outdoor PIR Trigger Success Rate vs Distance

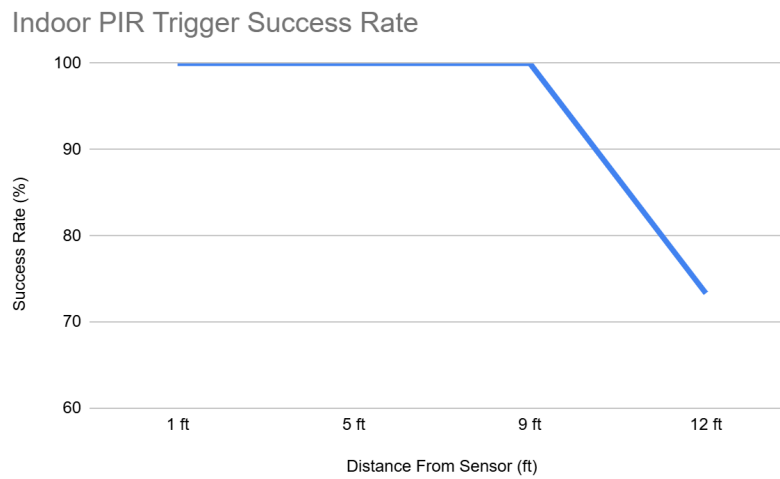


Figure 4.4.2 Indoor PIR Trigger Success Rate vs Distance

Sensor	Rated Range	Measured Performance	Result
Indoor	9 ft	100% ≤ 9 ft, 73% @ 12 ft	PASS
Outdoor	35 ft	100% ≤ 10 ft, 93% @ 20 ft, 100% @ 35 ft	PASS

Table 4.4.1 Motion Sensor Performance Summary

The indoor PIR sensor maintained 100% detection within its rated range of 9 ft. Detection accuracy decreased beyond this range, with reduced performance observed at 12 ft. The indoor sensor requires a higher target density for reliable detection, consistent with its intended use in enclosed environments where human motion is the primary trigger rather than smaller objects or low-motion targets.

The outdoor PIR sensor maintained reliable detection across its full rated range of 35 ft. A minor reduction in detection accuracy was observed at intermediate distance (20 ft), which is attributed to reduced motion sensitivity at longer ranges. At greater distances, larger or more pronounced motion is required to trigger detection, resulting in occasional missed events during low-motion conditions.

The results indicate that both sensors operate within their expected performance ranges, with detection behavior consistent with their intended indoor and outdoor applications.

4.5 Remote Monitoring and Battery Data Validation

Battery polling performance was evaluated to verify reliable transmission of battery state data from the system to the mobile application. The system was operated under normal conditions while periodically reporting battery percentage to the application.

4.5.1 Battery Poll Accuracy and Reliability

Battery data was polled at a fixed interval of 30 seconds over a continuous 30-minute test period. Each poll event was monitored to confirm successful transmission and correct display in the application interface.

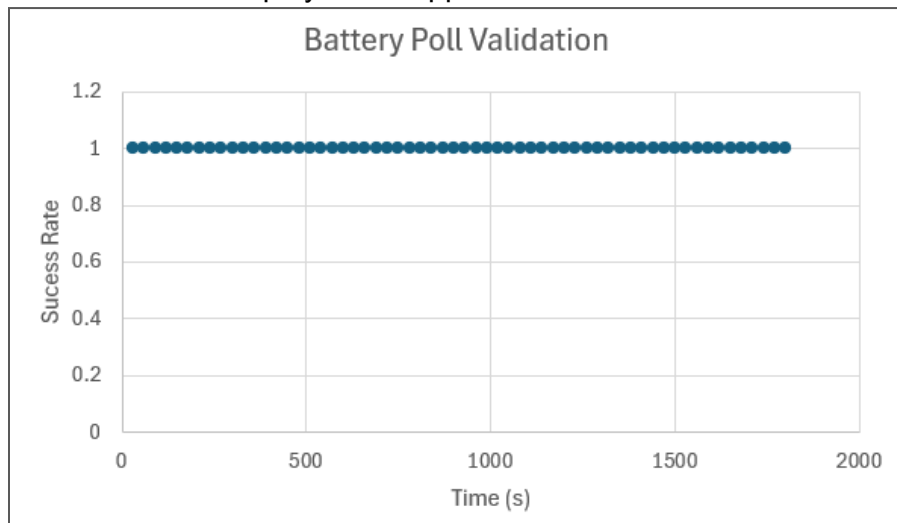


Figure 4.5.1 Battery Poll Success Rate Over Time

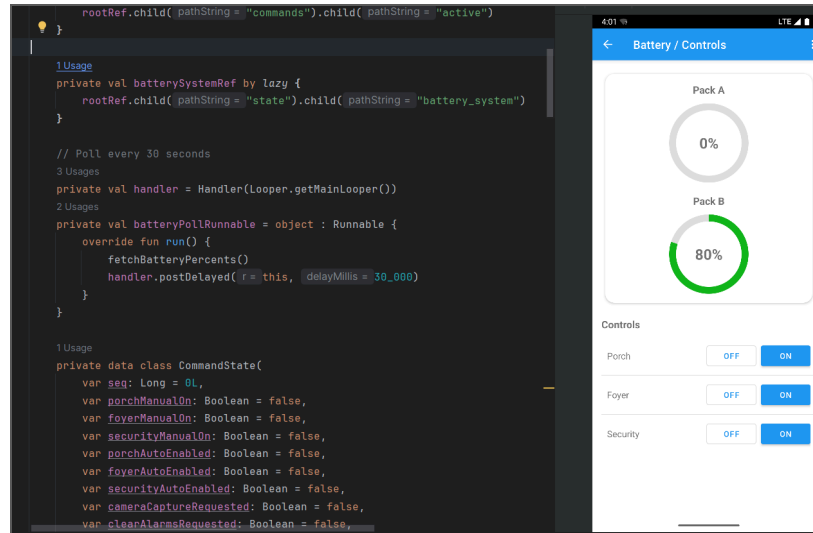


Figure 4.5.2 Mobile Application Battery Display Interface

A total of 60 polling events were executed during the test interval, with all events successfully received and displayed by the application. No dropped or missed updates were observed during the test period.

4.6 Security Response and Notification Performance

The end-to-end latency of the security response system was evaluated by measuring the time from motion detection by the outdoor PIR sensor to receipt of the corresponding notification and image in the mobile application. This test captures the full system response, including sensing, processing, communication, and application update.

4.6.1 Motion Detection to Application Notification Latency

Latency measurements were taken over 25 test events at varying distances. Each test recorded the total time required for the following sequence:

Motion Sensor → MCU → ESP32 → Database → Application Notification and Image Display

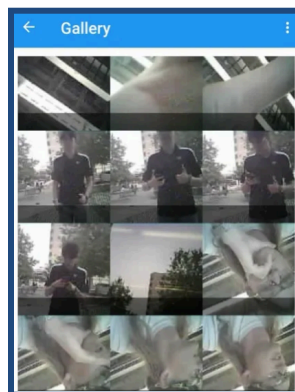


Figure 4.6.1 Security Event Image Capture and Application Display

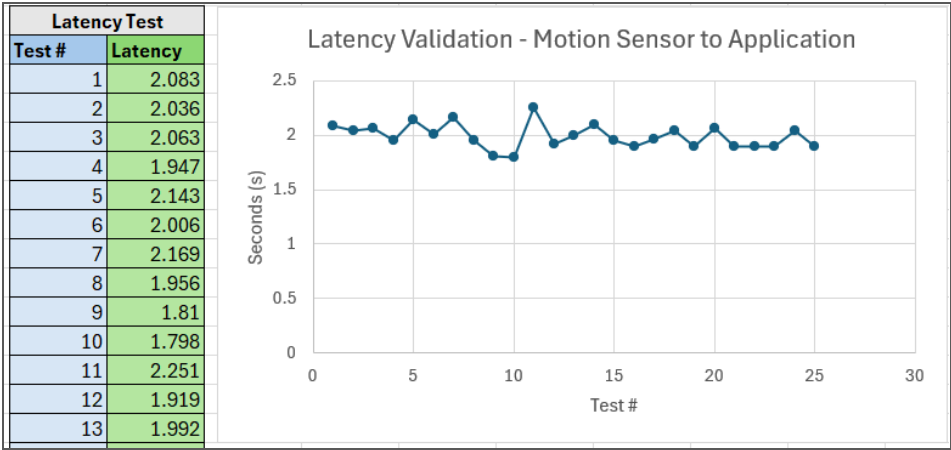


Figure 4.6.2 End-to-End Latency vs Test Number

Metric	Value
Number of Tests	25
Average Latency	1.98 s
Min Observed Latency	~1.8 s
Max Observed Latency	~2.2 s

Table 4.6.1 Latency Measurement Summary

The measured latency remained consistent across all test cases, with an average end-to-end response time of approximately 1.98 seconds. This includes sensor detection, system processing, database communication, and application update with image delivery. The results demonstrate reliable and repeatable system response, with consistent notification and image delivery to the application following motion detection.

4.7 Local Control Response and Latency

The local control response of the system was validated by measuring command execution latency and verifying correct system behavior across the HMI–MCU–lighting control pipeline. This test evaluates both functional correctness and response time of the local control interface.

4.7.1 HMI Command Execution and System Behavior

The validation checks performed during this test are summarized in Table 4.7.1. These checks verify that commands issued from the HMI are correctly transmitted, processed, executed, and reflected across the system.

#	Validation Check	Expected Outcome	Observed Result
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1	HMI command transmission over UART	Correct command string (e.g., PORCH=1) sent to MCU	TRUE
2	MCU command processing and execution	Corresponding light state updates correctly	TRUE
3	State synchronization to HMI	HMI display reflects system state changes	TRUE
4	Charge controller telemetry display on HMI	Battery percentage and active pack displayed correctly	TRUE

Table 4.7.1 HMI System Validation Checks

4.7.2 HMI Response Latency

Latency was measured for each lighting zone by recording the time from HMI input to lighting output response. A total of 15 tests were conducted per lighting zone (Porch, Foyer, Security).

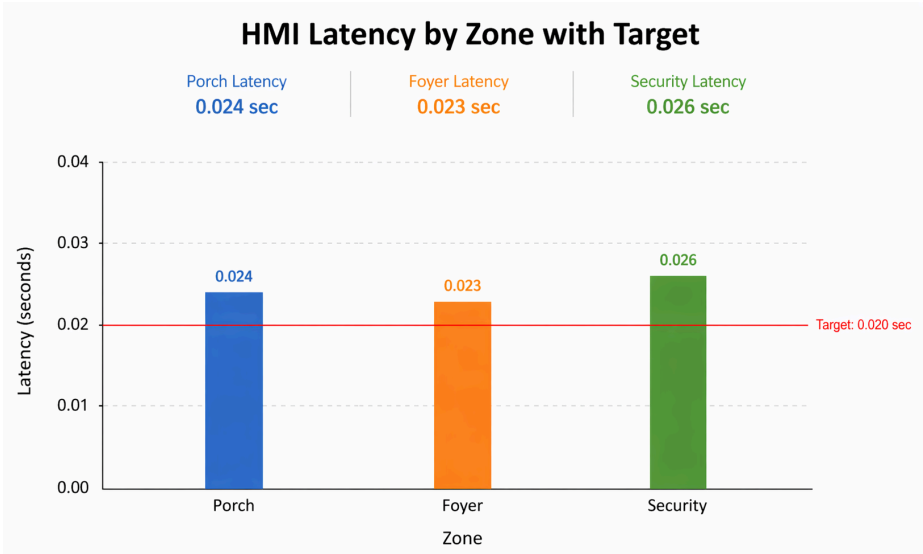


Figure 4.7.1 HMI Command Latency by Lighting Zone

The measured latencies for all zones remained within approximately 20–30 ms, indicating fast and consistent response for local control operations.

4.8 Sensor-Based End-to-End System Validation

The sensor-based automatic operation of the system was validated by testing the full response from environmental input to system output. This includes motion-triggered lighting behavior, notification response, and system behavior under different operating modes.

4.8.1 Sensor-Driven System Behavior

The validation checks performed are summarized in Table 4.8.1 and verify correct operation of automatic lighting, security response, and control logic.

#	Validation Check	Expected Outcome	Observed Result
1	Sensor-triggered light activation (Auto enabled)	Light turns ON when motion/dark condition is detected	Near-instant response (<0.05 s)
2	Motion-triggered notification (front door/security)	Camera triggers and notification is sent to application	Notification received successfully (~1–2 s delay)
3	Auto mode disabled behavior	System does not respond to sensor input when Auto mode is OFF	No light activation observed
4	Sensor hold and timeout behavior	Light remains ON during motion and turns OFF after timeout (~10 s)	Correct hold and shutdown timing observed

Table 4.8.1 Sensor-Based System Validation Checks

4.8.2 Sensor-to-Output Latency

Latency was measured from sensor trigger to lighting output for each zone. A total of 10 tests were conducted per zone (Porch, Foyer, Security), and the average latency was calculated from these measurements.

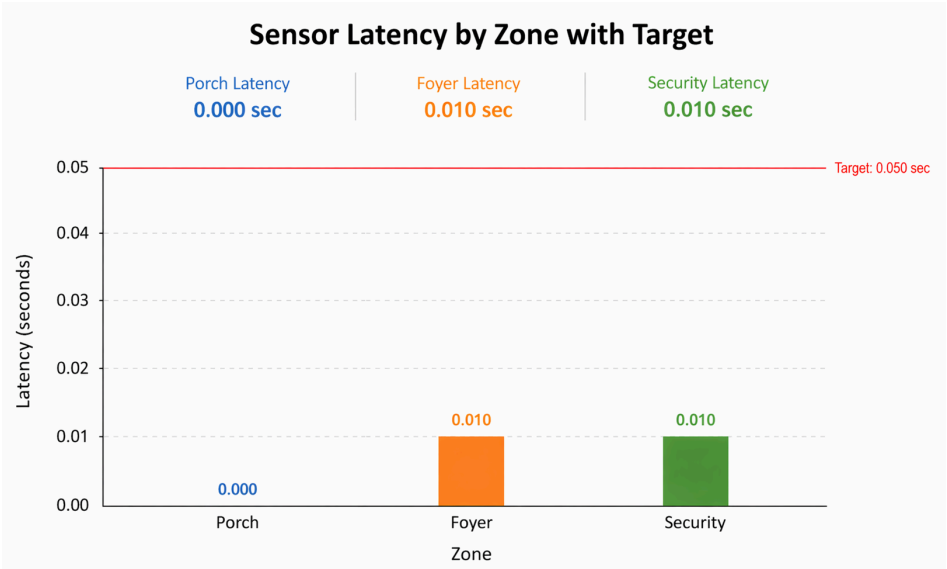


Figure 4.8.1 Sensor-Based Lighting Latency by Zone

The measured response times were within 0–10 ms across all zones, indicating near-instantaneous activation of lighting in automatic mode.

4.9 Application-Based End-to-End Control Validation

The remote control functionality of the system was validated by testing the complete control path from the mobile application to lighting output and back to application state update. This test evaluates both command execution and system feedback across the full communication pipeline.

4.9.1 Remote Control System Behavior

The validation checks performed are summarized in Table 4.9.1 and verify correct operation of command transmission, execution, and system feedback.

#	Check	Expected	Result
1	App writes command to database	Correct control value updates in database	Complete
2	Command reaches MCU and turns on light	Light turns ON/OFF from app input	~1–2 s delay
3	App updates after light changes	App reflects correct light state	~1–2 s delay

4	App displays battery/system data	Battery % and system data displayed correctly	Complete
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Table 4.9.1 Application Control Validation Checks

4.9.2 Application-to-Output Latency

Latency was measured from application command input to lighting response for each zone. A total of 10 tests were conducted per zone (Porch, Foyer, Security), and the average latency was calculated.

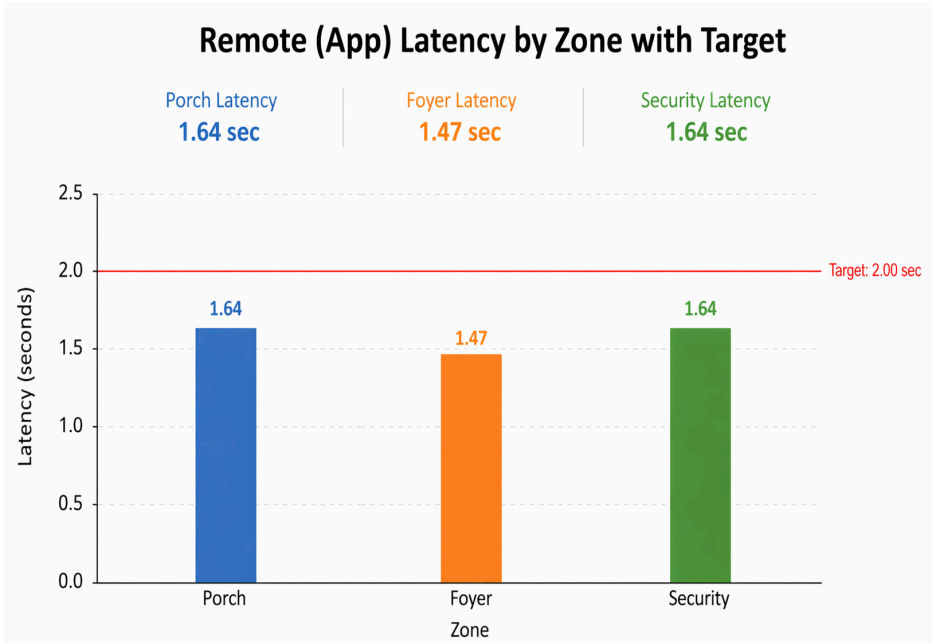


Figure 4.9.1 Remote (Application) Latency by Zone

The measured latency ranged from approximately 1.4 to 1.6 seconds across all zones, reflecting the delay introduced by database communication and network transmission.

4.10 Solar panel regulation under load

The validation results demonstrate that the system successfully integrates solar energy generation, battery storage, and load delivery. The system maintains stable load operation, properly manages battery charging based on state of charge, and ensures continuous energy flow under varying conditions.

4.10.1 Dual energy delivery (load + storage)

This figure shows that the system simultaneously powers the load while charging the battery. The total useful output power represents the combined energy delivered to both the load and storage.

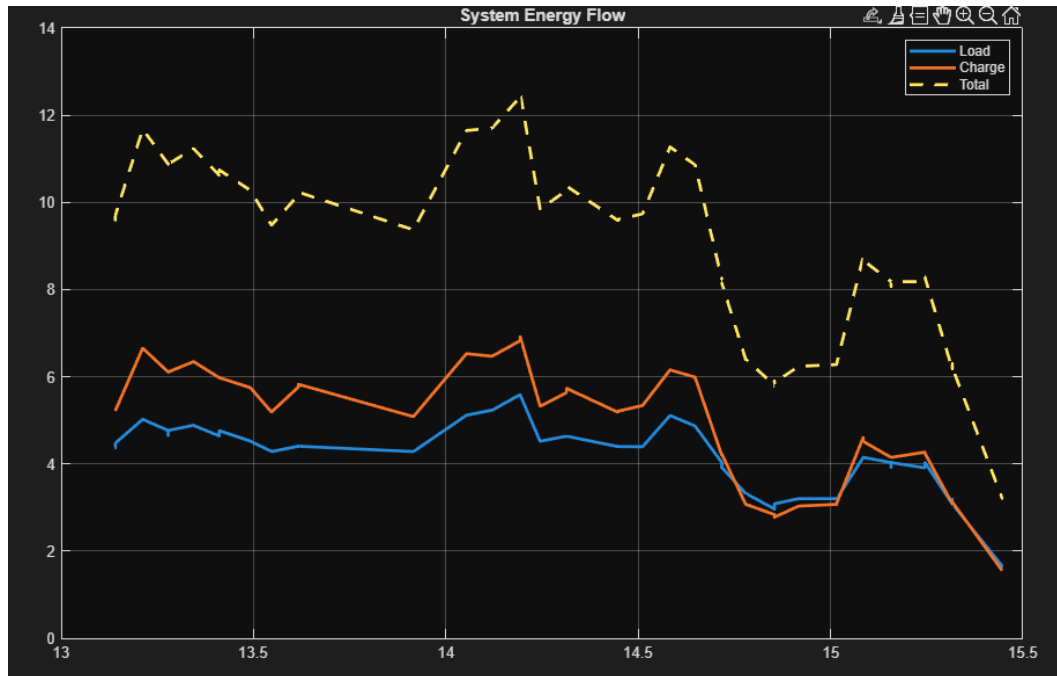


Figure 4.10.1 Load power, battery charging power, and total useful output power over time.

4.10.2 Correct energy storage behavior

Battery B shows a steady increase in SOC, indicating successful charging, while Battery A remains relatively stable as it powers the load.

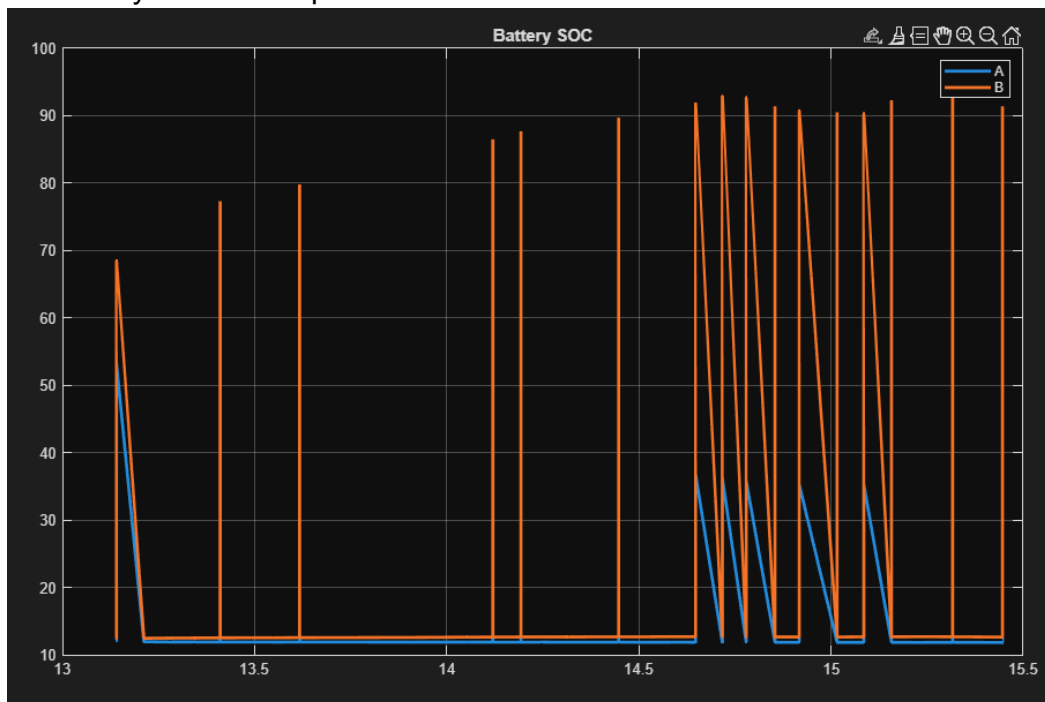


Figure 4.10.2 State of charge of Battery A and Battery B during operation.

4.10.3 Valid charging conditions

The solar panel voltage remains consistently higher than both battery voltages throughout the test period. This ensures that the system maintains valid charging conditions and allows continuous energy transfer from the solar source to the battery.

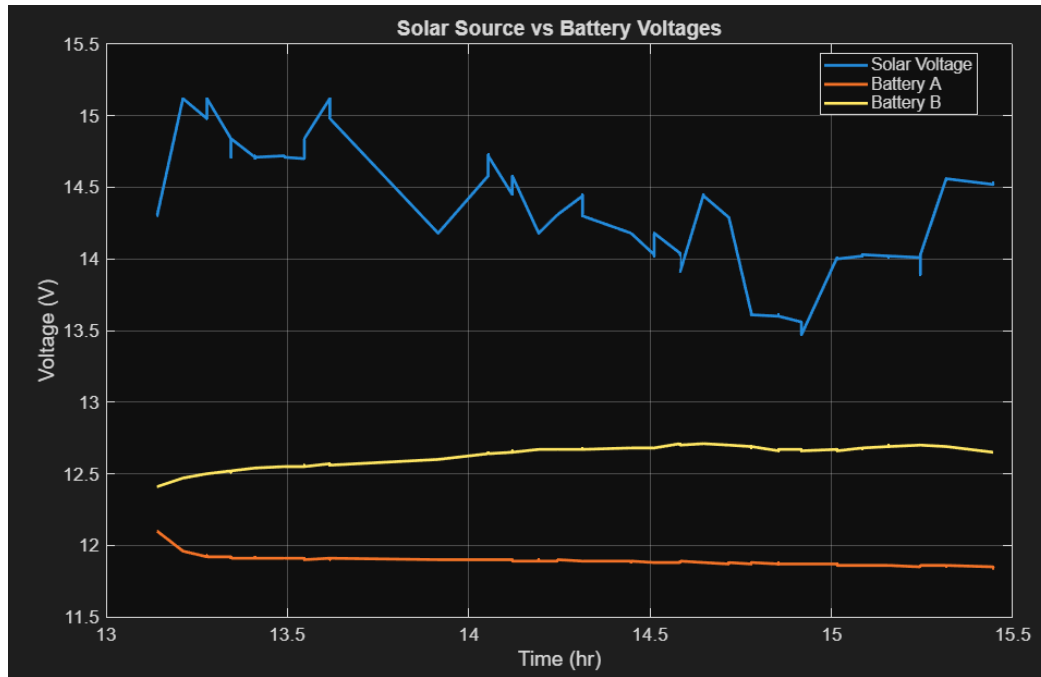


Figure 4.10.2 Solar panel voltage compared to battery voltages.

5. Overall System Results

The Solar Home Lighting System was successfully integrated into a unified platform capable of coordinated power management, user control, automated response, and remote monitoring. The system demonstrated reliable operation across all major functional areas, including power delivery, local and remote control, sensor-based automation, and security response.

As shown in Figure 5.1, the system was implemented on a model residential structure to demonstrate its intended real-world application. This configuration highlights the system's ability to provide lighting and security functionality in a residential environment, including sensor-based automation and distributed lighting control.



Figure 5.1 System Implementation on Model Residential Structure

Final Report

Solar Home Lighting

Validation results confirmed that the system maintains stable operation under realistic conditions. The power architecture provided consistent voltage regulation across the full battery range while supporting both lighting loads and control electronics. Local control through the HMI exhibited fast and reliable response, while sensor-based automation achieved near-instantaneous lighting activation with accurate detection performance within the rated operating range. Remote control and monitoring through the mobile application functioned correctly, with expected latency introduced by database and network communication.

The integrated security system demonstrated consistent end-to-end performance, successfully detecting motion, capturing images, and delivering notifications to the application with repeatable response times. Battery monitoring and data reporting were also validated, showing reliable communication between the system and the user interface.

A complete view of the integrated hardware is shown in Figure 5.2, which illustrates the full system assembly in its bench configuration. This view highlights the integration of the power system, control electronics, sensing components, and communication hardware into a single operational platform.



Figure 5.2 Fully Integrated System Hardware Bench Assembly

Final Report

Solar Home Lighting

While overall system performance met the intended design objectives, integration revealed sensitivity to power disturbances during battery switching events, which resulted in transient effects such as momentary resets and lighting flicker. These observations highlight areas for potential improvement in future iterations, such as enhanced filtering or energy buffering to improve power stability during transitions. Overall, the system meets its primary objective of providing a solar-powered residential lighting and security platform capable of autonomous operation, user interaction, and remote accessibility. The results demonstrate successful integration of all system components into a functional and reliable device, validating the effectiveness of the system architecture and design approach.